



भारत के विद्युत क्षेत्र का 1947 से 2025 तक का संवर्धन
**GROWTH OF ELECTRICITY SECTOR IN INDIA FROM 1947-
2025**

भारत सरकार
GOVERNMENT OF INDIA
विद्युत मंत्रालय
MINISTRY OF POWER
केन्द्रीय विद्युत प्राधिकरण
CENTRAL ELECTRICITY AUTHORITY
नई दिल्ली
NEW DELHI
नवम्बर, 2025
November, 2025

घनश्याम प्रसाद
अध्यक्ष
केंद्रीय विद्युत प्राधिकरण
भारत सरकार

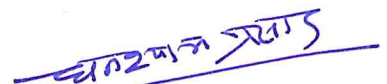


प्राक्कथन

केंद्रीय विद्युत प्राधिकरण द्वारा वार्षिक रूप से प्रकाशित पुस्तिका “ भारत में विद्युत क्षेत्र की संवृद्धि” विभिन्न वर्षों में भारतीय विद्युत क्षेत्र की संवृद्धि का विस्तृत परिदृश्य उपलब्ध कराती है। प्रासंगिक सूचना और सांख्यिकी आलेखों, पाई-चार्ट, मानचित्रों और सारणियों के रूप में प्रस्तुत की गई हैं। पंचवर्षीय योजनावार विकास के आरम्भ होने के बाद से भारतीय विद्युत क्षेत्र ने बहुत प्रगति की है। यह प्रकाशन महत्वपूर्ण विकास संकेतकों जैसे स्थापित विद्युत उत्पादन क्षमता, विद्युत ऊर्जा उत्पादन, पारेषण और संवितरण नेटवर्क, कैप्टिव विद्युत संयंत्रों और विद्युत खपत के पैटर्न आदि की संवृद्धि के विवरण उपलब्ध कराता है।

इस प्रकाशन के लिए आंकड़े देश में विद्युत क्षेत्र के विभिन्न संगठनों से प्राप्त किये गए हैं। प्रति व्यक्ति विद्युत खपत, खपत की विभिन्न श्रेणियों का प्रतिशतांक हिस्सा, पारेषण और संवितरण हानि आदि जैसी महत्वपूर्ण सांख्यिकी की कुछ विकसित और विकासशील राष्ट्रों के आंकड़ों के साथ तुलना की गई है। आशा है कि यह पुस्तिका भारतीय विद्युत क्षेत्र पर संक्षिप्त और लाभदायक सूचना उपलब्ध कराने में बहुत उपयोगी सिद्ध होगी।

नवम्बर, 2025


(घनश्याम प्रसाद)

**Ghanshyam Prasad
Chairperson
Central Electricity
Authority
Govt. of India**

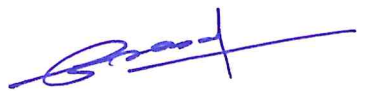


FOREWORD

The booklet “Growth of Electricity Sector in India” published annually by Central Electricity Authority provides panoramic view of growth of the Indian Power Sector over the years. The relevant information and statistics have been presented in the forms of graphs, pie-charts, maps and tables. The Indian Electricity Sector has come a long way ever since 5 year plan wise development was initiated. This publication provides details of the growth of vital development indicators like Installed Electricity Generating Capacity, Electrical Energy Generation, Transmission and Distribution Network, Captive Power Plants and pattern of Electricity Consumption etc.

The data for this publication has been sourced from various power sector entities in the country. The important statistics like per capita electricity consumption, percentage share of different categories of consumption, transmission and distribution losses etc., have been compared with the data of some of the developed and developing nations. I hope that this booklet will be very helpful in providing concise & useful information on Indian Electricity Sector.

November, 2025


(GHANSHYAM PRASAD)

A. Balan
Member (Planning)
Central Electricity Authority
Government of India



PREFACE

Power Sector in India has grown significantly since independence, both in the Installed Capacity and Transmission & Distribution (T&D) system. The total Power Generating Capacity of (utilities & non utilities) has increased from a meagre 1362 MW in 1947 to about 556.9 GW at the end of March, 2025. The Per Capita Electricity Consumption which was a mere 16.3 units in 1947, has increased to 1400 units in 2024-25. In the last few years, the shortages have reduced substantially and, in fact, at present we have unutilized generating capacity.

This booklet brought out by CEA is a regular feature, which contains vital information in the form of graphs, pie charts, maps and tables. The information covers 5 yearly (earlier called Plan period) pattern of growth accomplished for various important indicators like Installed Capacity, Hydroelectric Potential, state wise forecast of Electricity Demand of the country, Electrical Energy Generation, Transmission & Distribution network, Power Supply Position, Captive Power Plants, pattern of Electricity Consumption of the country as well as Per Capita Consumption. The booklet also contains maps showing state wise Installed Electricity Generating Capacity, thus affording a panoramic view of our country.

The Statistics from other countries i.e. Generation from various sources, Per Capita Consumption as well as category-wise Electricity Consumption are also shown in this booklet.

I am hopeful that this handy booklet would prove to be useful to all stakeholders.

November, 2025


(A. Balan)

ए. बालन
सदस्य (योजना)
केंद्रीय विद्युत प्राधिकरण
भारत सरकार



प्रस्तावना

स्वतंत्रता के बाद भारत में विद्युत क्षेत्र में स्थापित क्षमता और ट्रांसमिशन व वितरण (टी एंड डी) प्रणाली दोनों में उल्लेखनीय वृद्धि हुई है। कुल विद्युत उत्पादन क्षमता (यूटिलिटी और गैर-यूटिलिटी की) 1947 में 1362 मेगावाट से बढ़कर मार्च अंत, 2025 में लगभग 556.9 गीगा वाट हो गई है। प्रति व्यक्ति विद्युत खपत जो 1947 में केवल 16.3 यूनिट थी, 2024-25 में बढ़कर 1400 यूनिट हो गई है। पिछले कुछ वर्षों में, विद्युत कमी काफी हद तक कम हो गई हैं और वास्तव में, वर्तमान में, हमारे पास अप्रयुक्त उत्पादन क्षमता है।

सीईए द्वारा प्रकाशित यह पुस्तिका एक नियमित प्रस्तुति है जिसमें ग्राफ, पाई-चार्ट, नक्शों और तालिकाओं के रूप में महत्वपूर्ण जानकारी शामिल है। जानकारी में शामिल विभिन्न महत्वपूर्ण संकेतकों जैसे स्थापित क्षमता, जल विद्युत क्षमता, देश की विद्युत मांग का राज्यवार पूर्वानुमान, विद्युत ऊर्जा उत्पादन, ट्रांसमिशन और वितरण नेटवर्क, विद्युत आपूर्ति की स्थिति, कैपिटल विद्युत संयंत्र, बिजली संयंत्र, देश की विद्युत खपत के साथ-साथ प्रति व्यक्ति खपत का पैटर्न आदि विभिन्न महत्वपूर्ण संकेतकों के लिए पंच वर्षीय (जिसे पहले योजना अवधि कहा जाता था) विकास पैटर्न शामिल है। इस पुस्तिका में राज्यवार स्थापित बिजली उत्पादन क्षमता दर्शाने वाले मानचित्र भी शामिल हैं, जिससे हमारे देश का एक विशाल परिदृश्य दिखाई देता है।

इस पुस्तिका में अन्य देशों के आंकड़े जैसे कि विभिन्न स्रोतों से उत्पादन व प्रति व्यक्ति खपत के साथ-साथ श्रेणीवार विद्युत खपत भी दर्शाई गई है।

मुझे आशा है कि यह पुस्तिका सभी हितधारकों के लिए उपयोगी साबित होगी।

नवम्बर, 2025


(ए. बालन)

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Table 1
Growth of Electricity Sector in India
Utilities

Sl. No.	As on / during financial year ending with	Installed Capacity (MW)	No.of villages electrified	Length of T & D Lines (Ckt. kms.)(#)	Per Capita Consumption (\$) (kWh)
1	31.12.1947	1,362	N.A.	23,238	16
2	31.12.1950	1,713	3,061	29,271	18
3	31.03.1956(End of the 1st Plan)	2,886	7,294	85,427	31
4	31.03.1961(End of the 2nd Plan)	4,653	21,754	1,57,887	46
5	31.03.1966 (End of the 3rd Plan)	9,027	45,148	5,41,704	74
6	31.03.1969(End of the 3 Annual Plans)	12,957	73,739	8,86,301	98
7	31.03.1974(End of the 4th Plan)	16,664	1,56,729	15,46,097	126
8	31.03.1979(End of the 5th Plan)	26,680	2,32,770	21,45,919	172
9	31.03.1980(End of the 2 Annual Plans)	28,448	2,49,799	23,51,609	172
10	31.03.1985(End of the 6th Plan)	42,585	3,70,332	32,11,956	229
11	31.03.1990(End of the 7th Plan)	63,636	4,70,838	44,07,501	329
12	31.03.1992(End of the 2 Annual Plans)	69,065	4,87,170	45,74,200	348
13	31.03.1997(End of the 8th Plan)	85,795	4,98,836	51,41,413	465
14	31.03.2002(End of the 9th Plan)	1,05,046	5,12,153	60,30,148	559
15	31.03.2007 (End of 10th Plan)	1,32,329	4,82,864	69,39,894	672
16	31.03.2012 (End of 11th Plan)	1,99,877	5,56,633	87,26,092	884
17	31.03.2017 (End of 12th Plan)	3,28,146	5,92,972	1,06,86,448	1,122
18	31.03.2018	3,45,631	5,97,121	1,19,58,511	1,149
19	31.03.2019	3,57,871	5,97,464	1,26,82,649	1,181
20	31.03.2020	3,71,334	5,97,464	1,32,79,315	1,208
21	31.03.2021	3,83,521	5,97,464	1,36,79,477	1,161
22	31.03.2022	3,99,497	5,97,464	1,39,79,990	1,255
23	31.03.2023	4,16,059	5,97,464	1,44,23,639	1,331
24	31.03.2024	4,41,970	5,97,464	1,47,06,316	1,400
25	31.03.2025	4,75,212	5,97,464	1,50,86,410	1460@

Note:

1. N.A.: Not Available.

2. (@) Estimated.

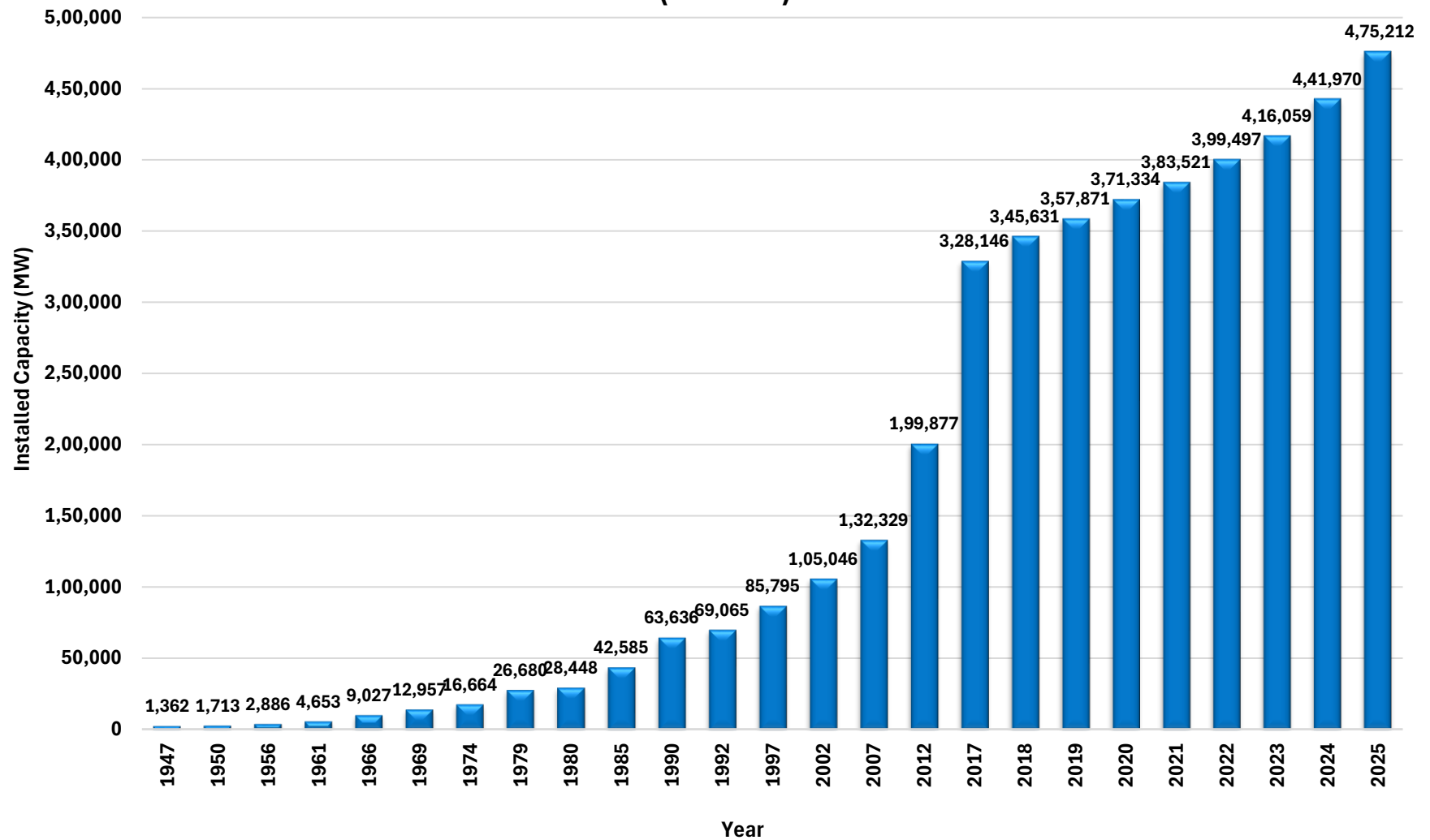
3. (\$) Per Capita Consumption=Gross Electrical Energy availability/ Mid year Population

4. (#) Includes 440 Volts Distribution Lines.

5. Rural electrification is based on revised definition of village electrification in the year 2004 and based on 2011 Census.

Installed Generation Capacity (Utilities)

Chart : 1

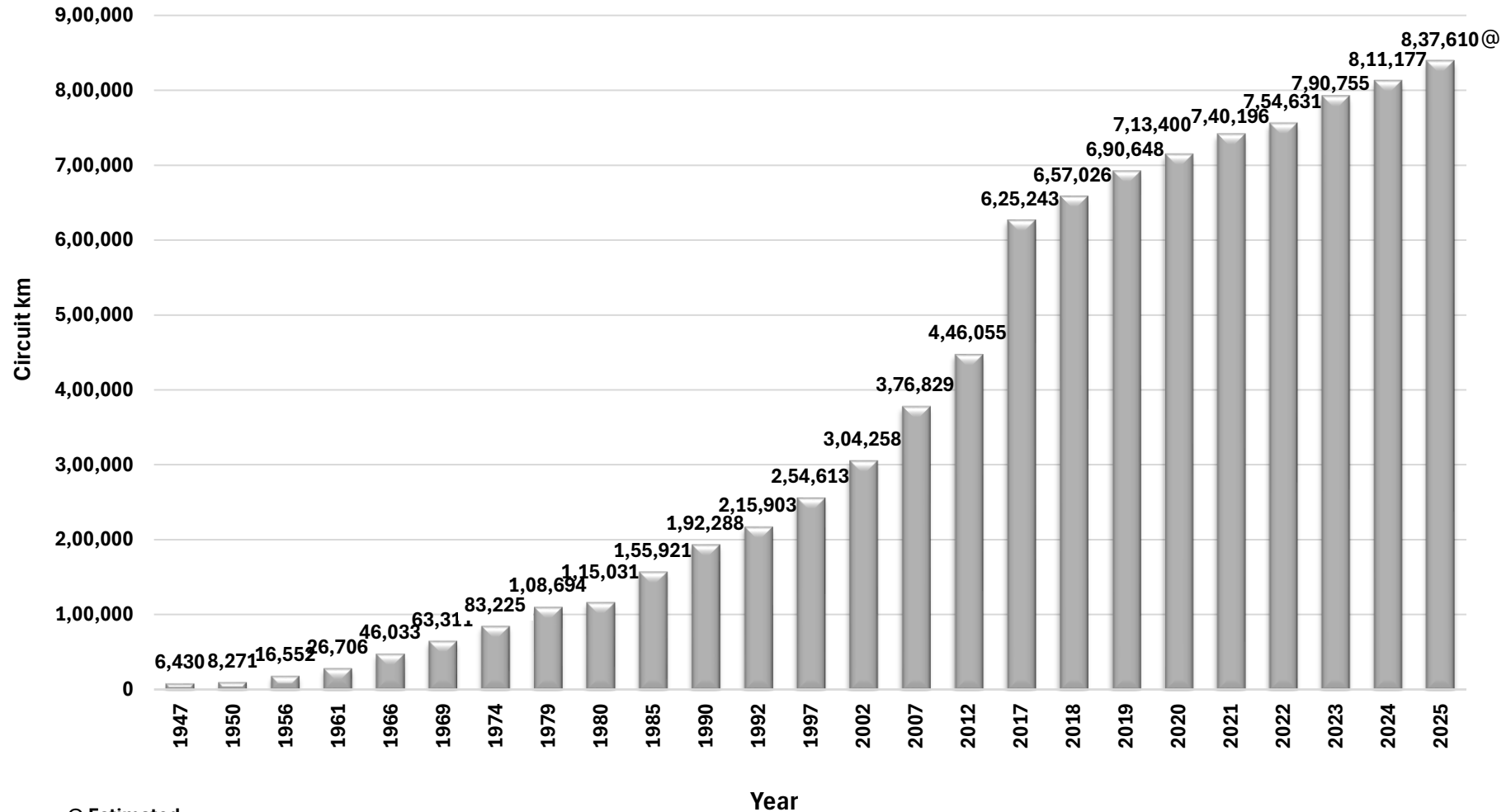


Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

Length of Transmission Lines (66 KV and above)

Chart : 2A



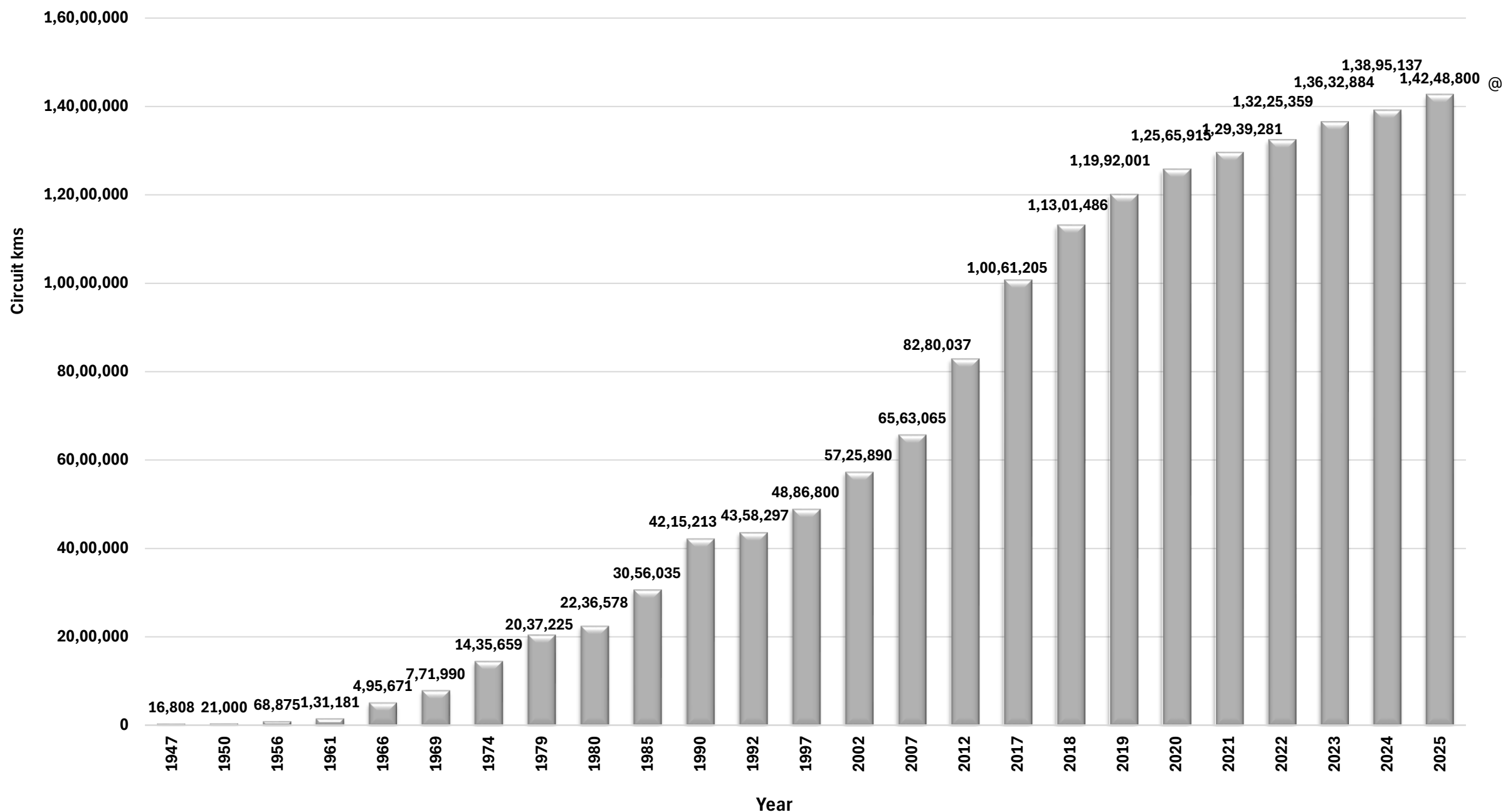
@ Estimated

Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

Chart : 2B

Length of Transmission & Distribution Lines (below 66 kV)

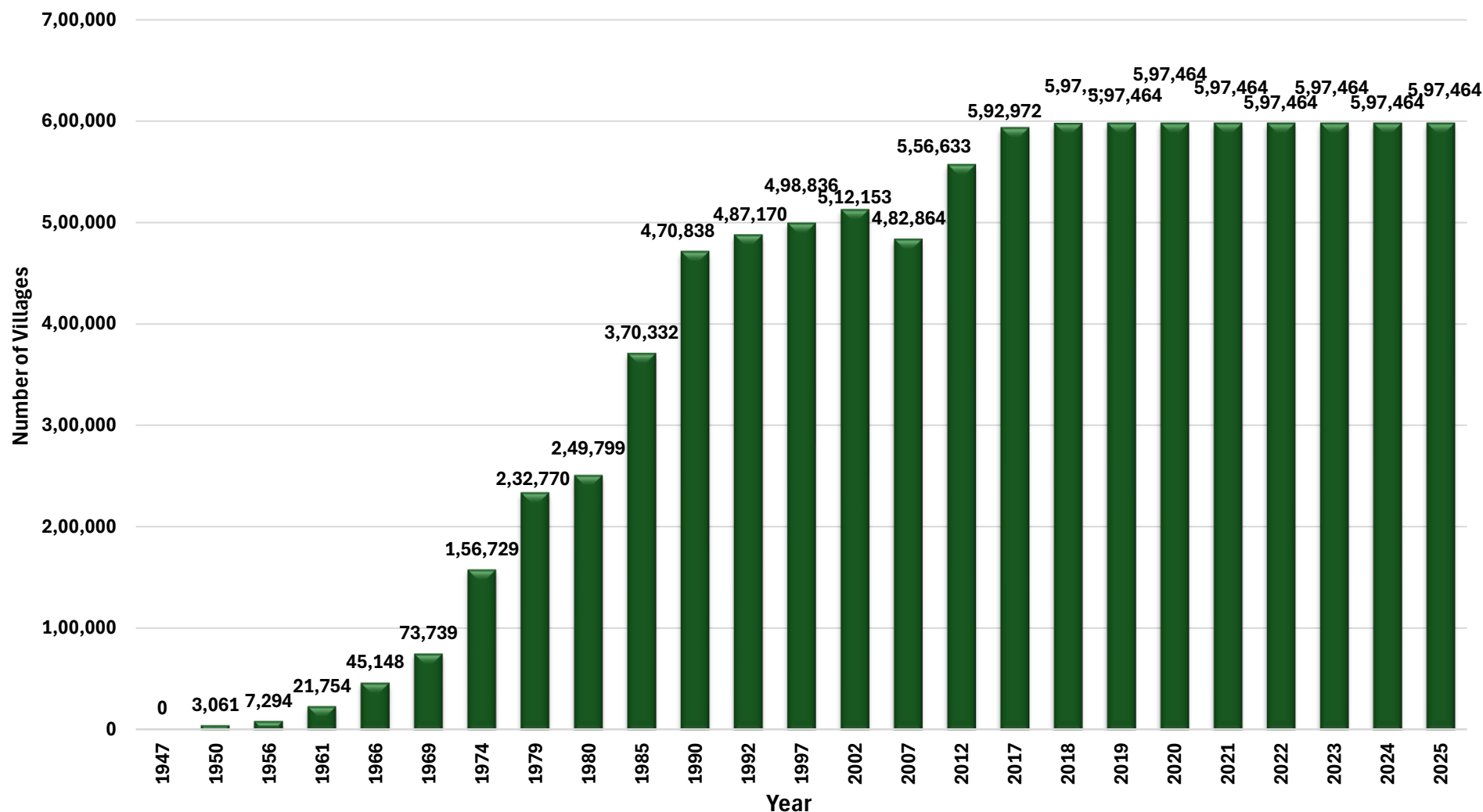


@ Estimated

Note: 1. Figures for the years 1947 & 1950 are as on 31st December.
2. Figures for the year 1955-56 & onward are as on 31st March.

Number of Villages Electrified in the Country

Chart : 3

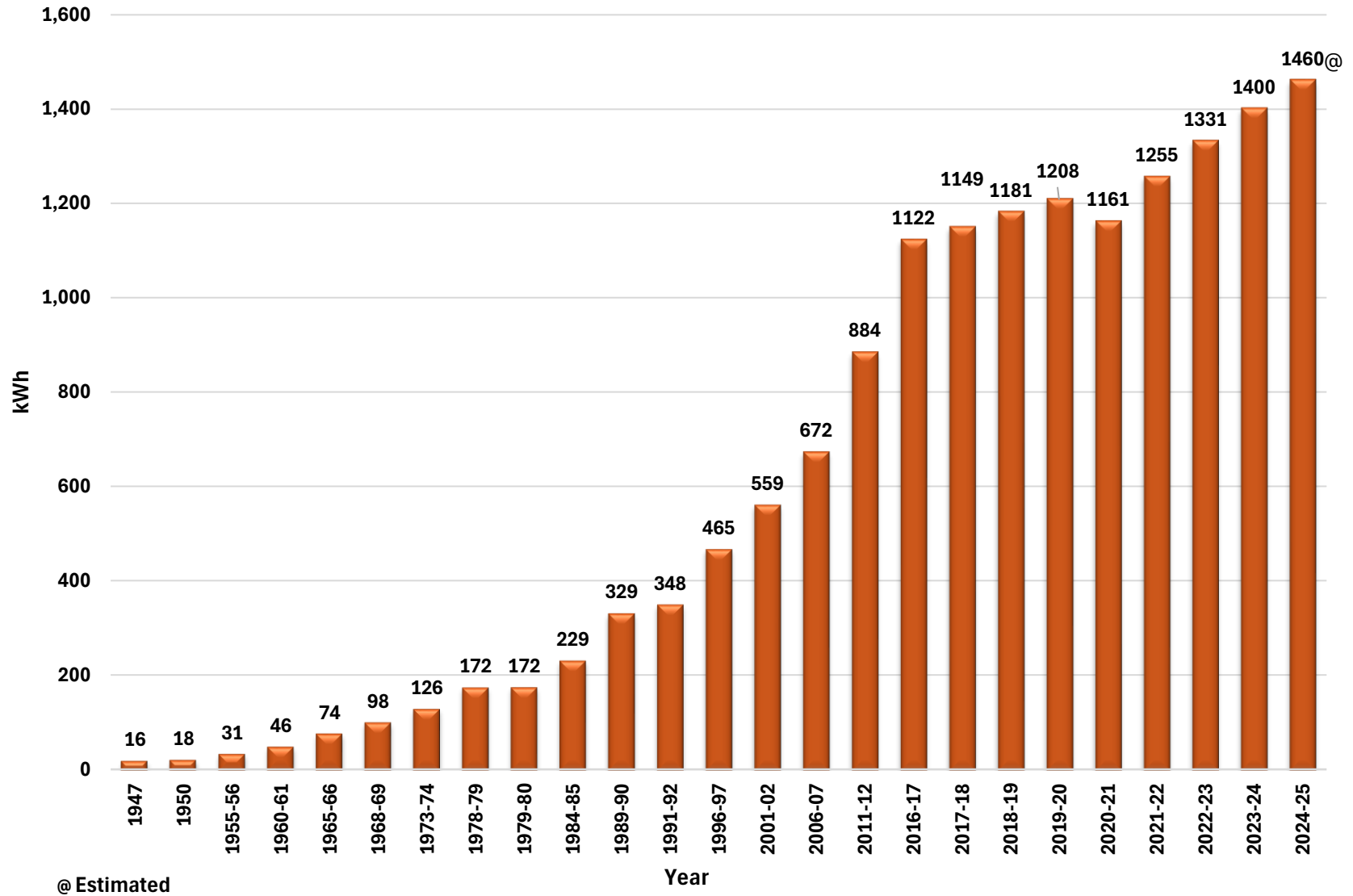


Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

Per Capita Consumption of Electricity

Chart : 4

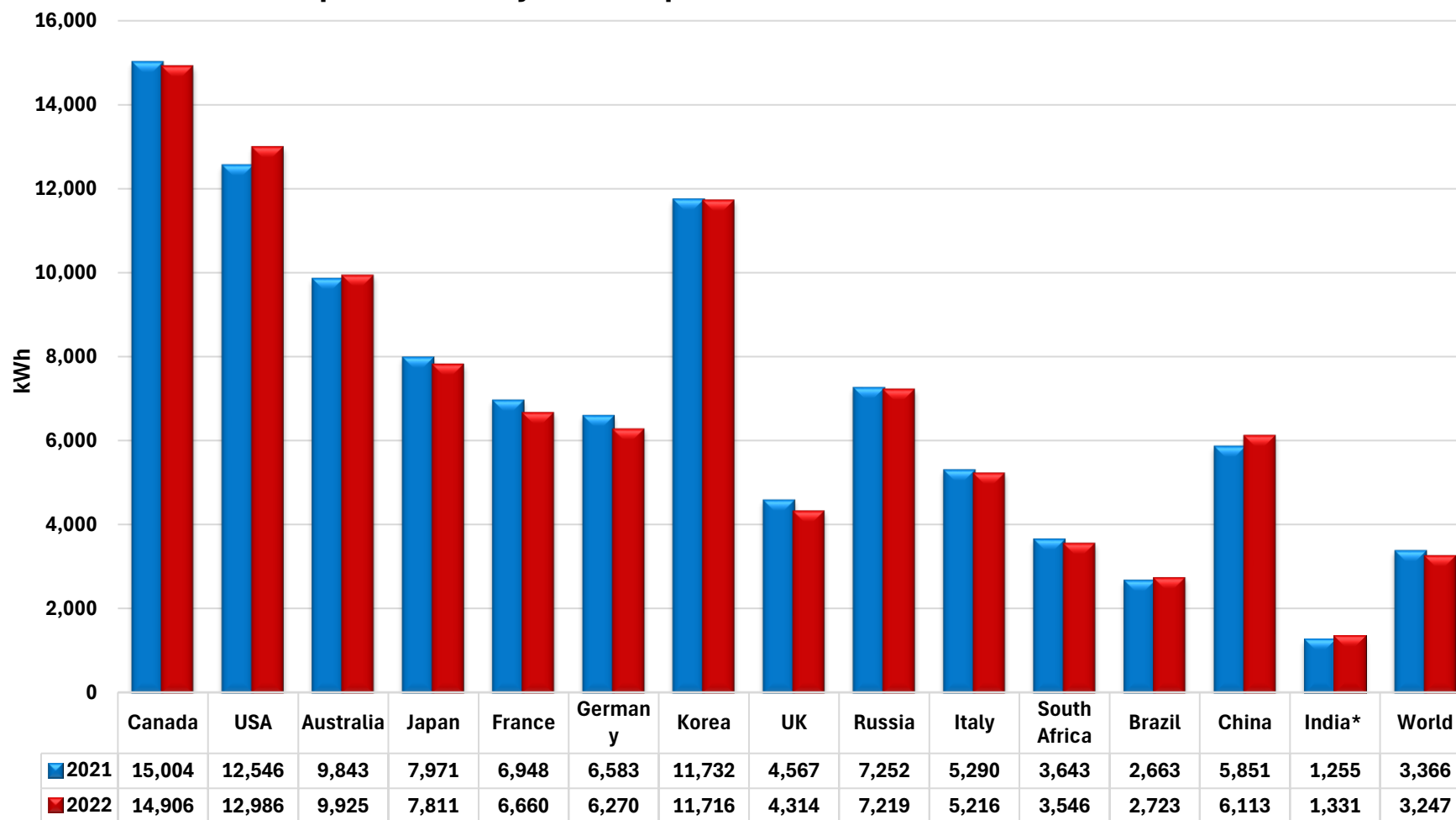


Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

Chart : 5

Per Capita Electricity Consumption in Various Countries in 2021 & 2022



Source : IEA

* Per capita consumption = (gross electrical energy availability / mid year population)

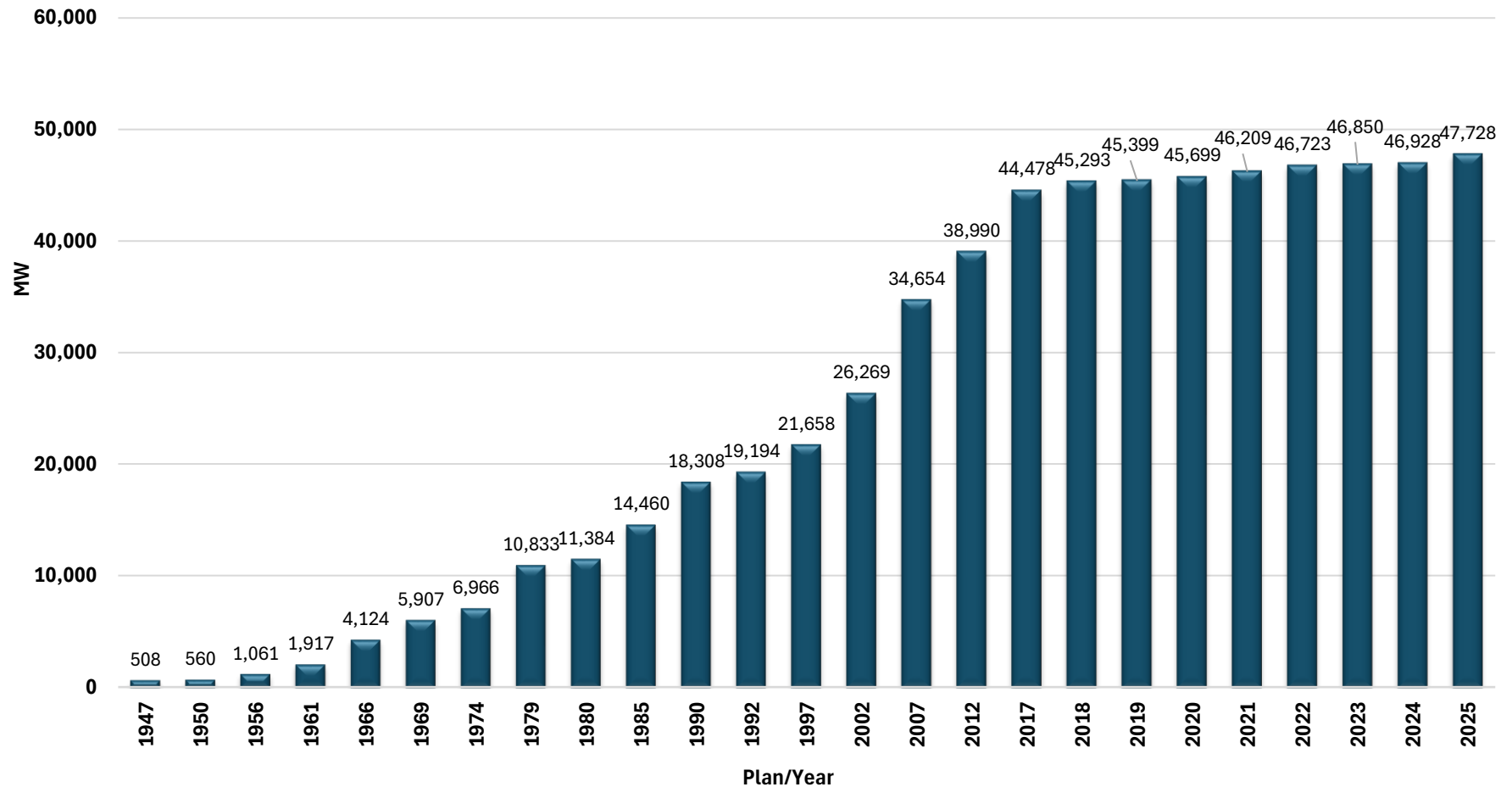
Table 2
Installed Generation Capacity - Mode-wise
(Utilities)

Sl. No.	As on	Thermal				Nuclear	RES			Total
		Coal/Lignite	Gas	Diesel	Total		Hydro	RES (MNRE)	RES Total	
1	31.12.47	756	0	98	854	0	508	0	508	1,362
2	31.12.50	1,004	0	149	1,153	0	560	0	560	1,713
3	31.03.56 (End of 1st Plan)	1,597	0	228	1,825	0	1,061	0	1,061	2,886
4	31.03.61 (End of 2nd Plan)	2,436	0	300	2,736	0	1,917	0	1,917	4,653
5	31.03.66 (End of 3rd Plan)	4,417	134	352	4,903	0	4,124	0	4,124	9,027
6	31.03.69 (End of 3 Annual Plans)	6,640	134	276	7,050	0	5,907	0	5,907	12,957
7	31.03.74 (End of 4th Plan)	8,652	165	241	9,058	640	6,966	0	6,966	16,664
8	31.03.79 (End of 5th Plan)	14,875	168	164	15,207	640	10,833	0	10,833	26,680
9	31.03.80 (End of Annual Plan)	15,991	268	165	16,424	640	11,384	0	11,384	28,448
10	31.03.85 (End of 6th Plan)	26,311	542	177	27,030	1,095	14,460	0	14,460	42,585
11	31.03.90 (End of 7th Plan)	41,237	2,343	165	43,745	1,565	18,308	18	18,326	63,636
12	31.03.92 (End of 2 Annual Plans)	44,791	3,095	168	48,054	1,785	19,194	32	19,226	69,065
13	31.03.97 (End of 8th Plan)	54,154	6,562	294	61,010	2,225	21,658	902	22,560	85,795
14	31.03.02 (End of 9th Plan)	62,131	11,163	1,135	74,429	2,720	26,269	1,628	27,897	1,05,046
15	31.03.07 (End of 10th Plan)	71,121	13,692	1,202	86,015	3,900	34,654	7,760	42,414	1,32,329
16	31.03.12 (End of 11th Plan)	1,12,022	18,381	1,200	1,31,603	4,780	38,990	24,504	63,494	1,99,877
17	31.03.17 (End of 12th Plan)	1,92,163	25,329	838	2,18,330	6,780	44,478	58,558	1,03,037	3,28,146
18	31.03.2018	1,97,172	24,897	838	2,22,907	6,780	45,293	70,651	1,15,945	3,45,631
19	31.03.2019	2,00,705	24,937	638	2,26,279	6,780	45,399	79,412	1,24,811	3,57,871
20	31.03.2020	2,05,135	24,955	510	2,30,600	6,780	45,699	88,255	1,33,955	3,71,334
21	31.03.2021	2,09,295	24,924	510	2,34,729	6,780	46,209	95,803	1,42,012	3,83,521
22	31.03.2022	2,10,700	24,900	510	2,36,109	6,780	46,723	1,09,885	1,56,608	3,99,497
23	31.03.2023	2,11,855	24,824	589	2,37,269	6,780	46,850	1,25,160	1,72,010	4,16,059
24	31.03.2024	2,17,589	25,038	589	2,43,217	8,180	46,928	1,43,645	1,90,573	4,41,970
25	31.03.2025	2,21,813	24,533	589	2,46,935	8,180	47,728	1,72,368	2,20,096	4,75,212

RES:-Renewable Energy Sources includes Hydro capacity of 25.00 MW and below as reported by MNRE.

Installed Generation Capacity Hydro Electric Power Stations (25 MW & above)

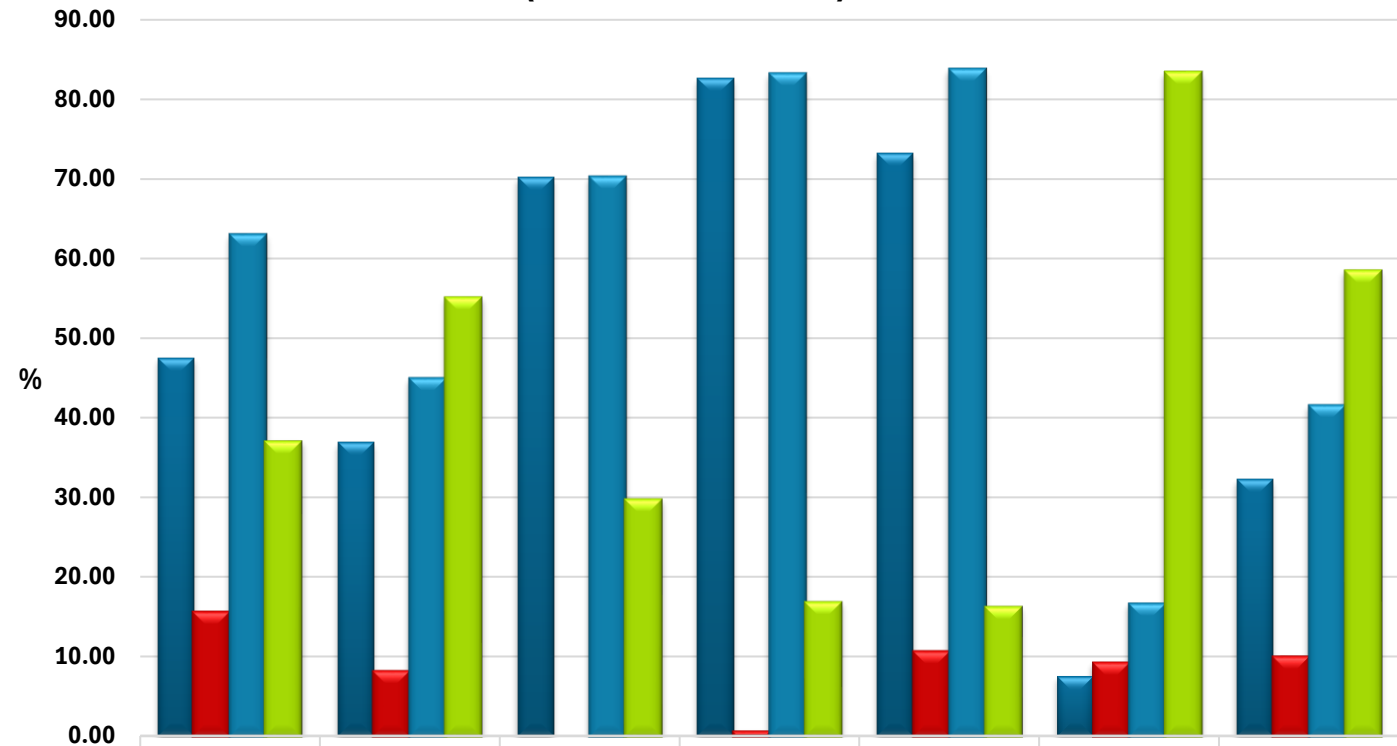
Chart : 6



Note: 1. Figures for the years 1947 & 1950 are as on 31st December.
2. Figures for the year 1955-56 & onward are as on 31st March.

Hydro Electric Potential Development - Basin-wise **Installed Capacity above 25 MW** **% wise (As on 31.03.2025)**

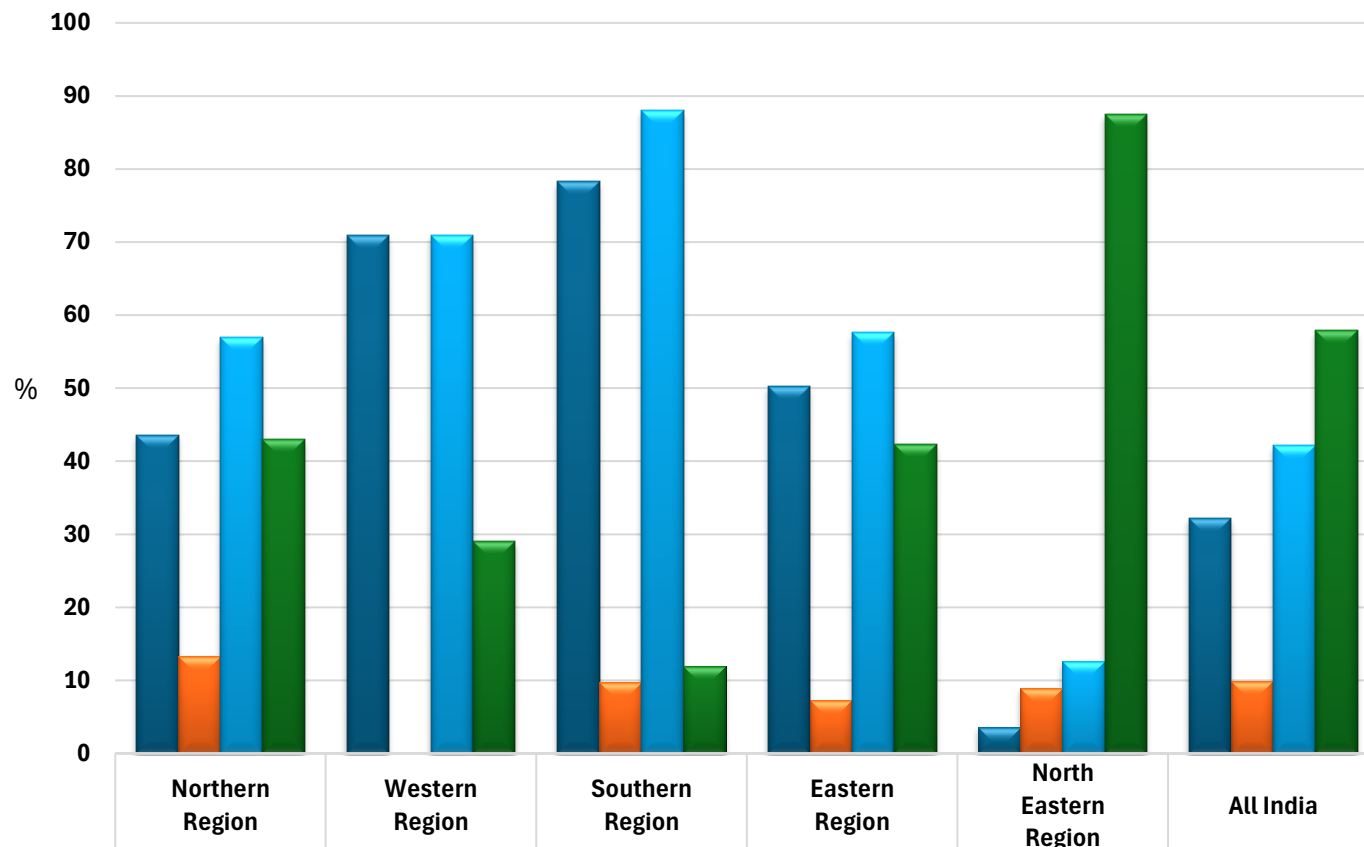
Chart : 6A



■ % Capacity Developed / Under Operation	47.51	36.86	70.24	82.61	73.2	7.47	32.22
■ % Cap. Under Construction	15.48	8.11	0	0.57	10.56	9.15	9.92
■ % Cap. Dev. + Under Con.	62.99	44.91	70.24	83.18	83.76	16.62	41.54
■ % Balance Potential	37.01	55.09	29.76	16.82	16.24	83.38	58.46

Hydro Electric Potential Development - Region-wise Installed Capacity above 25 MW % wise (As on 31.03.2025)

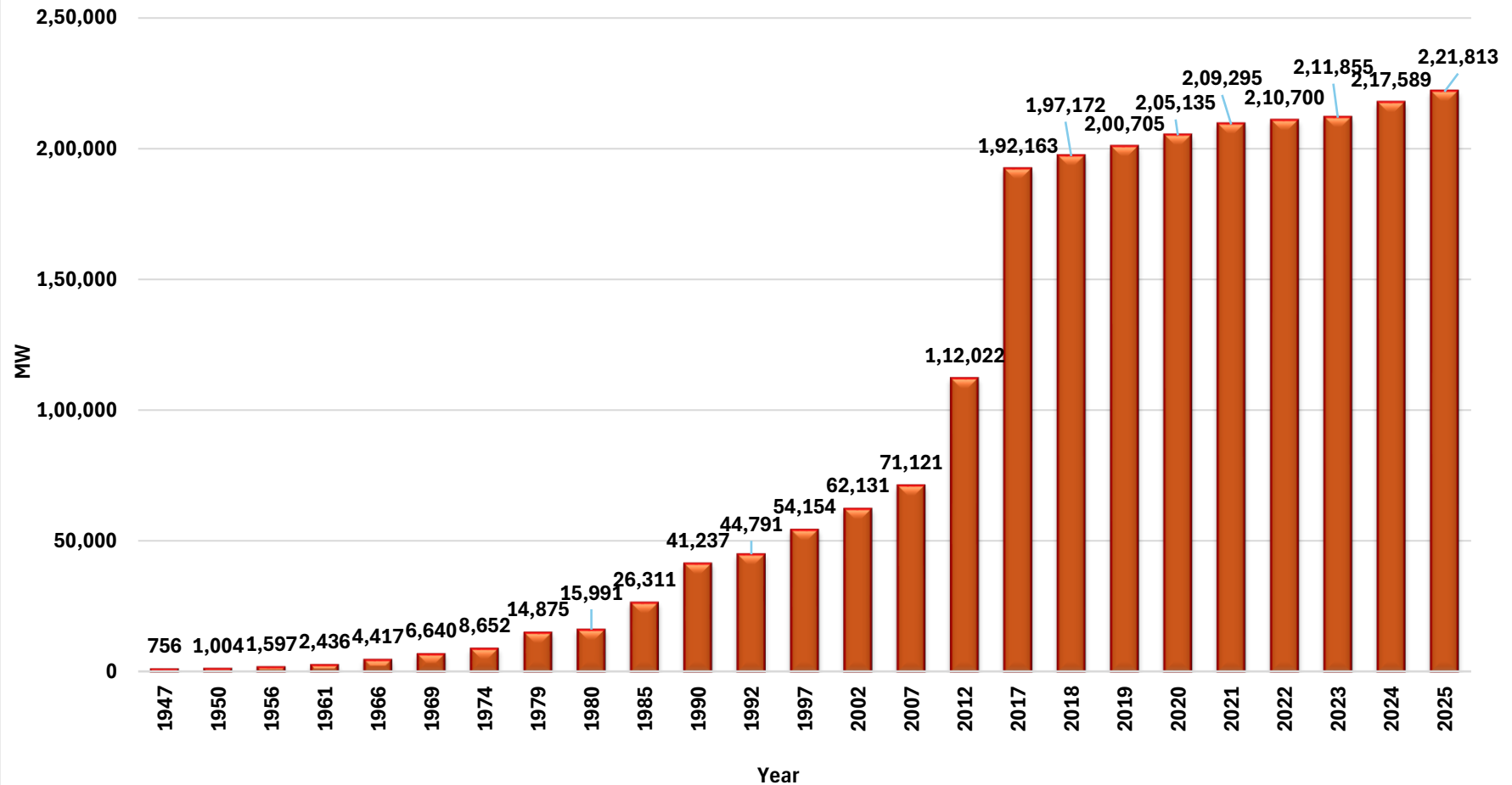
Chart : 6B



■ % Capacity Developed /Under Operation	43.59	70.96	78.29	50.30	3.62	32.22
■ %Cap. Under Construction	13.34	0.00	9.78	7.32	8.94	9.92
■ %Cap. Dev. +Under Construction	56.93	70.96	88.07	57.62	12.56	42.14
■ % Capacity yet to be Developed	43.07	29.04	11.93	42.38	87.44	57.86

Installed Generation Capacity Coal/ Lignite-based Plants

Chart : 7A

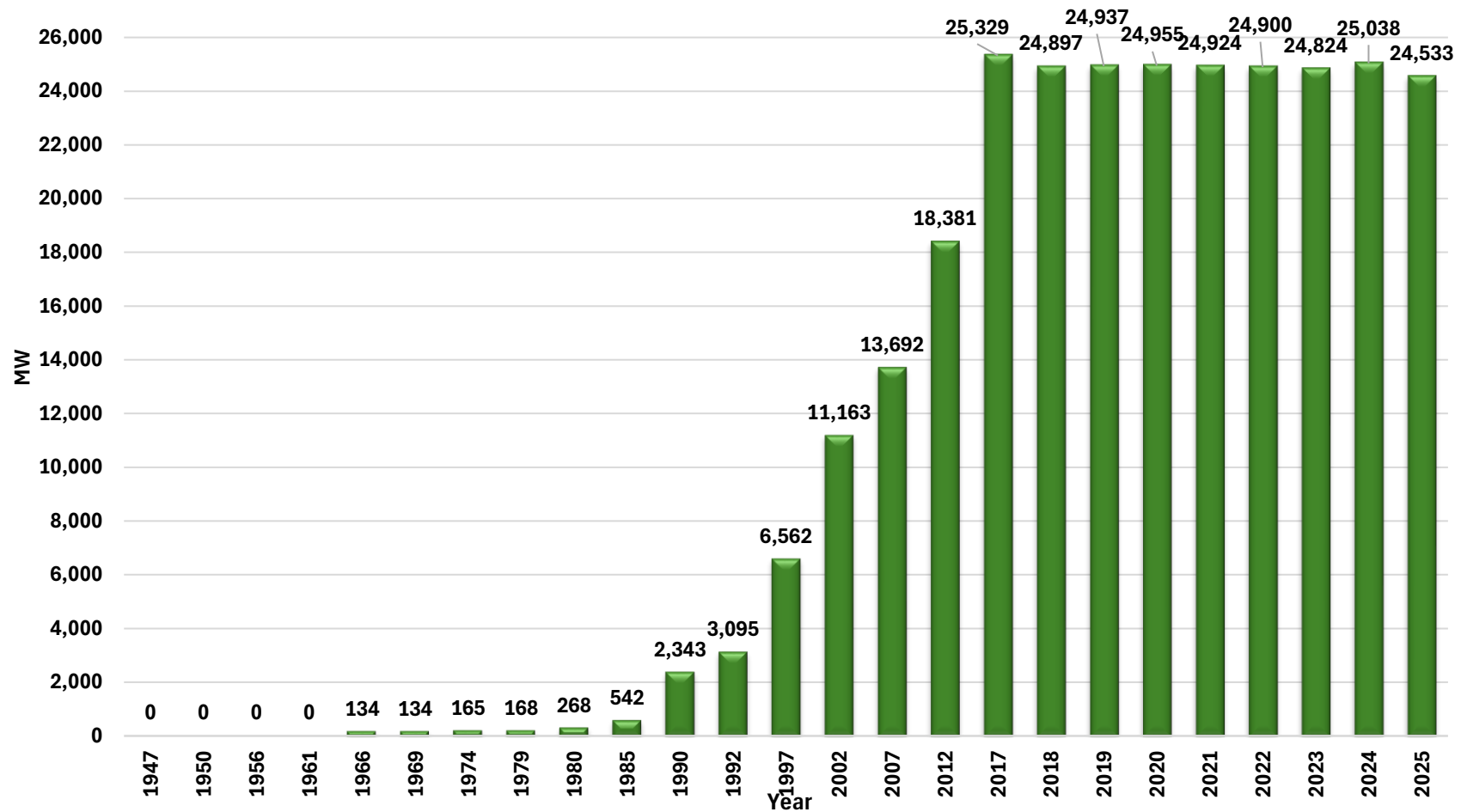


Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

Installed Generation Capacity Gas/ Liquid-based Plants

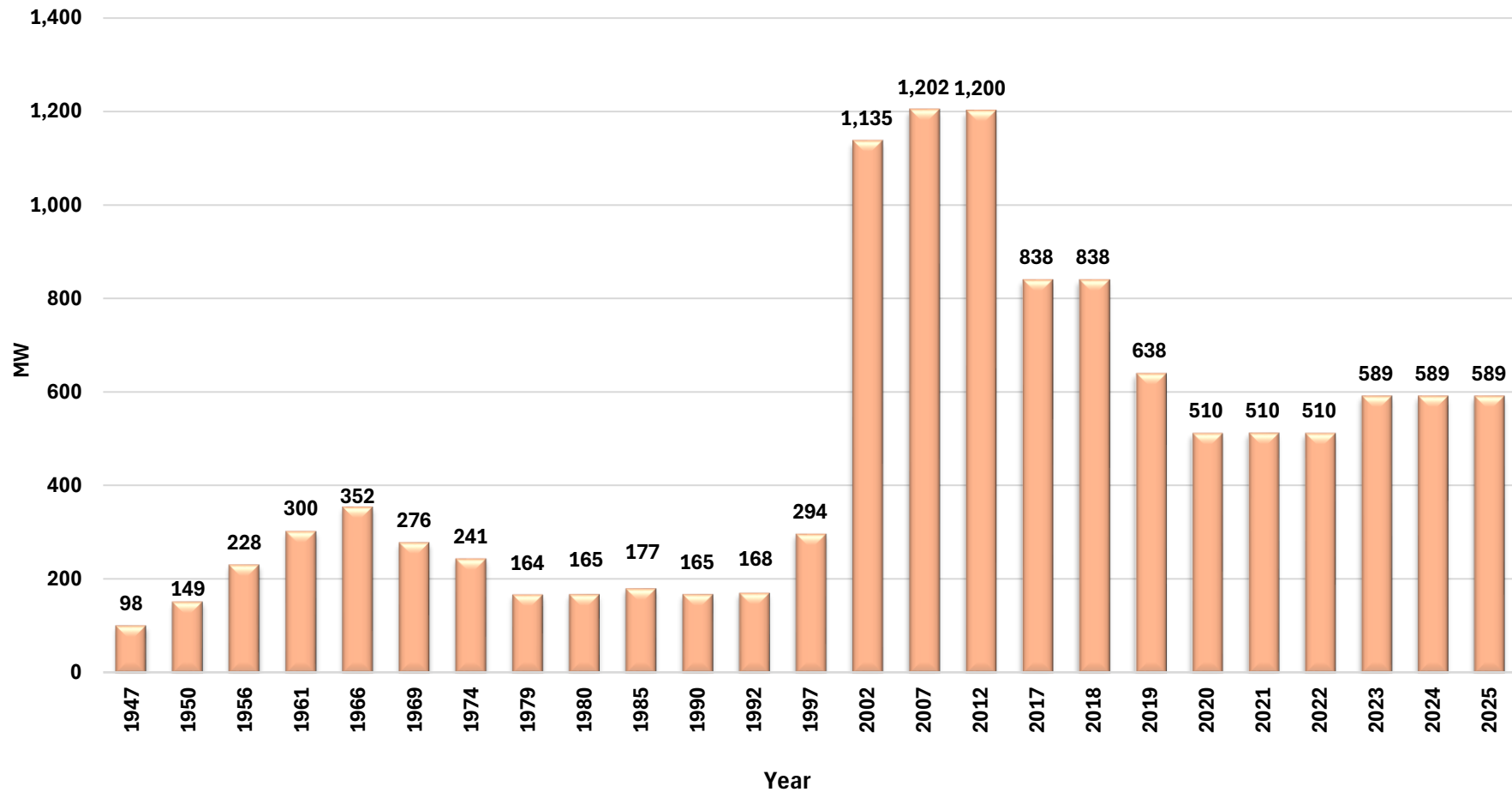
Chart : 7B



Note: 1. Figures for the years 1947 & 1950 are as on 31st December.
2. Figures for the year 1955-56 & onward are as on 31st March.

Installed Generation Capacity Diesel-based Plants

Chart : 7C



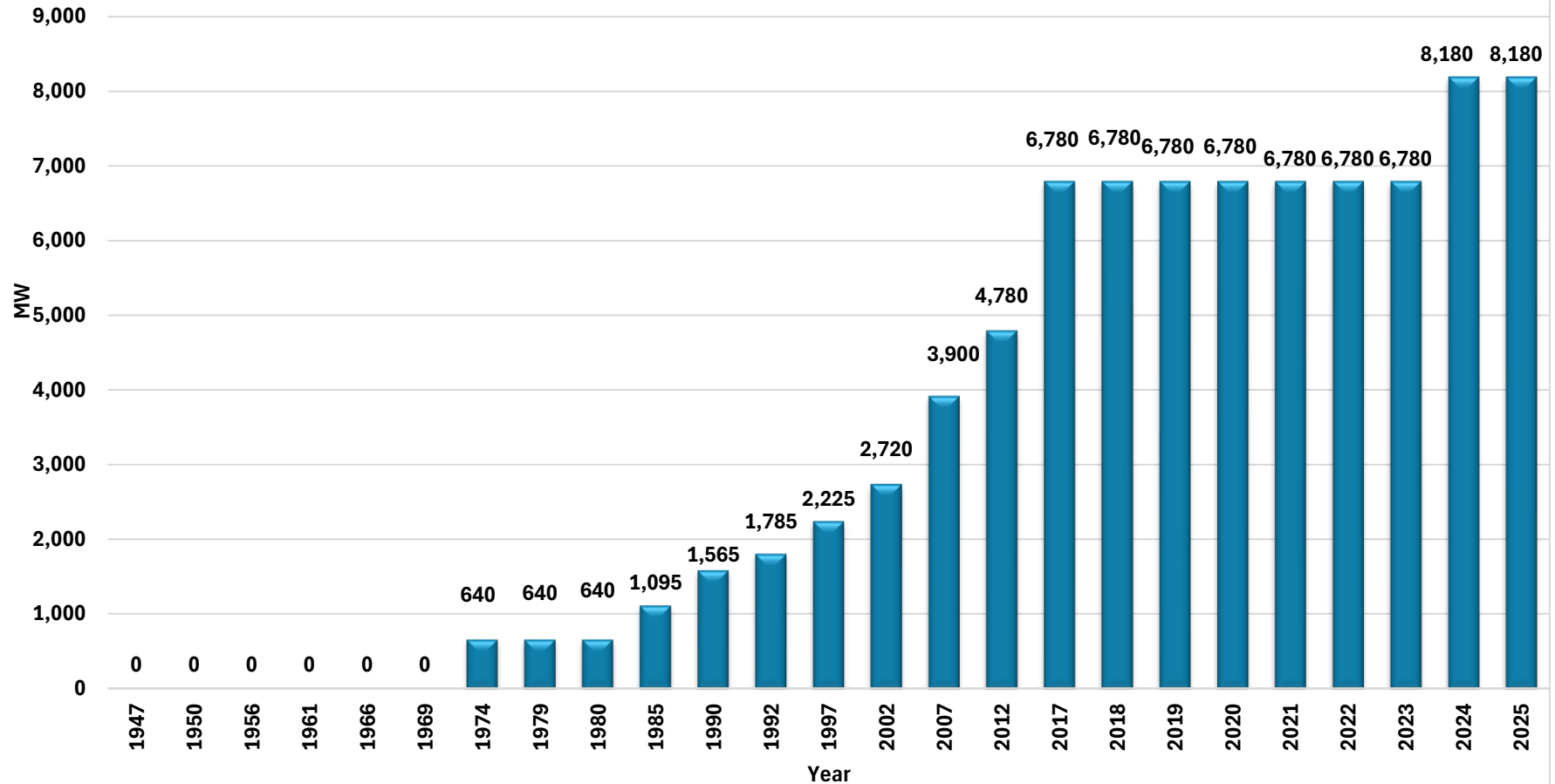
Note: 1. Figures for the years 1947 & 1950 are as on 31st December.

2. Figures for the year 1955-56 & onward are as on 31st March.

3. Diesel based Capacity includes capacity of plants having capacity 25 MW & above and Capacity of DG sets group located in Lakshadweep and Andaman & Nicobar Islands

Installed Generation Capacity Nuclear-based Plants

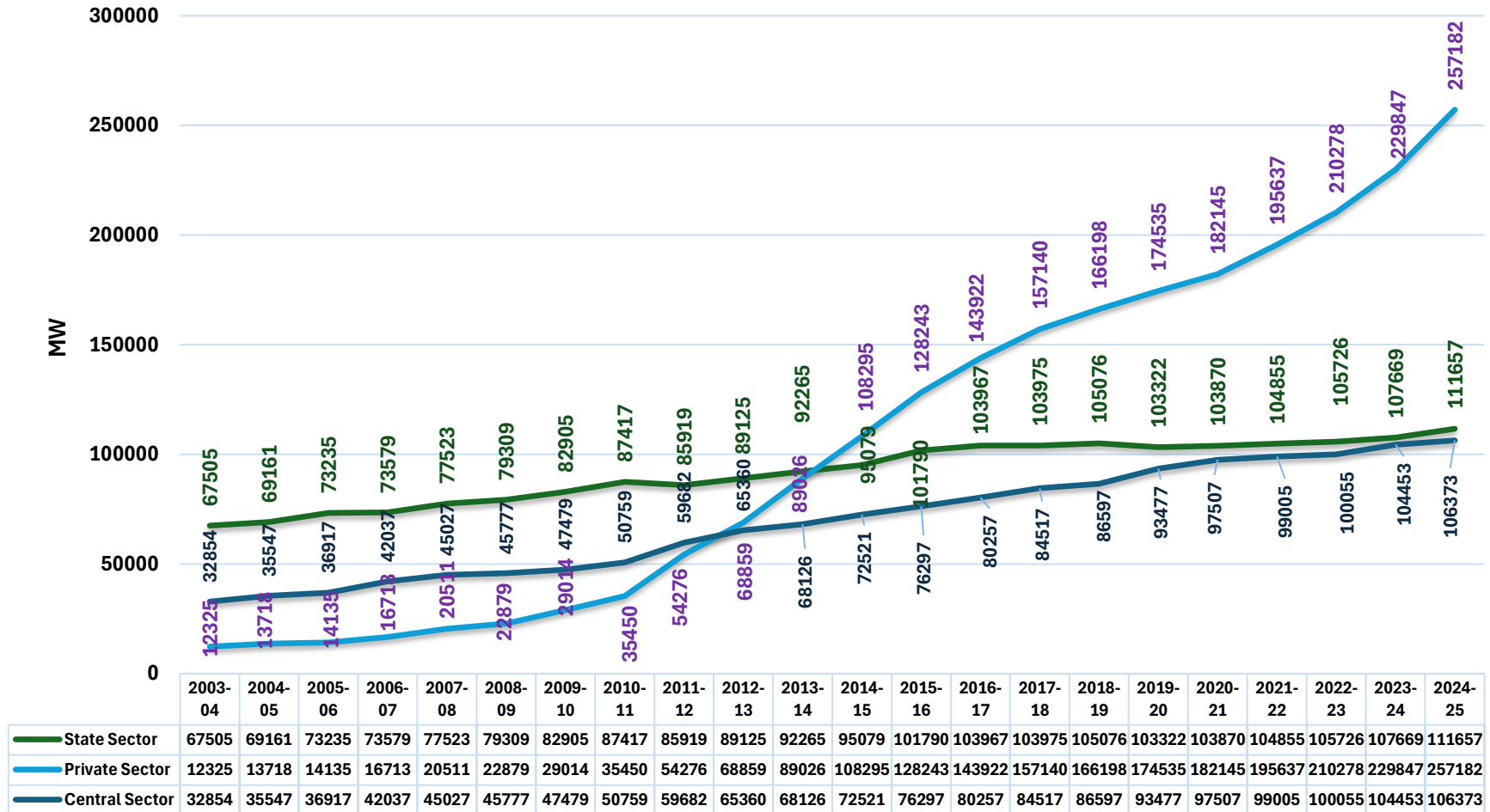
Chart : 8



Note: 1. Figures for the years 1947 & 1950 are as on 31st December.
2. Figures for the year 1955-56 & onward are as on 31st March.

Installed Generation Capacity - Sector-wise(MW)
2003-04 to 2024-25

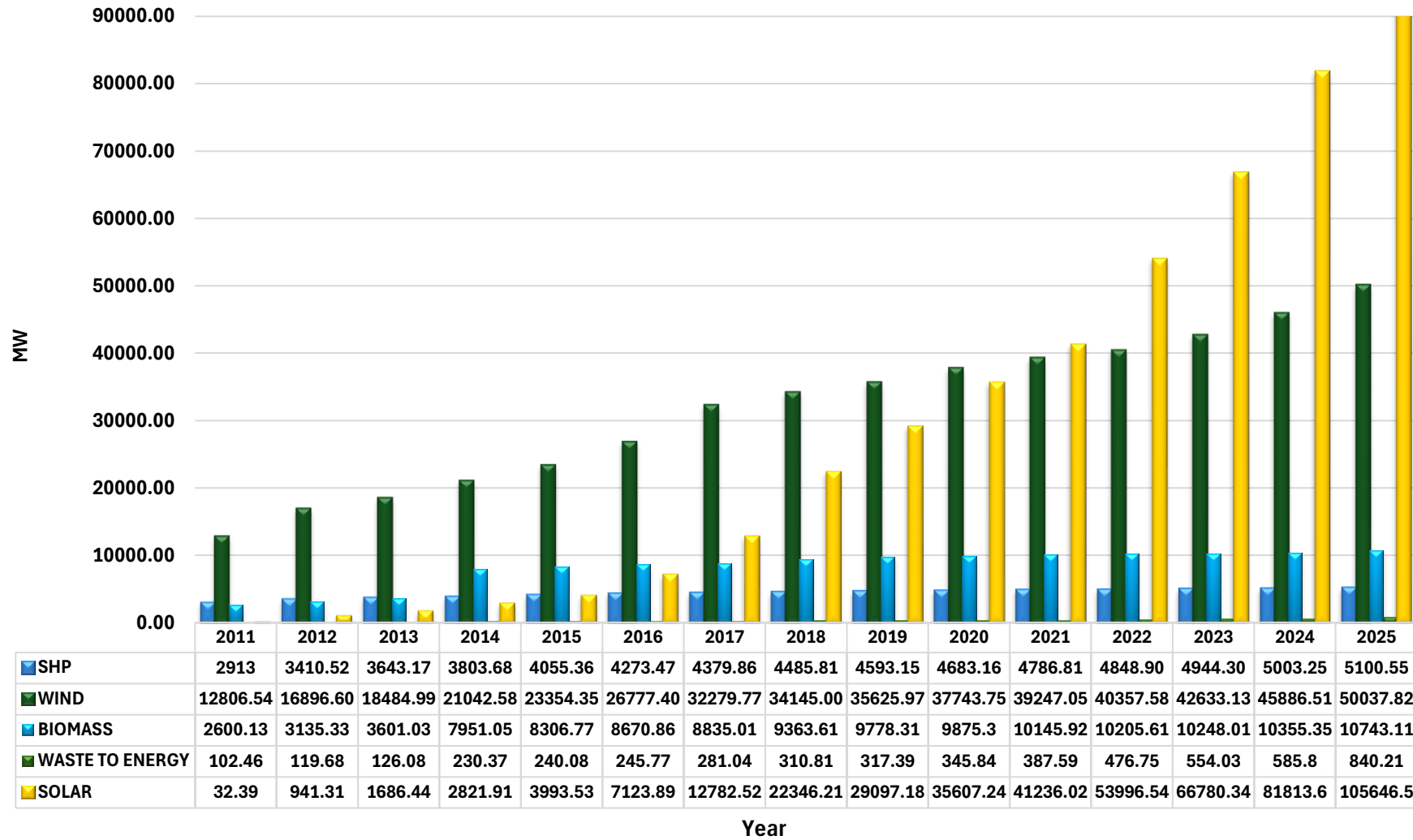
Chart : 8A



Note: DVC & RRMR Installed Capacity is considered under Central Sector

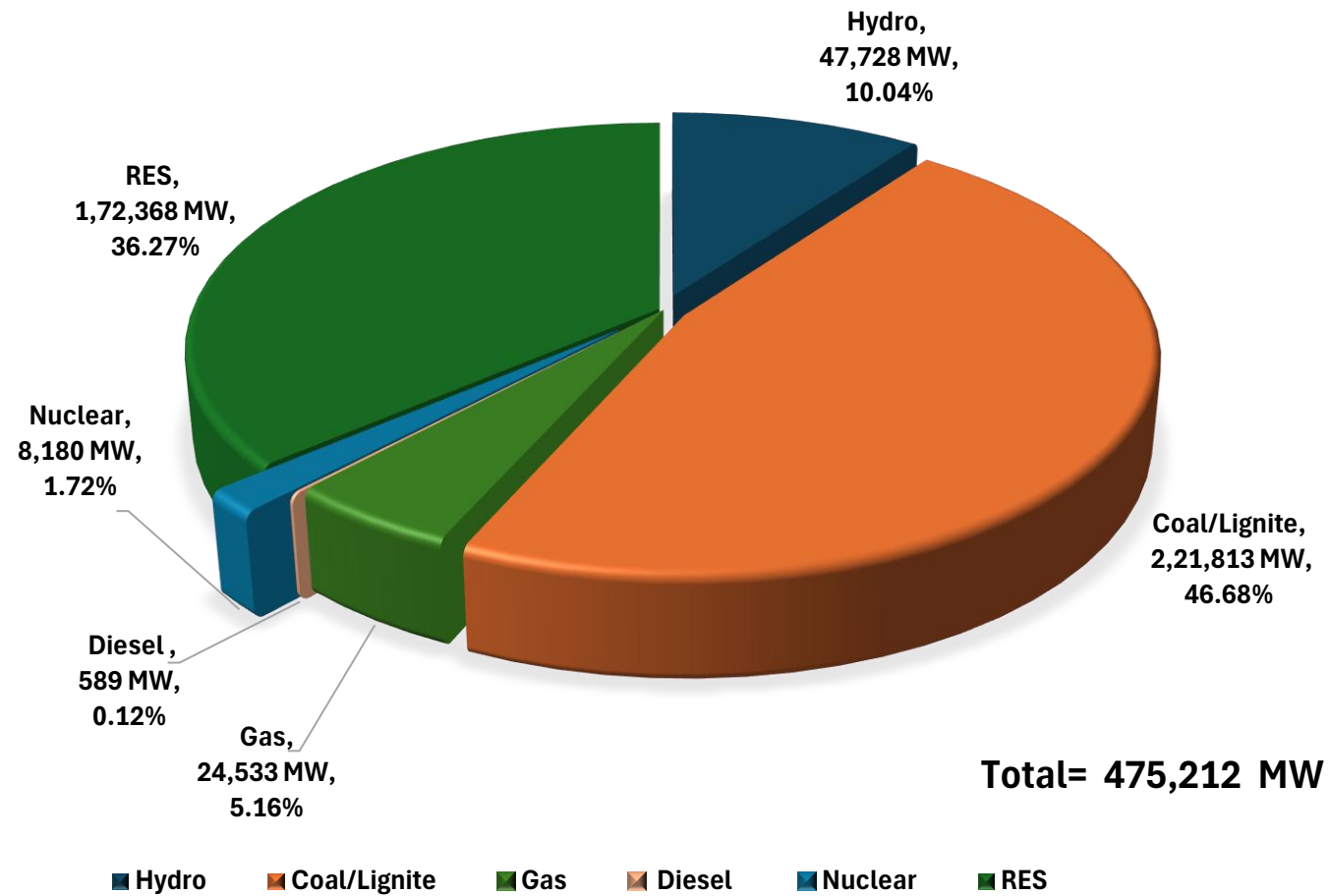
**Installed Generation Capacity of R.E.S. - Mode-wise/Year-wise
during last 15 years**

Chart :8 B



Pie Chart : 9

**All India Installed Generation capacity
(Utilities)
as on 31.03.2025**



**All India Installed Generation Capacity
(Utilities)
(As on 31.03.2024)**

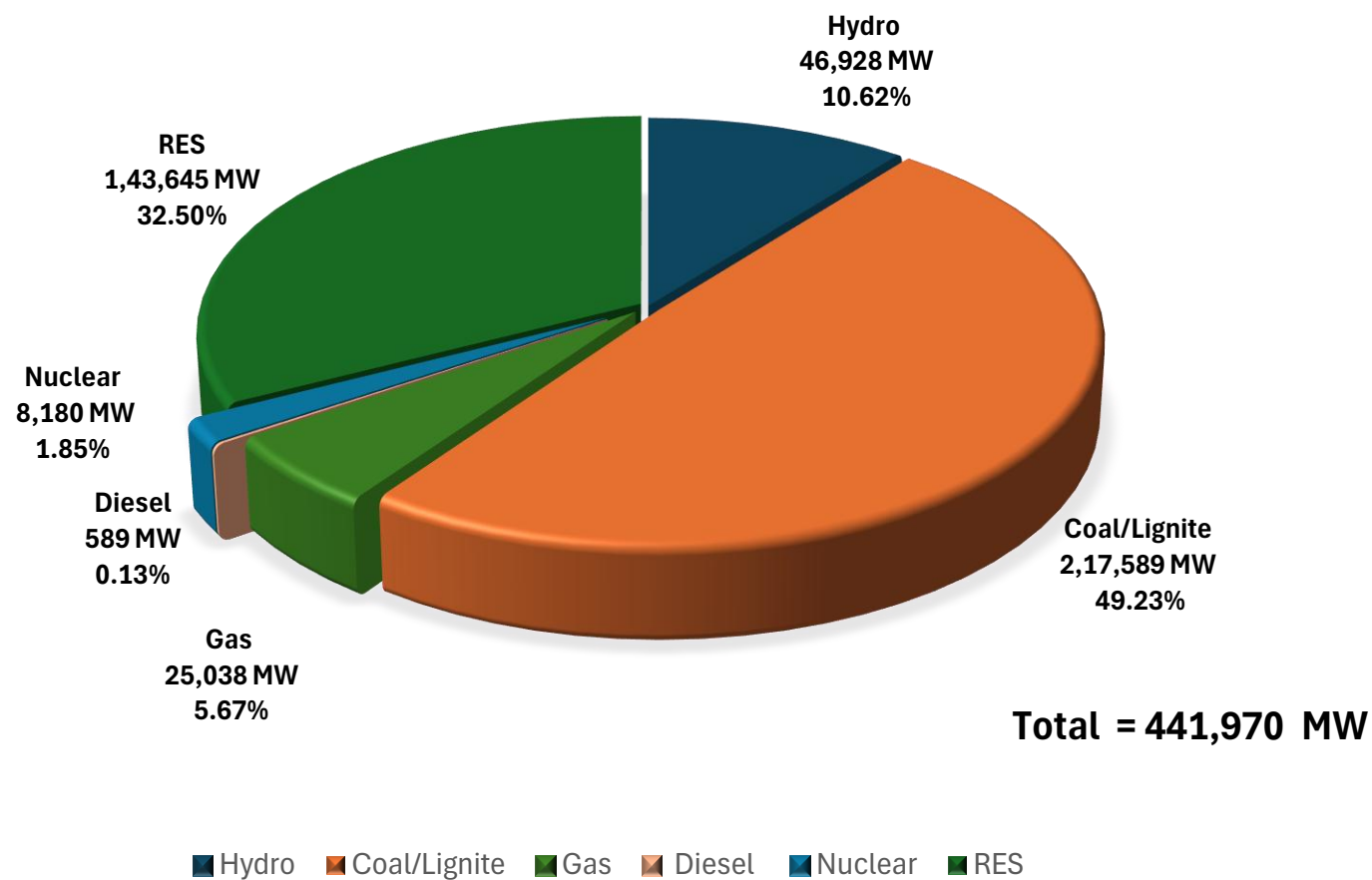


Table 3
Gross Electricity Generation in India- Mode-wise
(Utilities)

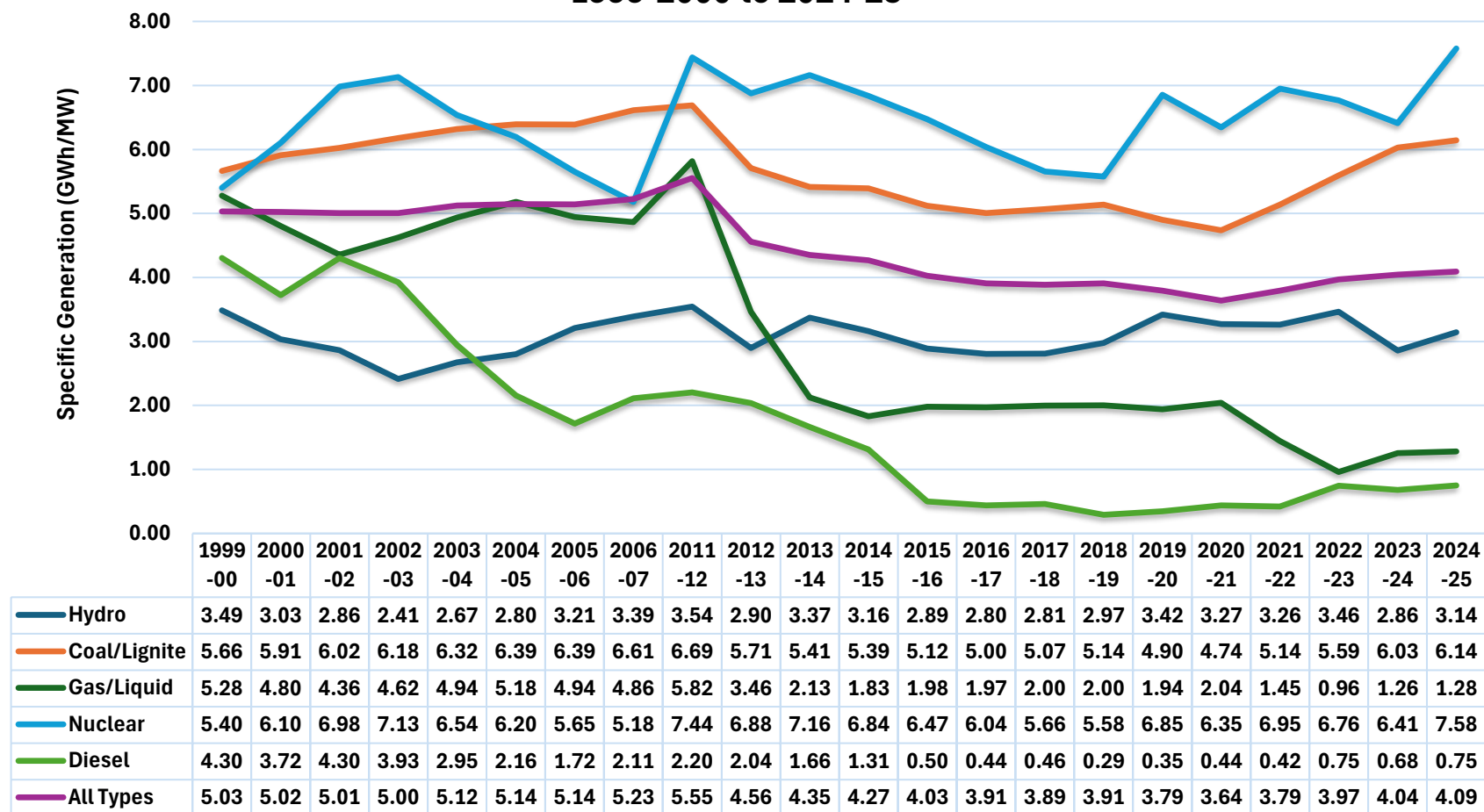
SL. No	During financial year ending with	Thermal				Hydro	Nuclear	RES	Total
		Coal/ Lignite	Gas	Diesel	Total				
1	1947	1,733	0	144	1,877	2,195	0	0	4,073
2	1950	2,587	0	200	2,787	2,519	0	0	5,106
3	1955-56 (End of the 1st Plan)	5,367	0	233	5,600	4,295	0	0	9,662
4	1960-61 (End of the 2nd Plan)	9,100	0	368	9,468	7,837	0	0	16,937
5	1965-66(End of the 3rd Plan)	17,765	69	324	18,158	15,225	0	0	32,990
6	1968-69(End of the 3 Annual Plans)	26,711	124	194	27,029	20,723	0	0	47,434
7	1973-74(End of the 4th Plan)	34,853	343	125	35,321	28,972	2,396	0	66,689
8	1978-79(End of the 5th Plan)	52,024	515	55	52,594	47,159	2,770	0	1,02,523
9	1979-80(End of the Annual Plan)	55,720	500	53	56,273	45,478	2,876	0	1,04,627
10	1984-85 (End of the 6th Plan)	96,957	1,834	45	98,836	53,948	4,075	0	1,56,859
11	1989-90 (End of the 7th Plan)	1,72,643	5,962	85	1,78,690	62,116	4,625	6	2,45,438
12	1991-92(End of the 2 Annual Plans)	1,97,163	11,450	95	2,08,708	72,757	5,525	38	2,87,029
13	1996-97(End of the 8th Plan)	2,89,378	26,985	679	3,17,042	68,901	9,071	876	3,95,889
14	2001-02(End of the 9th Plan)	3,70,884	47,099	4,317	4,22,300	73,579	19,475	2,085	5,17,439
15	2006-07(End of the 10th Plan)	4,61,794	64,157	2,539	5,28,490	1,13,502	18,802	9,860	6,70,654
16	2011-12(End of the 11th Plan)	6,12,497	93,281	2,649	7,08,427	1,30,511	32,287	51,226	9,22,451
17	2016-17(End of the 12th Plan)	9,44,022	49,094	401	9,93,516	1,22,378	37,916	81,548	12,35,358
18	2017-18	9,86,591	50,208	348	10,37,146	1,26,123	38,346	1,01,839	13,03,455
19	2018-19	10,22,265	49,834	215	10,72,314	1,34,894	37,813	1,26,759	13,71,779
20	2019-20	9,94,197	48,443	199	10,42,838	1,55,769	46,472	1,38,337	13,83,417
21	2020-21	9,81,443	50,944	224	10,32,611	1,50,300	43,029	1,47,248	13,73,187
22	2021-22	10,78,581	36,016	214	11,14,811	1,51,627	47,112	1,70,912	14,84,463
23	2022-23	11,82,096	23,885	409	12,06,390	1,62,099	45,861	2,03,555	16,17,905
24	2023-24	12,94,852	31,296	401	13,26,549	1,34,054	47,937	2,25,835	17,34,375
25	2024-25	13,31,867	31,580	443	13,63,890	1,48,634	56,681	2,55,009	18,24,214

Note:

* Provisional

2. RES Generation during 2014-2015 is as per actual generation received from utilities.

**Specific Generation by All India Power Stations Mode-wise
(Utilities)
1999-2000 to 2024-25**



Years

Electricity Generation Hydro Electric Power Stations (25 MW & above)

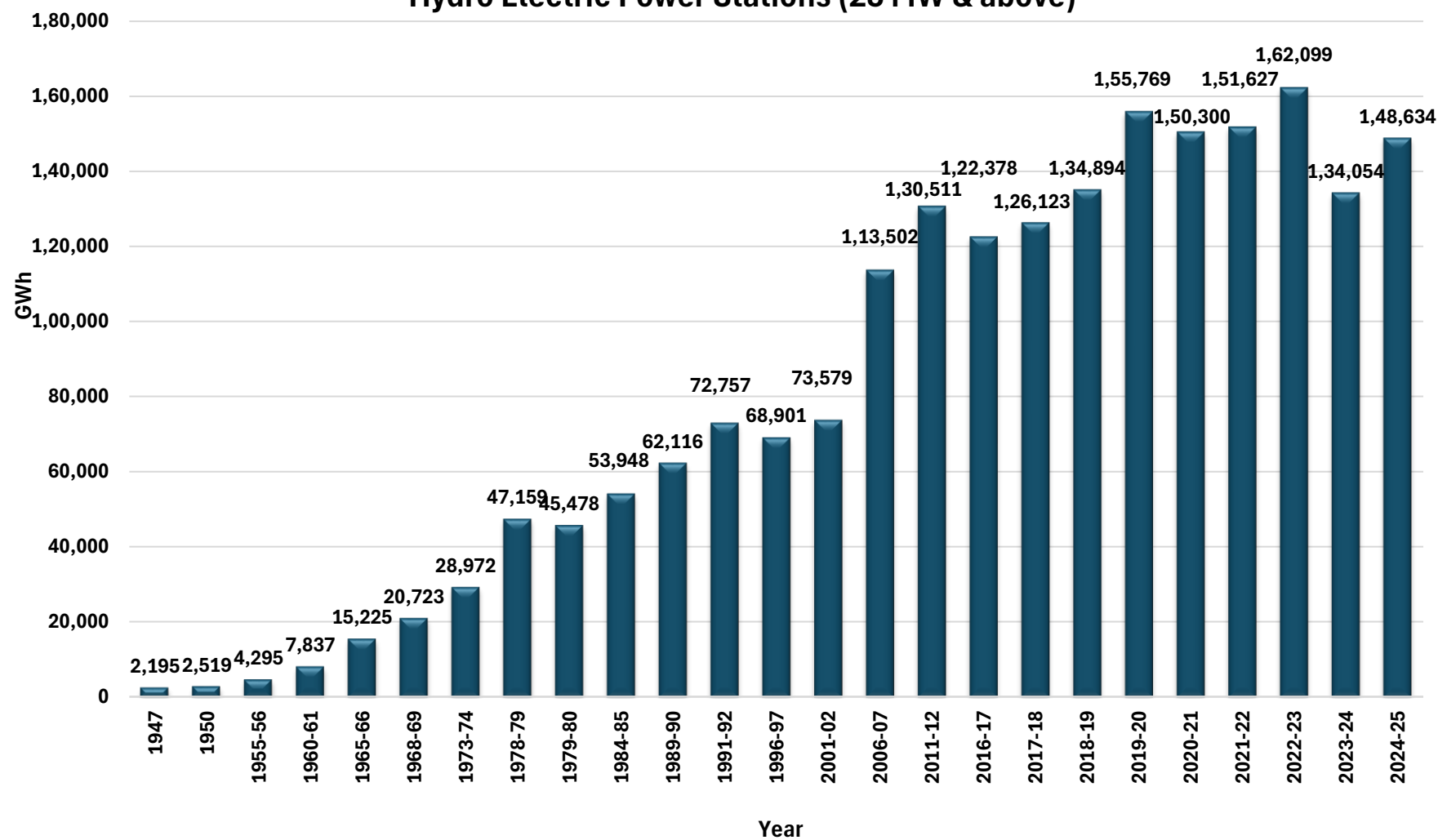
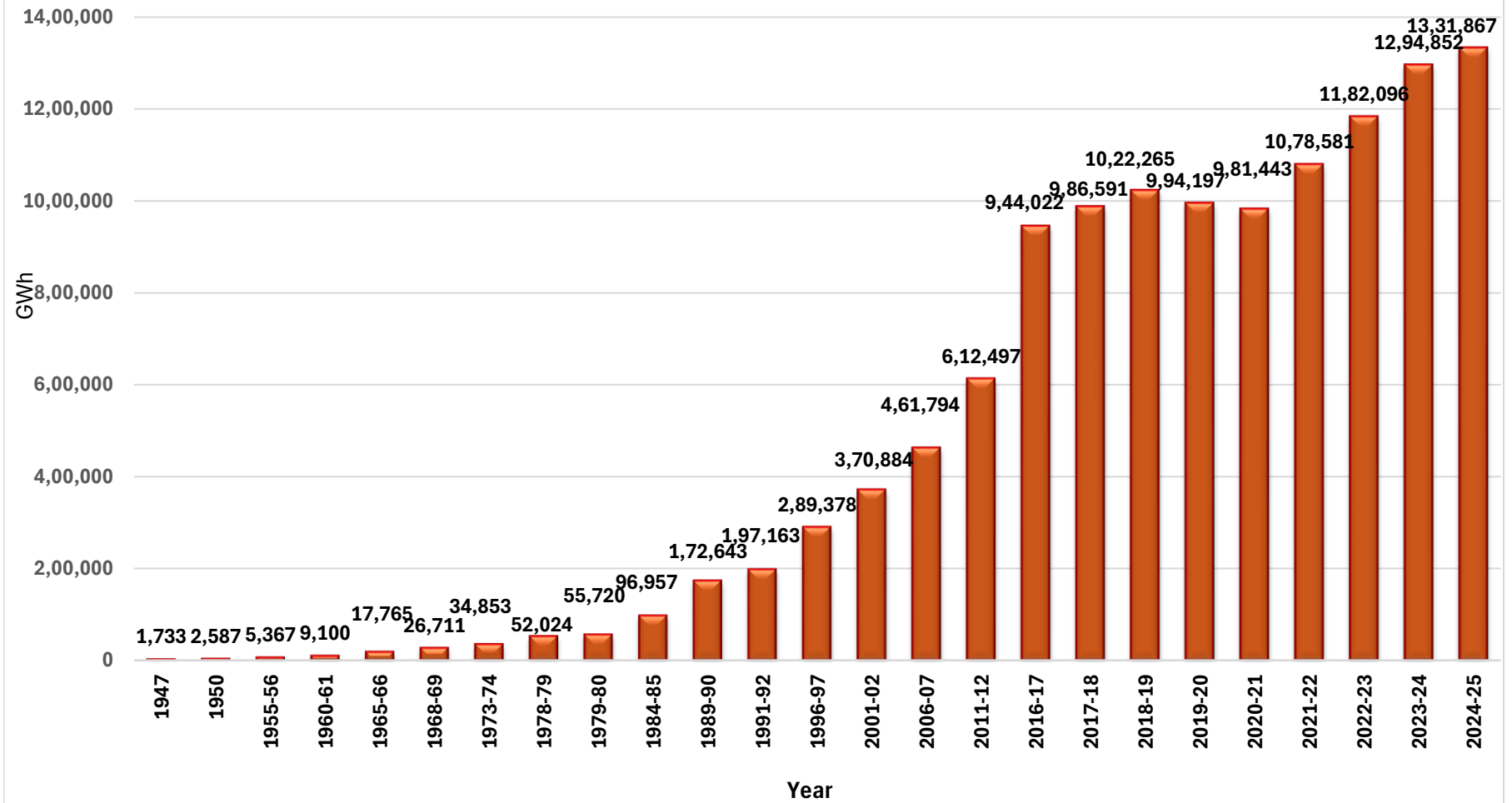


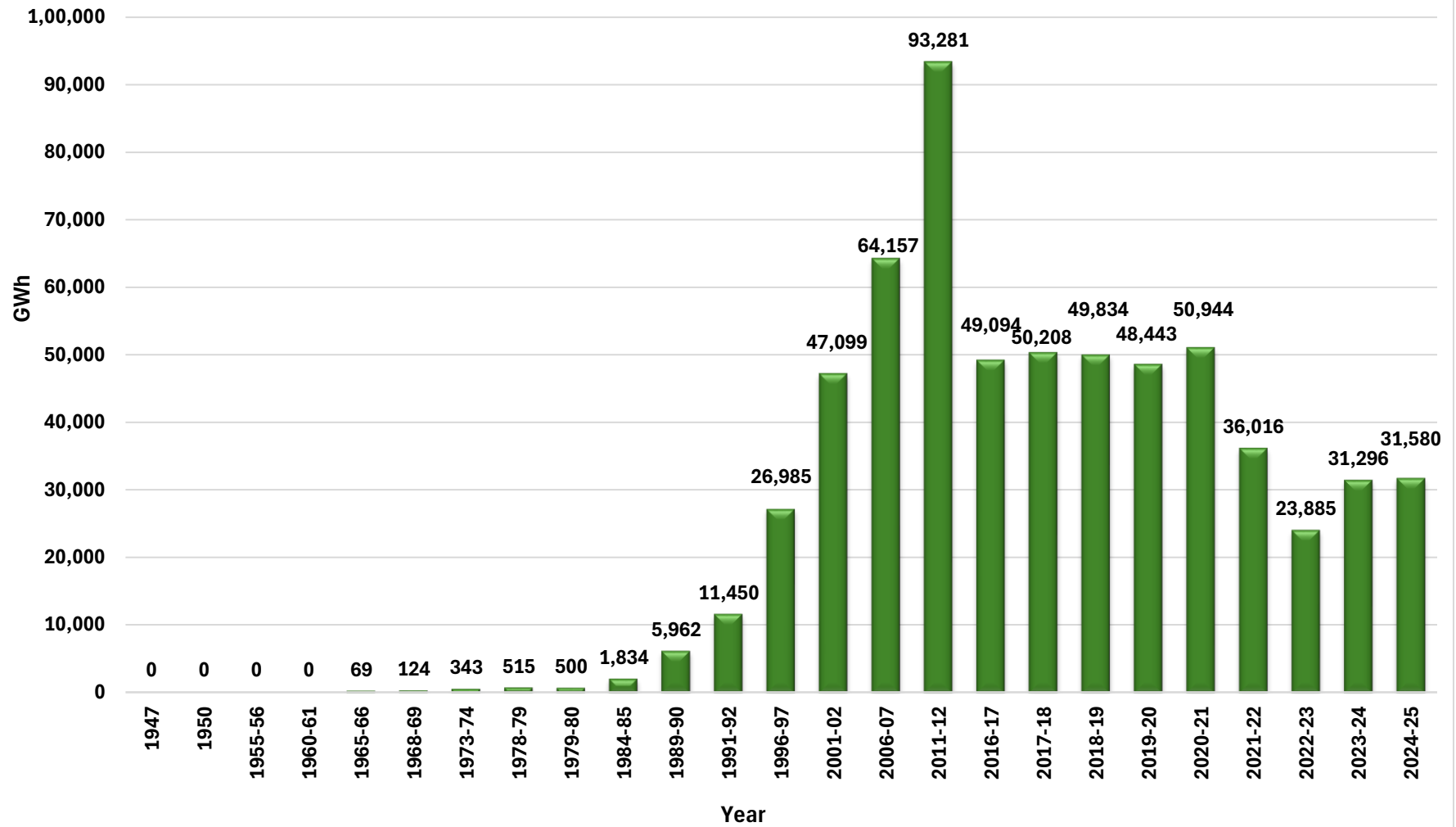
Chart : 12A

Electricity Generation Coal / Lignite-based Power Plants

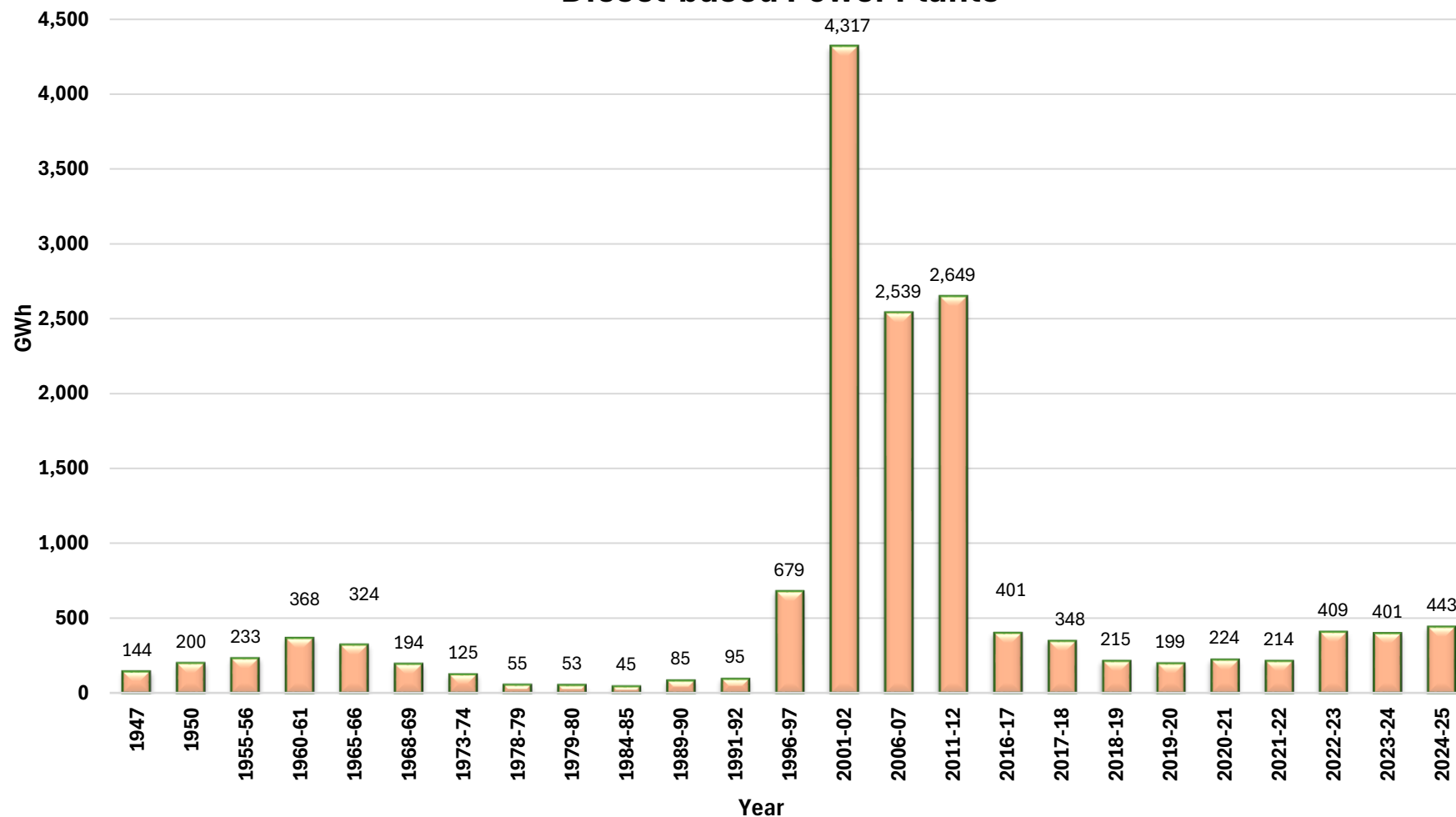


Electricity Generation Gas/ Liquid Fuel-based Power Plants

Chart : 12B

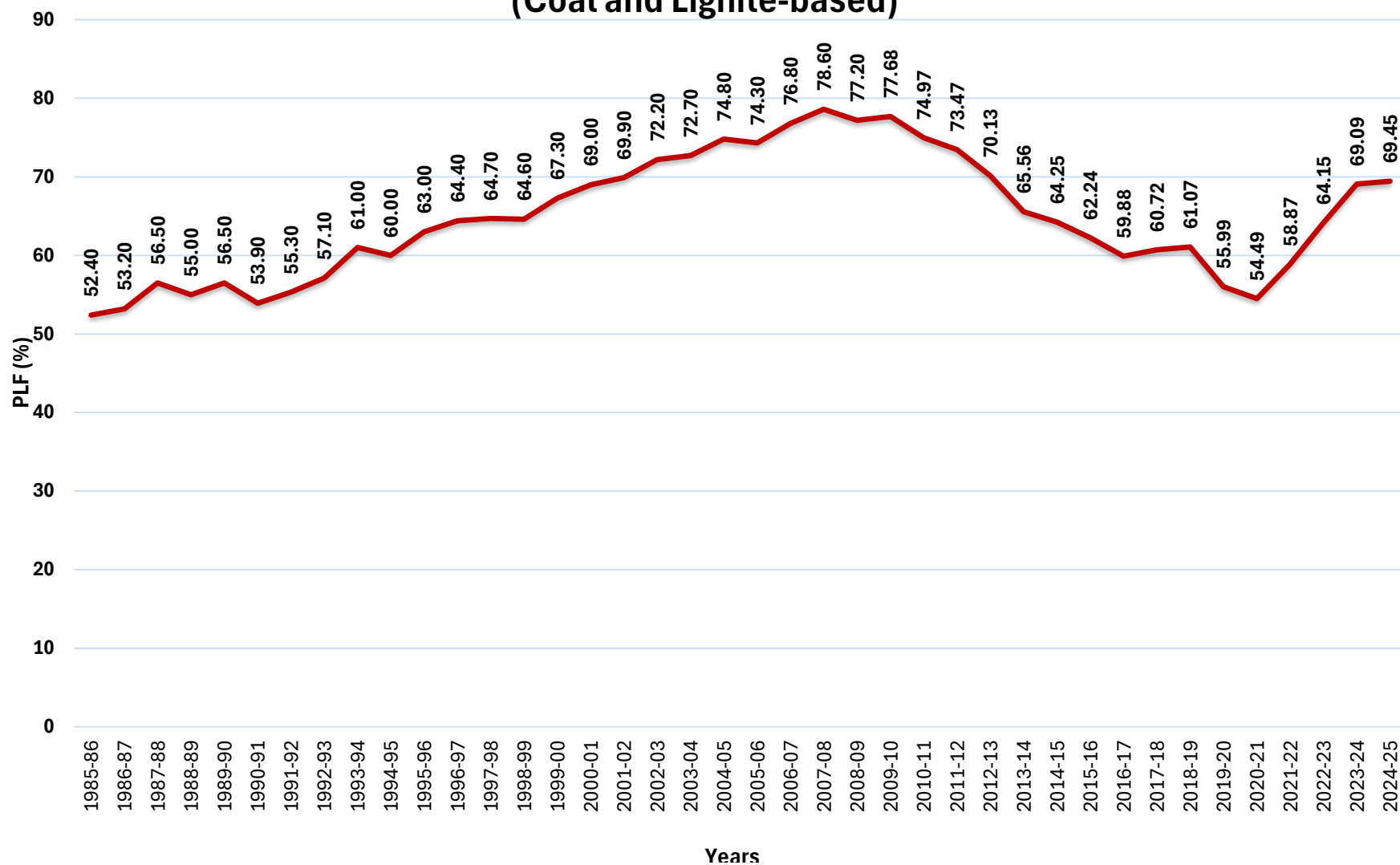


Electricity Generation Diesel-based Power Plants

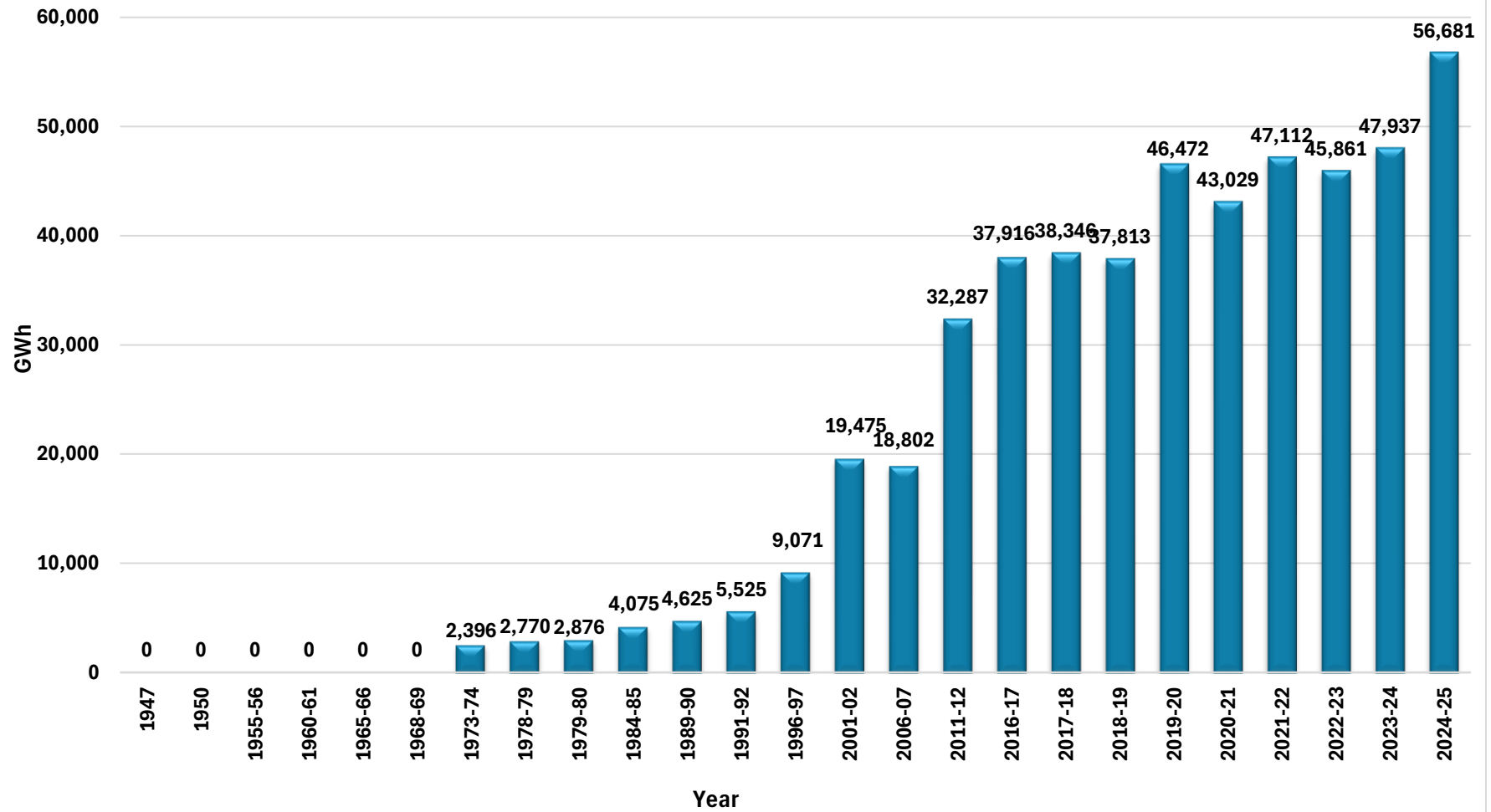


Note: Generation included from Diesel based plants (25 MW & above capacity) generation & generation from DG sets group located in Lakshadweep and Andaman & Nicobar Islands

All India PLF (%) of Thermal Power Stations (Coal and Lignite-based)



Electricity Generation Nuclear-based Power Plants



All India Electricity Generation (Utilities)

Chart : 15

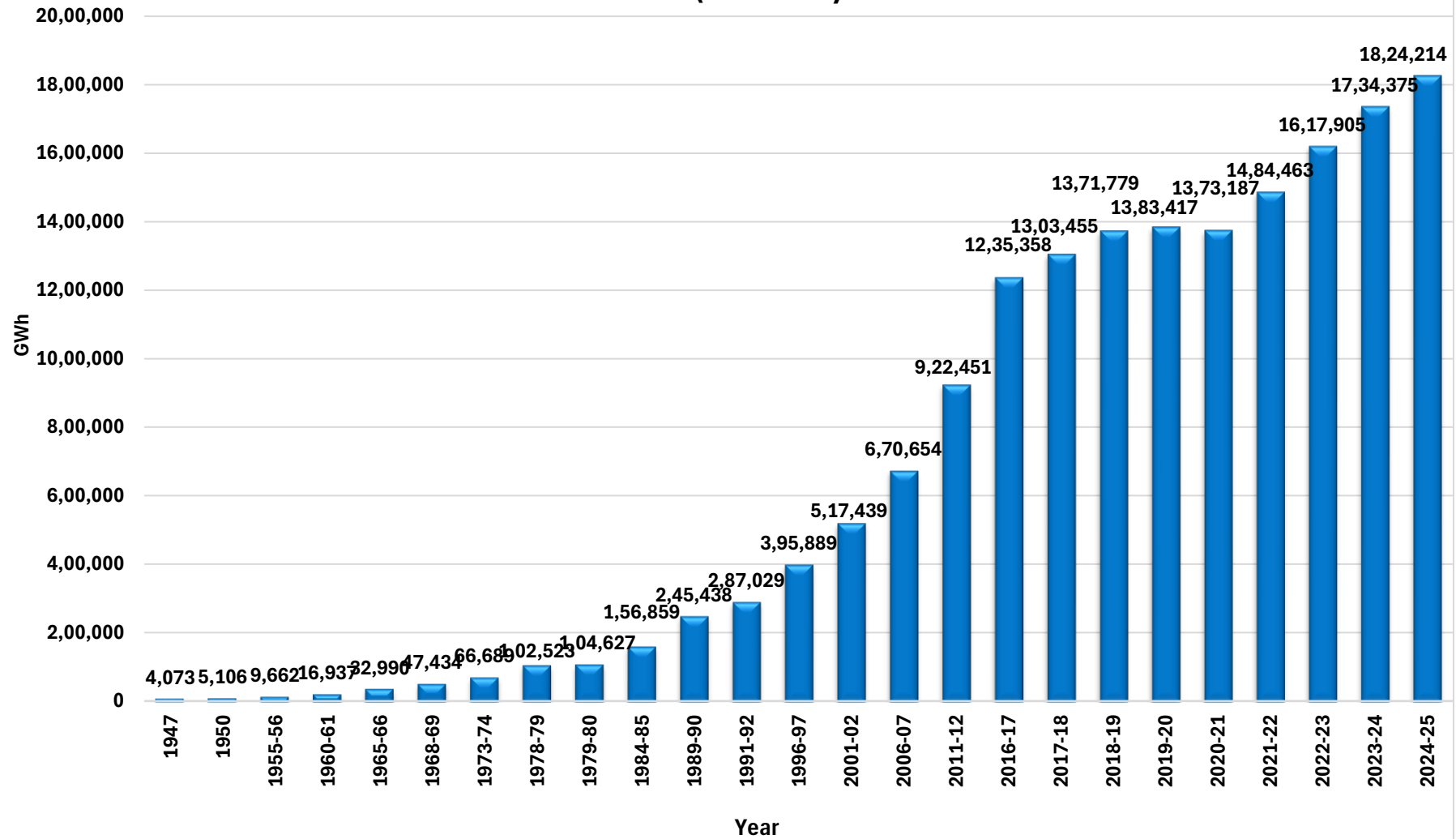
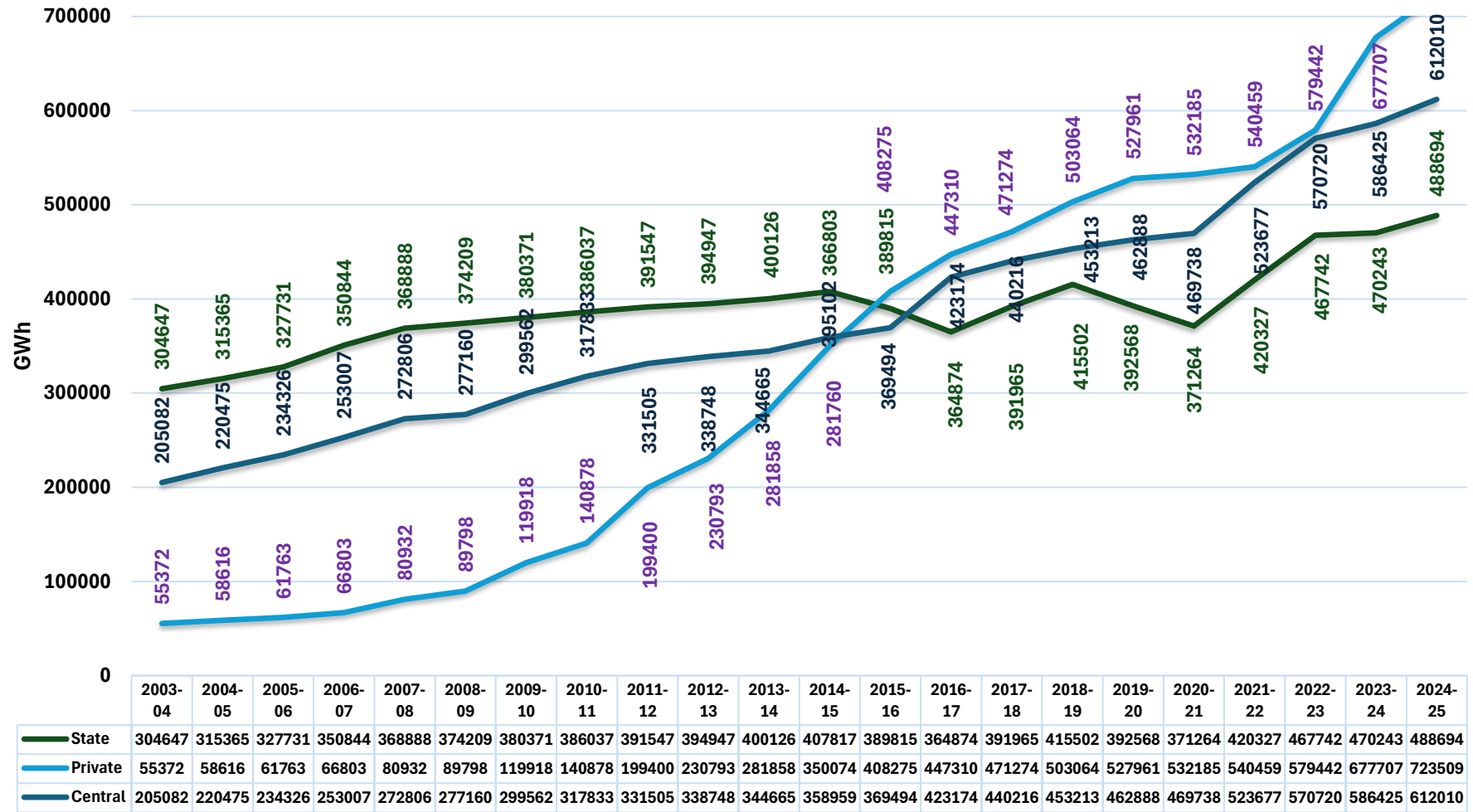


Chart : 15A

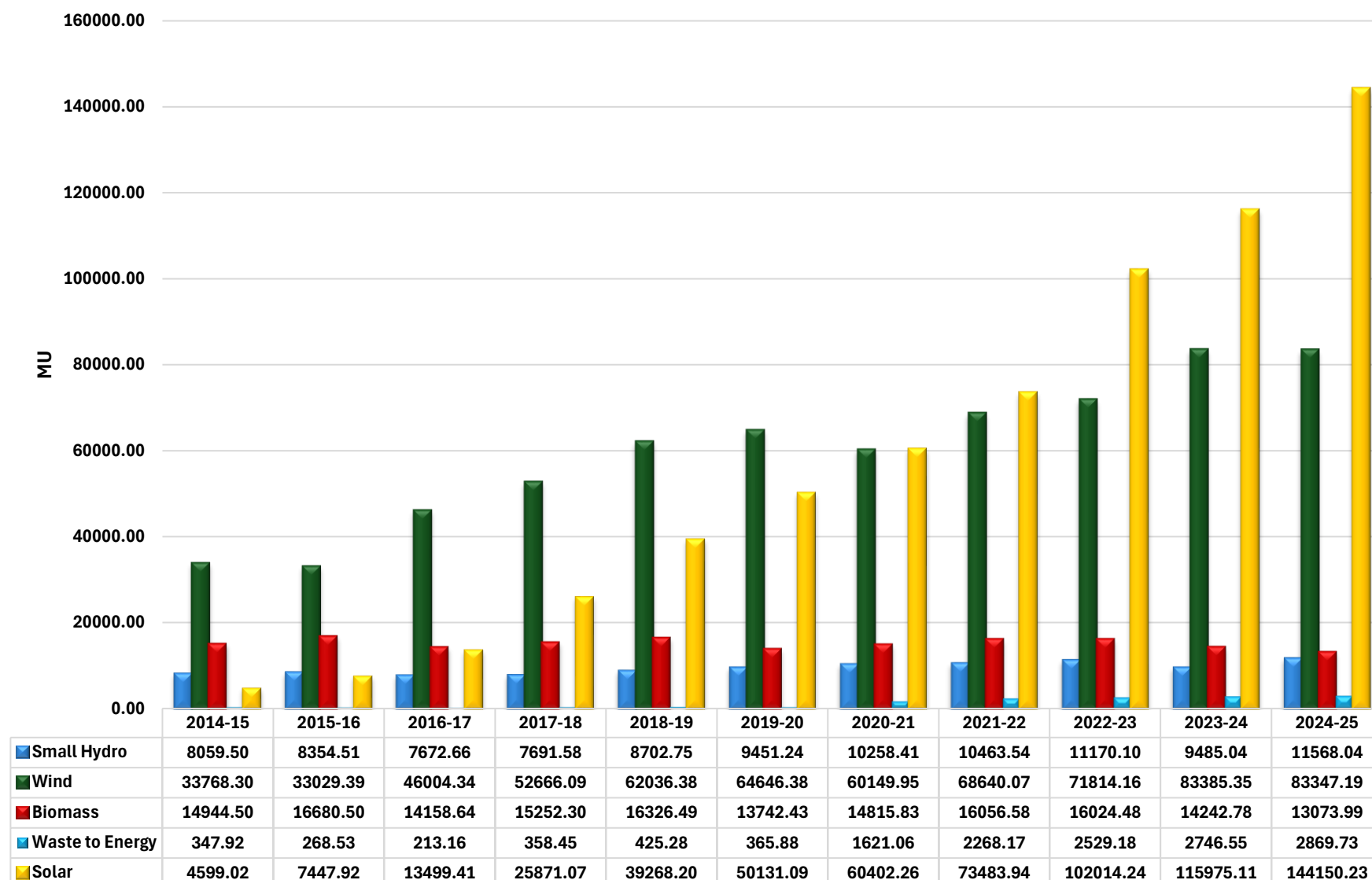
Electricity Generation Sector-wise (GWh) 2003-04 to 2024-25



Note: DVC & RRMR Generation is considered under Central Sector

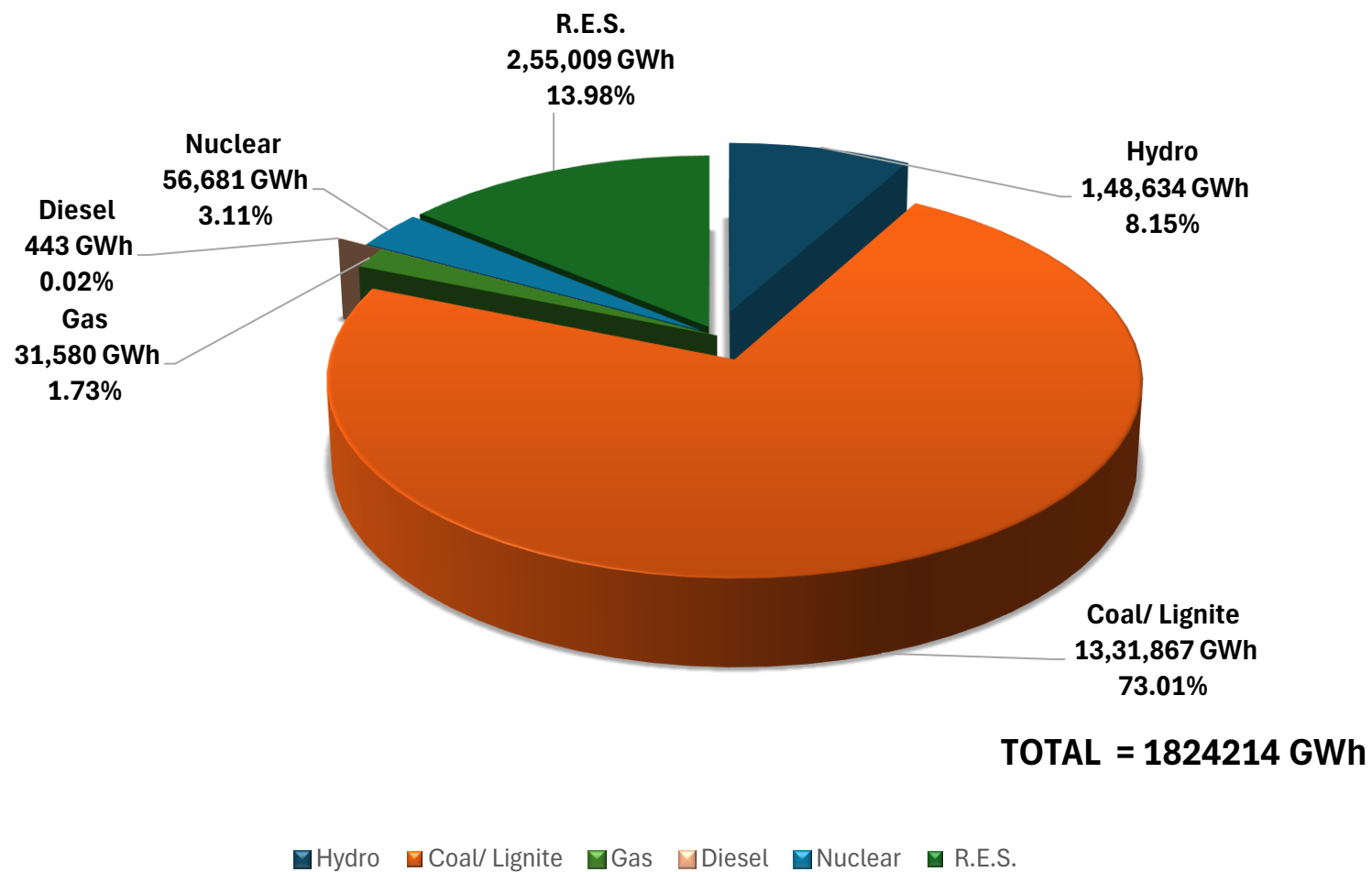
Electricity Generation from R.E.S. - Mode-wise/ Year-wise during last 10 years

Chart: 15B



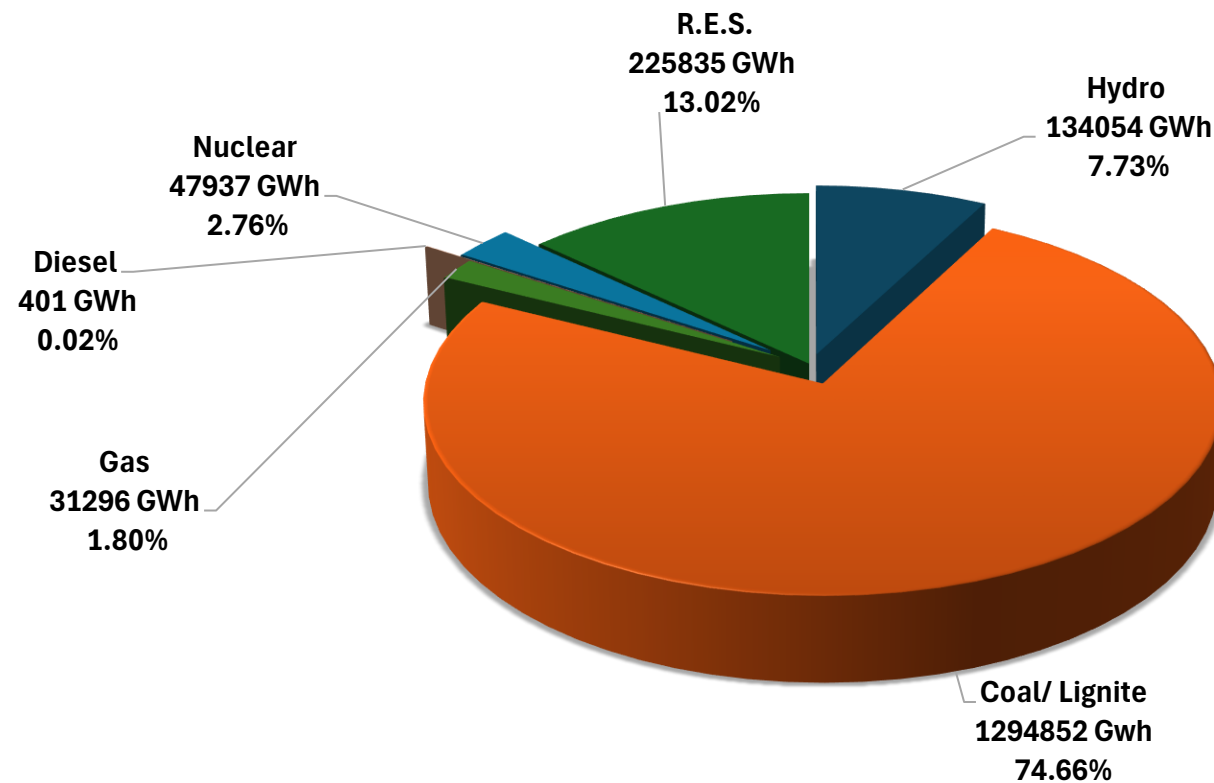
**Electricity Generation in India
Mode-wise -Utilities
2024-25**

Chart : 16



**Electricity Generation in India
Mode-wise -Utilities
2023-24**

Chart : 16A

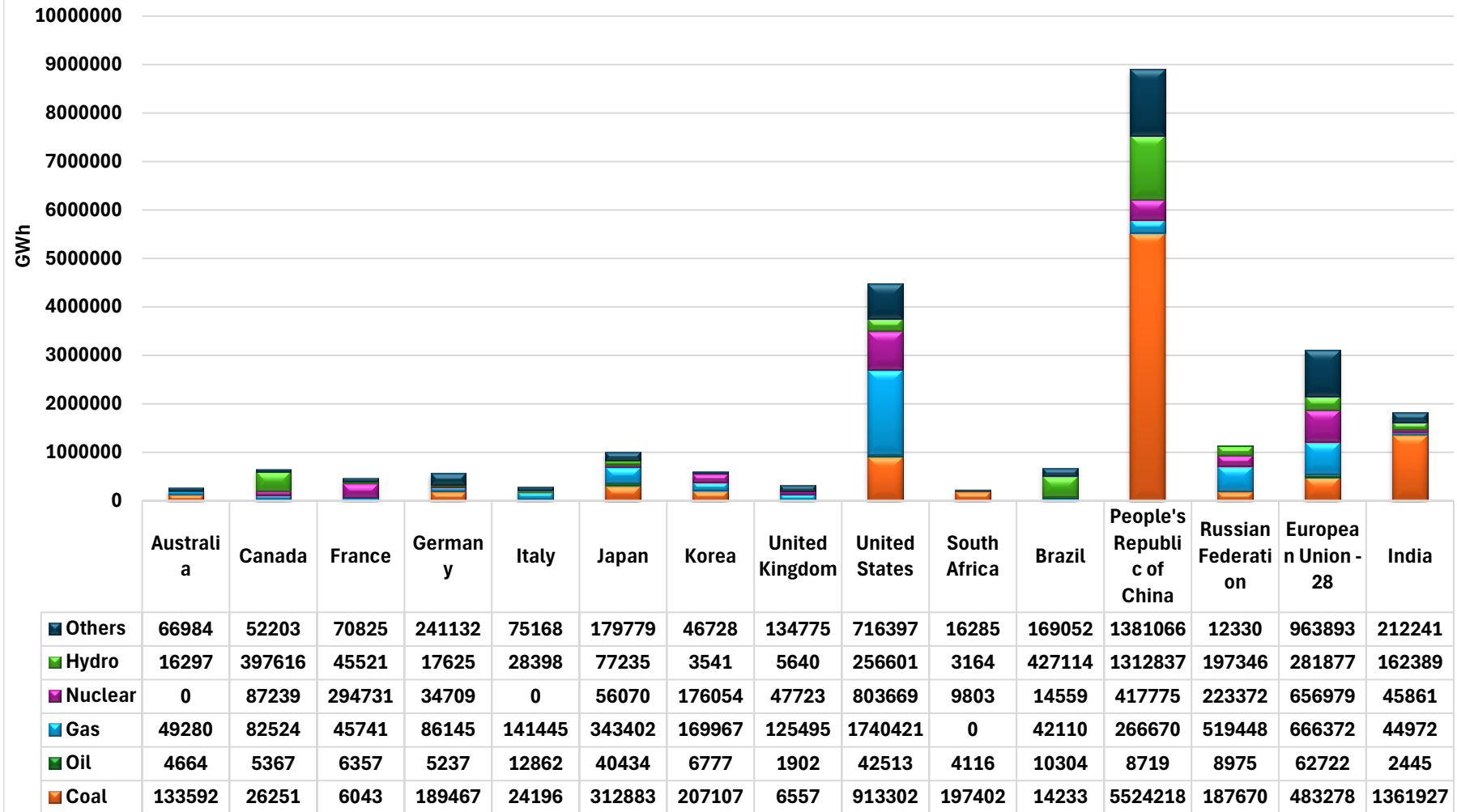


TOTAL 1734375 GWh

■ Hydro ■ Coal/ Lignite ■ Gas ■ Diesel ■ Nuclear ■ R.E.S.

**Electricity Generation in various Countries Mode-Wise
(2022)**

Chart : 17



Source : - International Energy Agency (IEA) except India

Electricity Generation (% Share) Mode-Wise in Various Countries
(2022)

Chart : 18

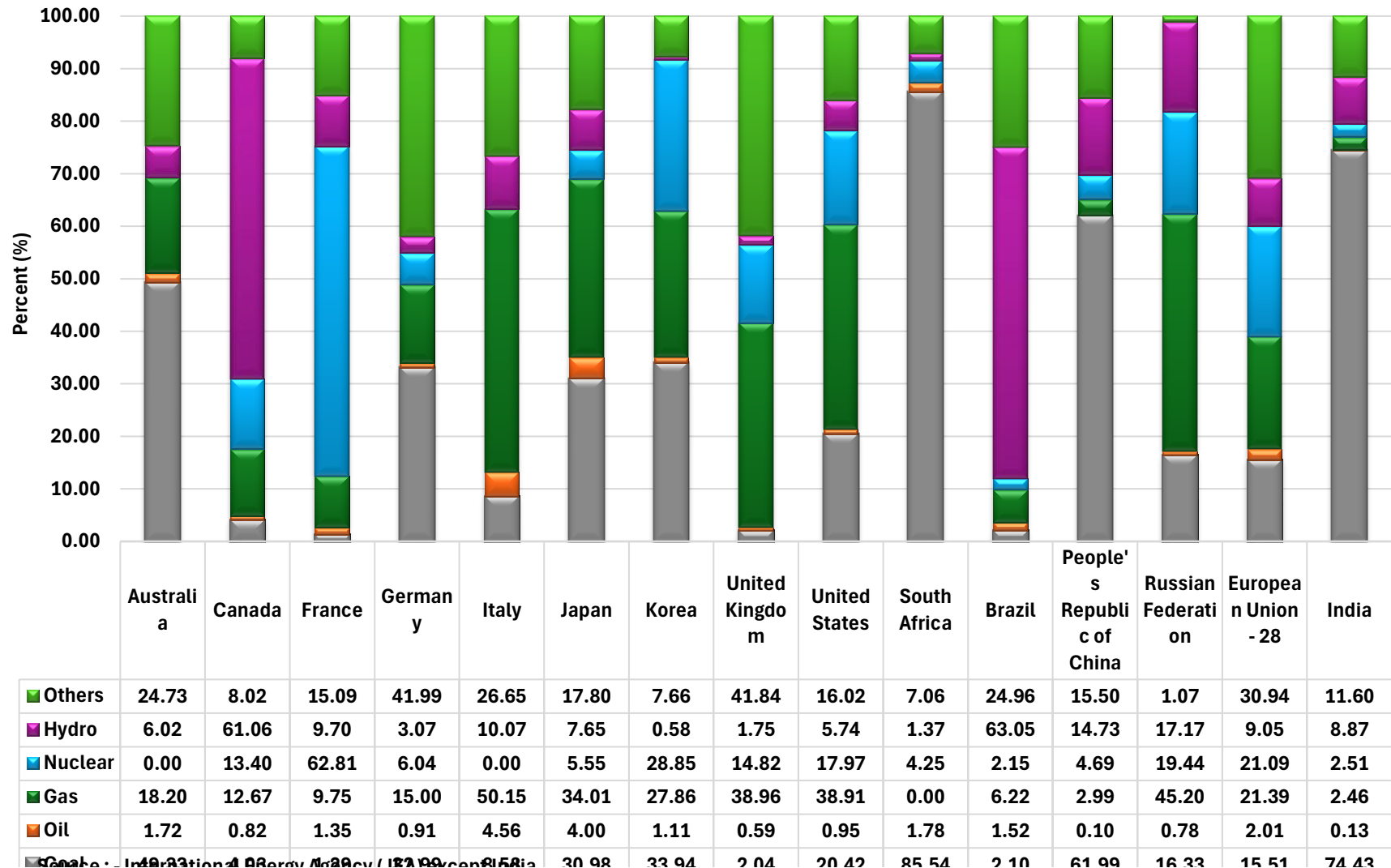


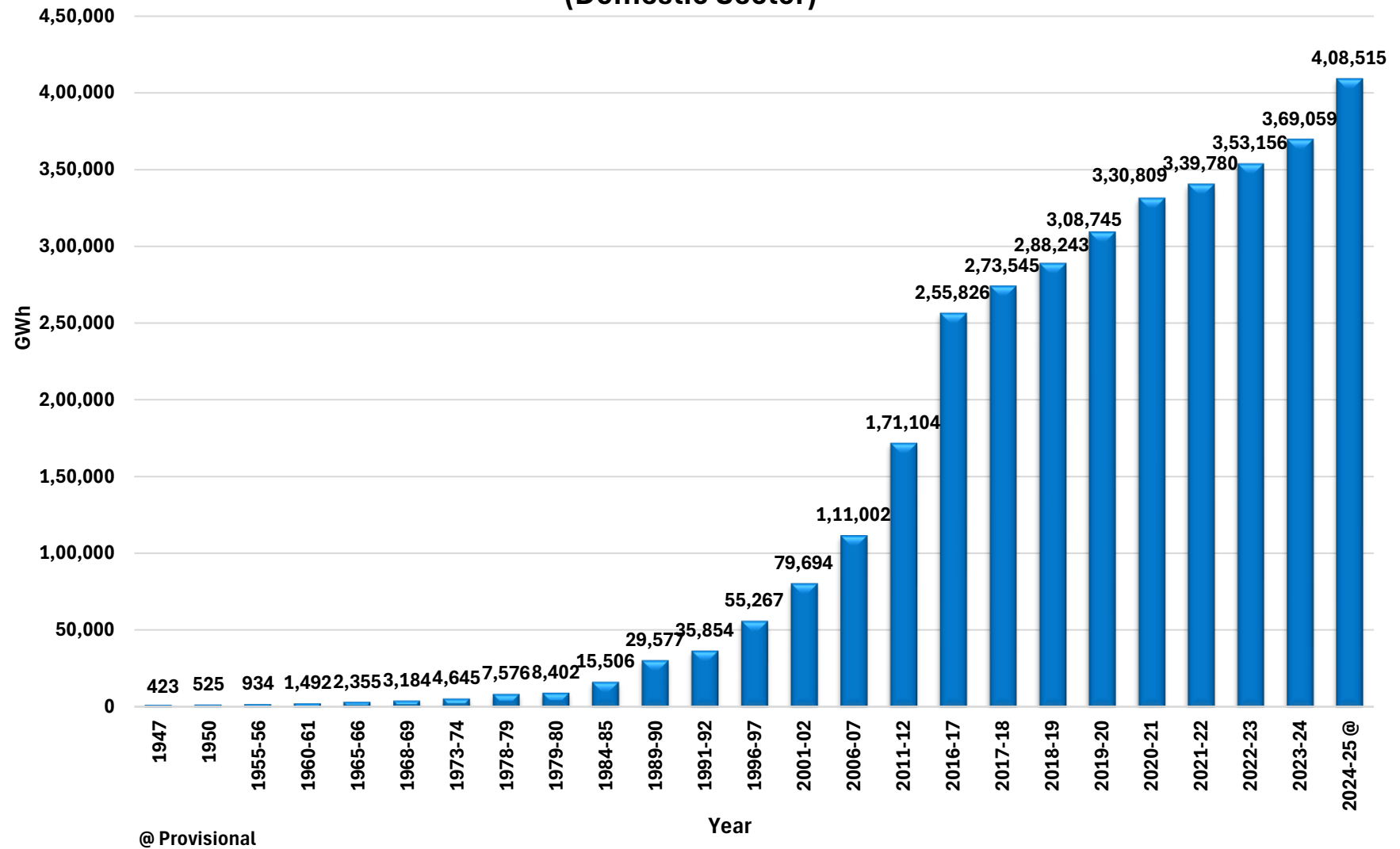
Table: 4
Category-wise Electricity Consumption in India
(Utilities & Non- Utilities)

Sl. No.	During financial year ending with	(GWh)											
		Domestic	% to Total	Commercial	% to Total	Industrial	% to Total	Traction	% to Total	Agriculture	% to Total	Misc.	% to Total
1	1947	423	10.11	178	4.26	2,960	70.78	277	6.62	125	2.99	219	5.24
2	1950	525	9.36	309	5.51	4,057	72.32	308	5.49	162	2.89	249	4.44
3	1955-56 (End of the 1st Plan)	934	9.20	546	5.38	7,514	74.03	405	3.99	316	3.11	435	4.29
4	1960-61 (End of the 2nd Plan)	1,492	8.88	848	5.05	12,547	74.67	454	2.70	833	4.96	630	3.75
5	1965-66(End of the 3rd Plan)	2,355	7.73	1,650	5.42	22,596	74.19	1,057	3.47	1,892	6.21	905	2.97
6	1968-69(End of the 3 Annual Plans)	3,184	7.69	2,126	5.14	29,931	72.31	1,247	3.01	3,465	8.37	1,439	3.48
7	1973-74(End of the 4th Plan)	4,645	8.36	2,988	5.38	37,791	68.02	1,531	2.76	6,310	11.36	2,292	4.13
8	1978-79(End of the 5th Plan)	7,576	9.02	4,330	5.15	54,440	64.81	2,186	2.60	12,028	14.32	3,445	4.10
9	1979-80(End of the 2 Annual Plans)	8,402	9.85	4,657	5.46	53,206	62.35	2,301	2.70	13,452	15.76	3,316	3.89
10	1984-85 (End of the 6th Plan)	15,506	12.45	6,937	5.57	73,520	59.02	2,880	2.31	20,961	16.83	4,765	3.83
11	1989-90 (End of the 7th Plan)	29,577	15.16	9,548	4.89	1,00,373	51.45	4,070	2.09	44,056	22.58	7,474	3.83
12	1991-92(End of the 2 Annual Plans)	35,854	15.51	12,032	5.20	1,10,844	47.94	4,520	1.96	58,557	25.33	9,394	4.06
13	1996-97(End of the 8th Plan)	55,267	17.53	17,519	5.56	1,39,253	44.17	6,594	2.09	84,019	26.65	12,642	4.01
14	2001-02(End of the 9th Plan)	79,694	21.27	24,139	6.44	1,59,507	42.57	8,106	2.16	81,673	21.80	21,551	5.75
15	2006-07(End of 10th Plan)	1,11,002	21.12	40,220	7.65	2,41,216	45.89	10,800	2.05	99,023	18.84	23,411	4.45
16	2011-12 (End of 11th Plan)	1,71,104	21.79	65,381	8.33	3,52,291	44.87	14,206	1.81	1,40,960	17.95	41,252	5.25
17	2016-17 (End of 12th Plan)	2,55,826	24.11	89,825	8.46	4,40,206	41.48	15,683	1.48	1,91,151	18.01	68,493	6.45
18	2017-18	2,73,545	24.35	93,755	8.35	4,68,613	41.71	17,433	1.55	1,99,247	17.74	70,834	6.31
19	2018-19	2,88,243	23.82	98,228	8.12	5,19,196	42.91	18,837	1.56	2,13,409	17.64	72,058	5.96
20	2019-20	3,08,745	24.74	1,06,047	8.50	5,32,820	42.69	19,148	1.53	2,11,295	16.93	70,031	5.61
21	2020-21	3,30,809	26.89	86,950	7.07	5,08,776	41.36	14,668	1.19	2,21,303	17.99	67,701	5.50
22	2021-22	3,39,780	25.80	97,121	7.38	5,56,481	42.26	21,935	1.67	2,28,451	17.35	72,996	5.54
23	2022-23	3,53,156	24.52	1,17,231	8.14	5,93,895	41.23	30,028	2.08	2,43,852	16.93	1,02,149	7.09
24	2023-24	3,69,059	23.96	1,28,623	8.35	6,40,626	41.60	33,918	2.20	2,75,485	17.89	92,876	6.03
25	2024-25 @	4,08,515	25.17	1,35,163	8.33	6,55,562	40.39	34,500	2.13	2,60,276	16.04	1,28,953	7.95

(@) Provisional

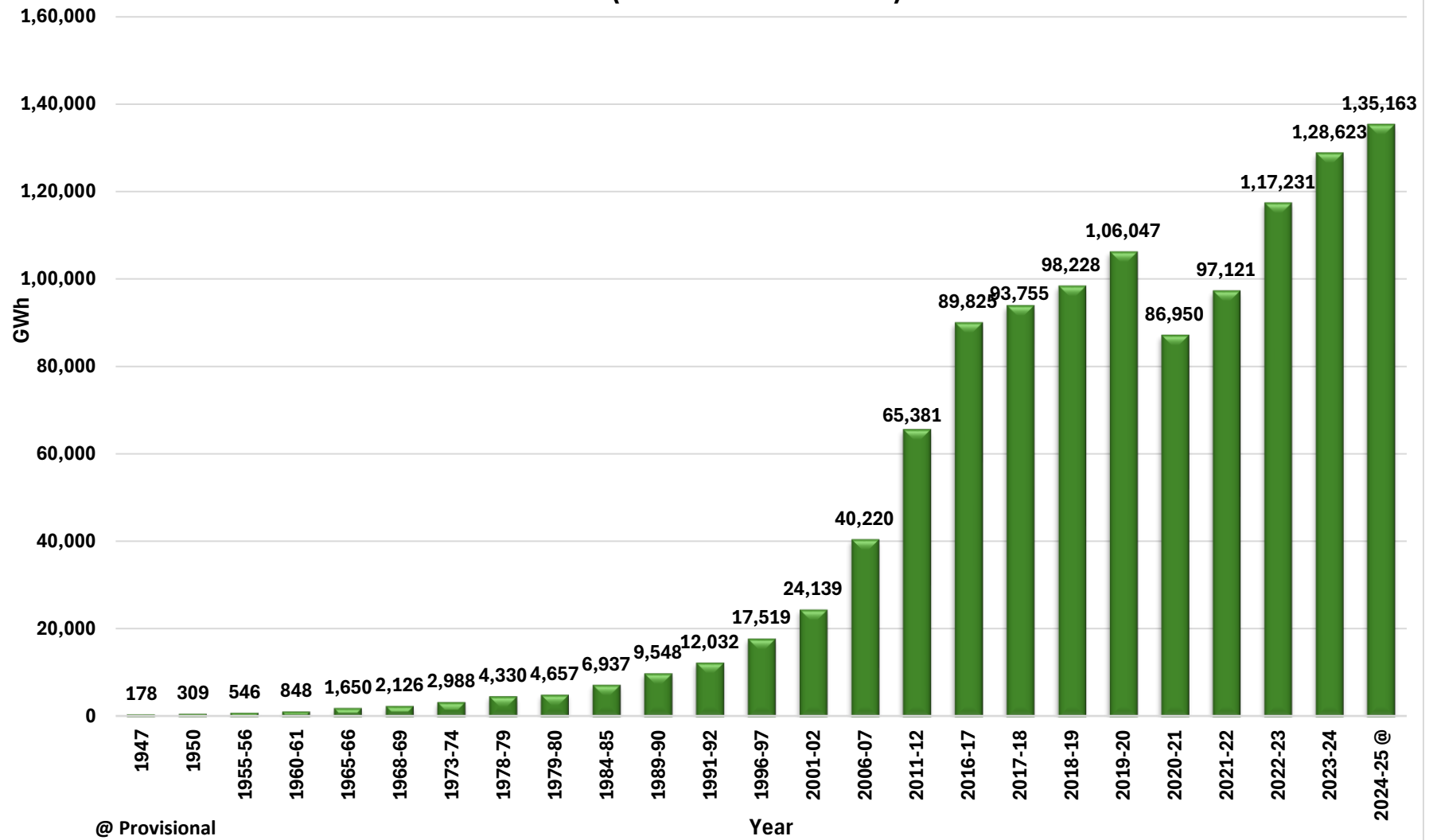
Electricity Consumption (Domestic Sector)

Chart : 19



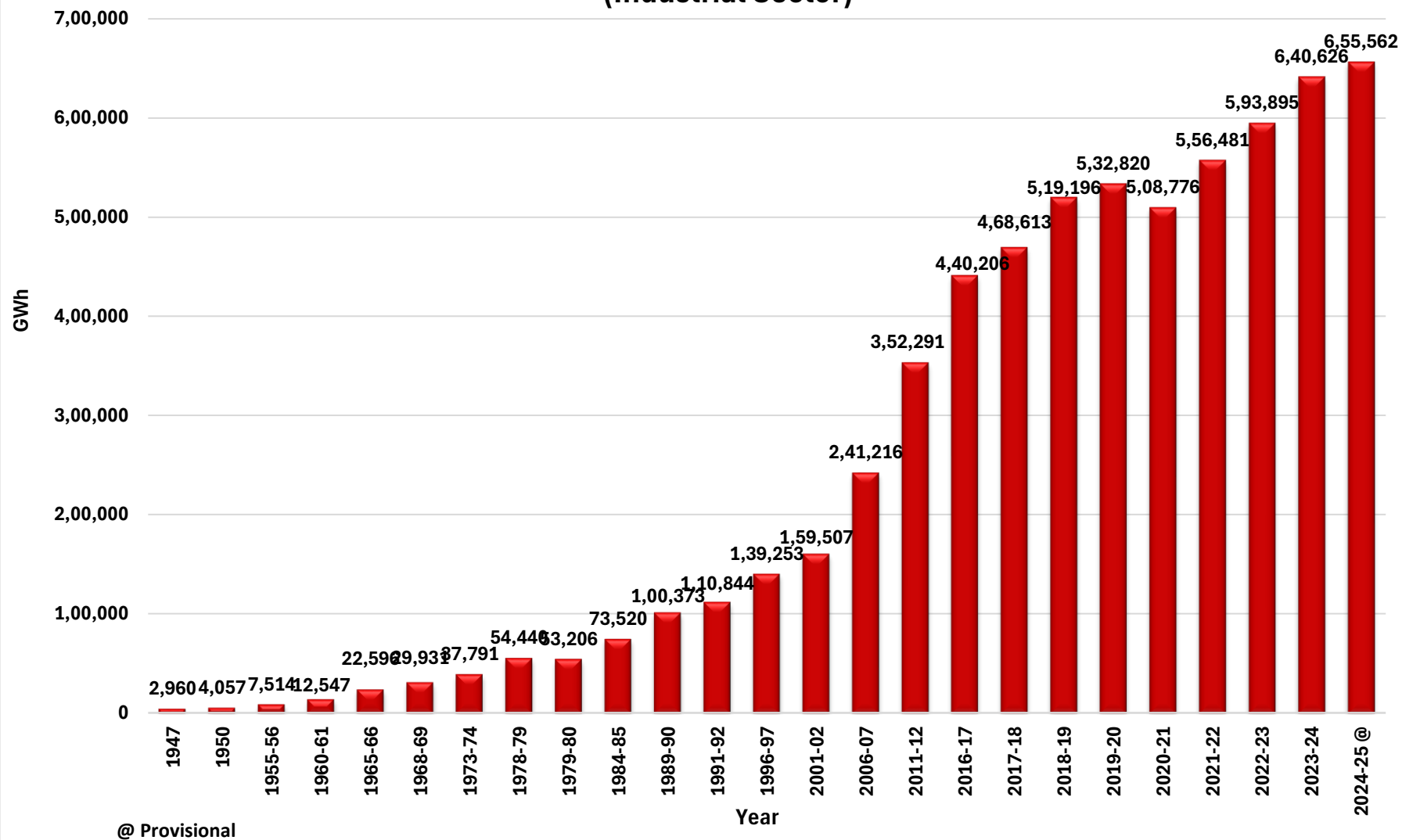
Electricity Consumption (Commercial Sector)

Chart : 20



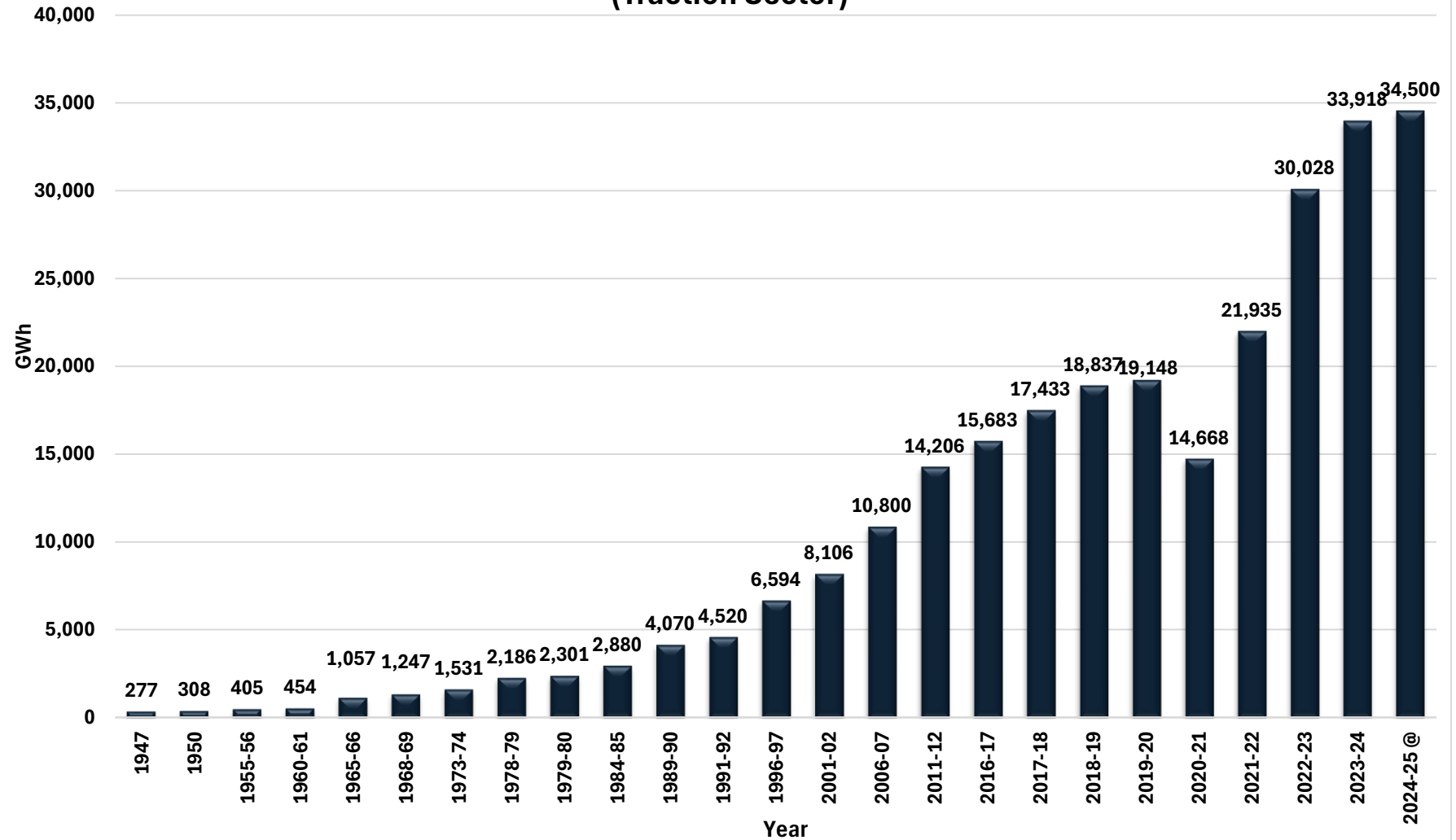
Electricity Consumption (Industrial Sector)

Chart : 21



Electricity Consumption (Traction Sector)

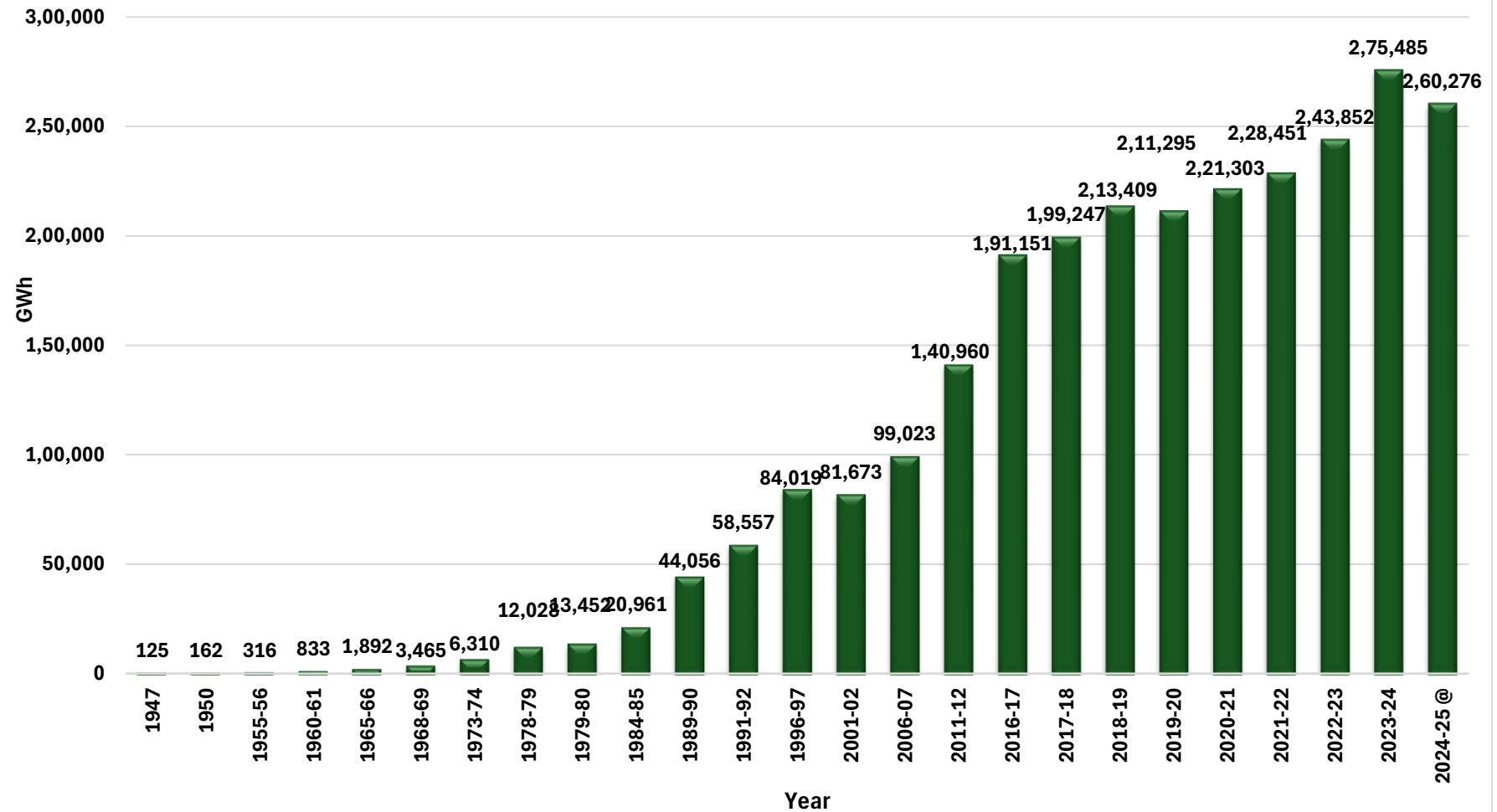
Chart : 22



@ Provisional

Electricity Consumption (Agriculture Sctor)

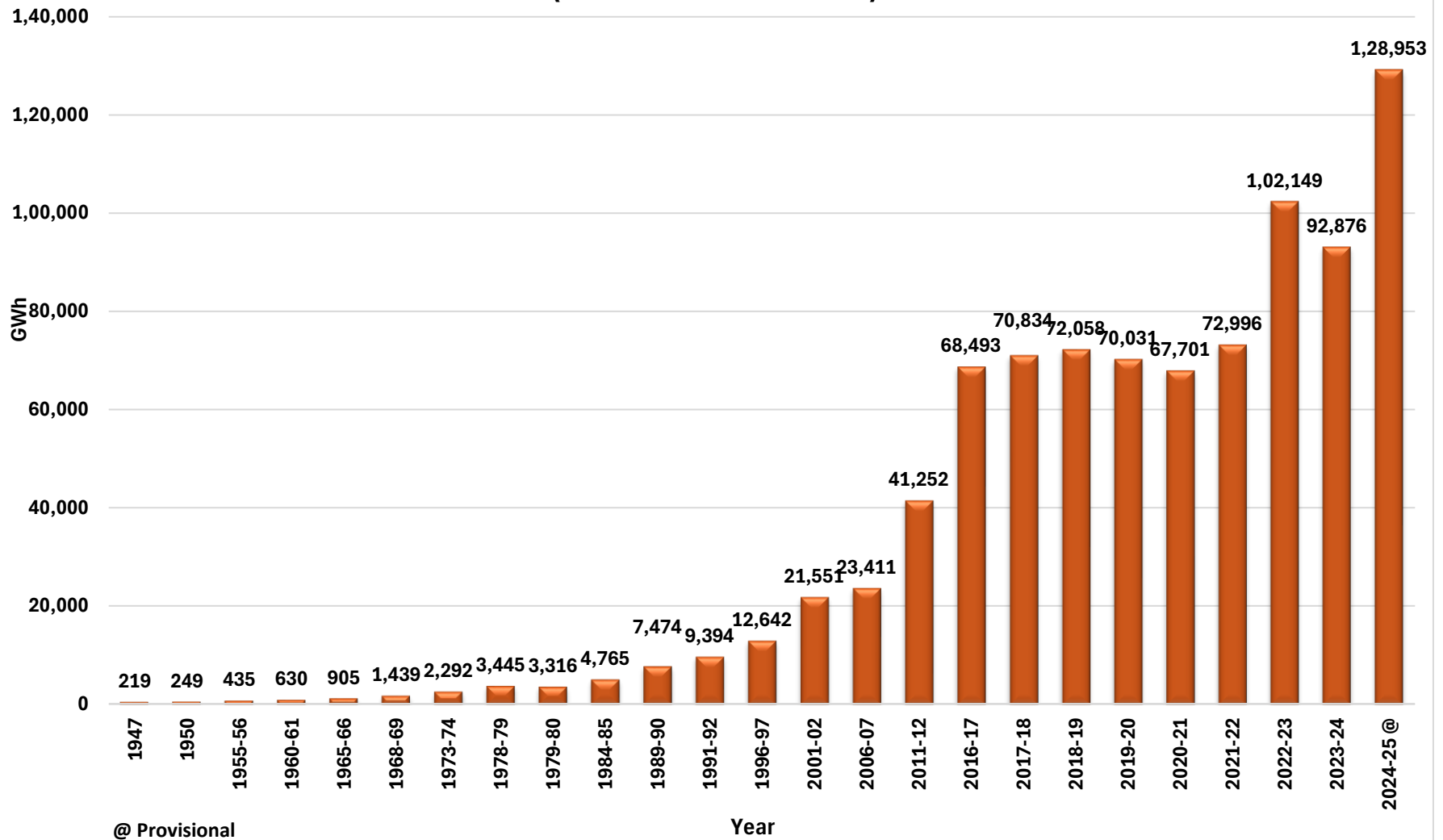
Chart : 23



@ Provisional

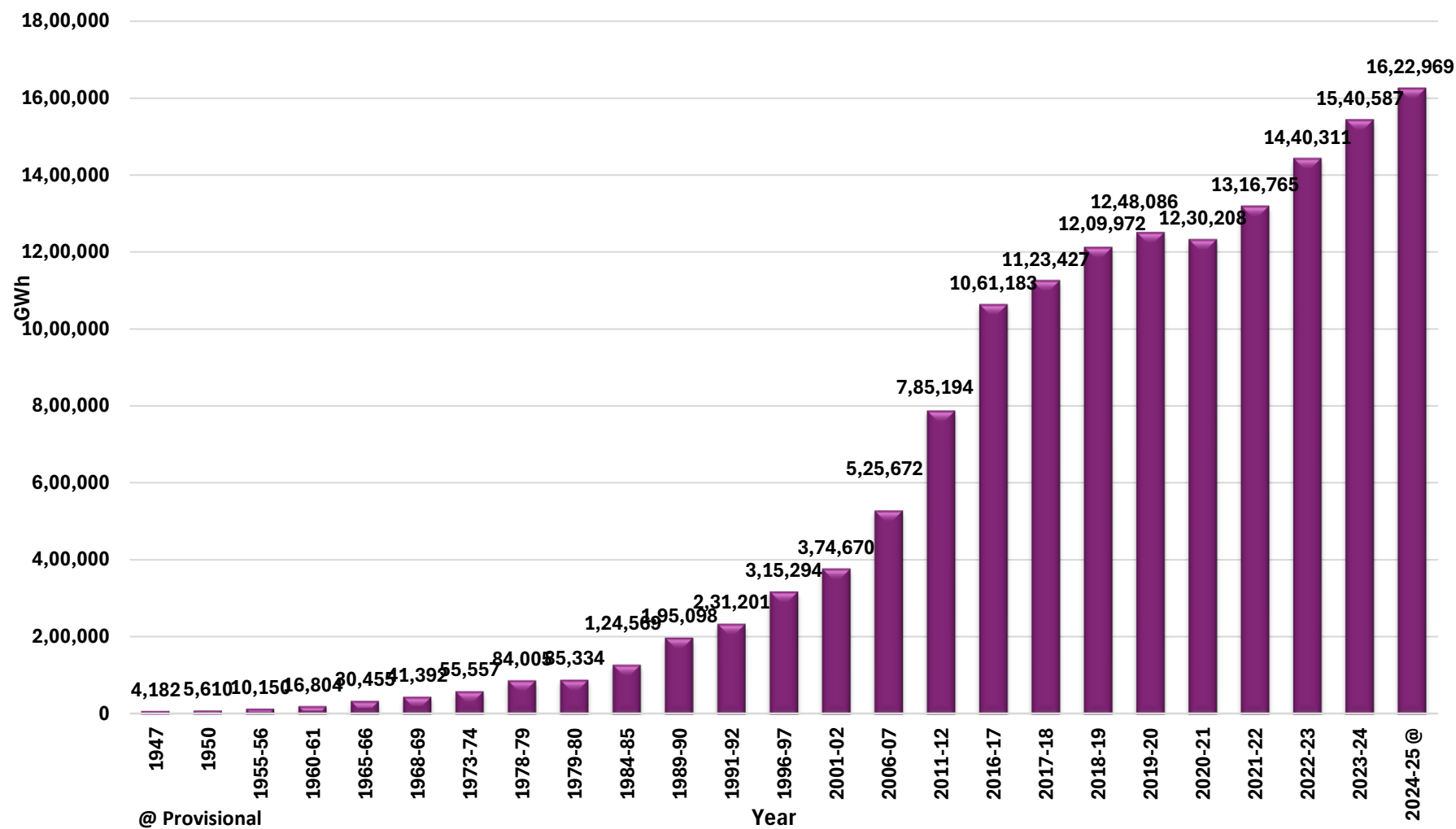
Electricity Consumption (Miscellaneous Sector)

Chart : 24



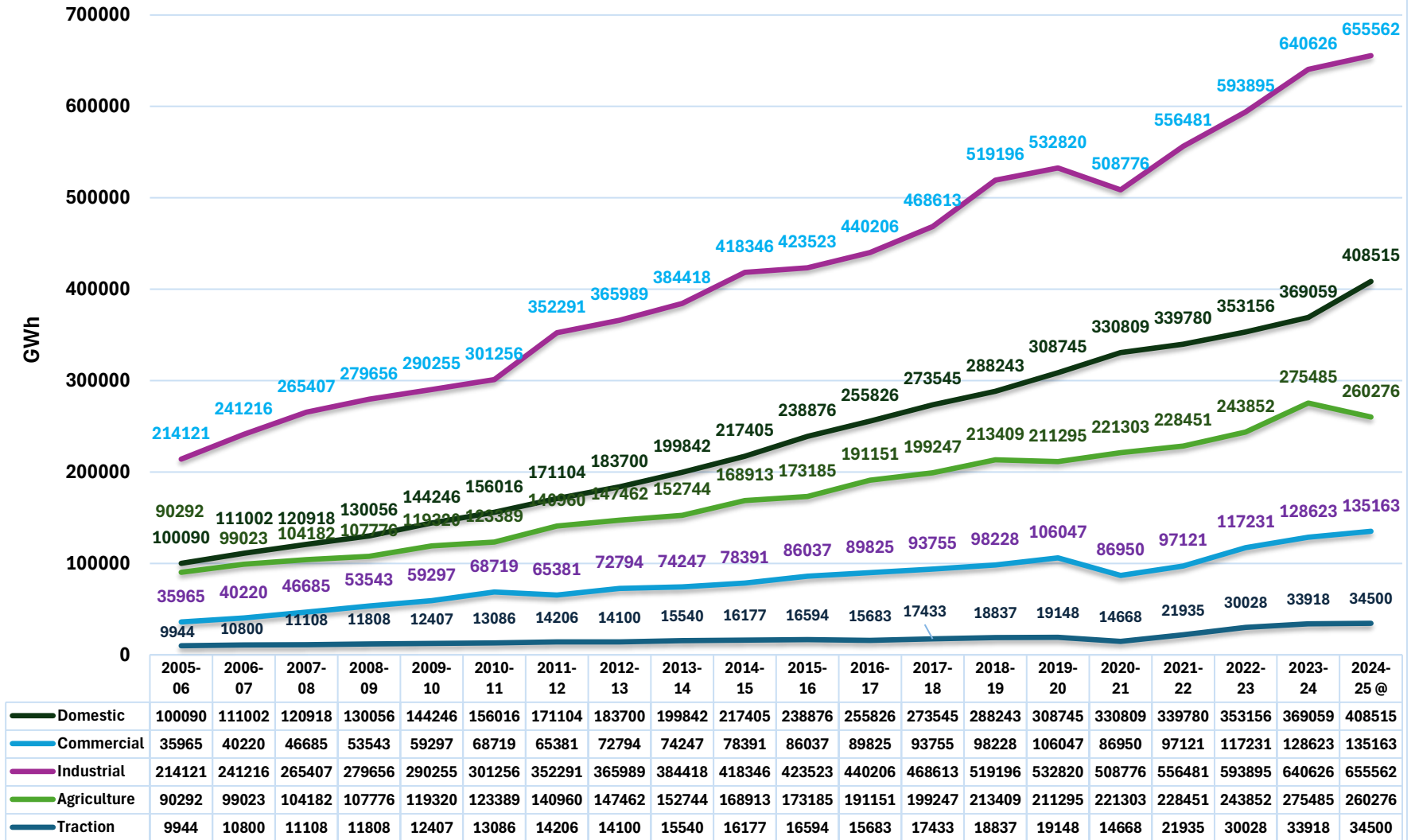
All India Total Electricity Consumption (Utilities & Non-Utilities)

Chart : 25

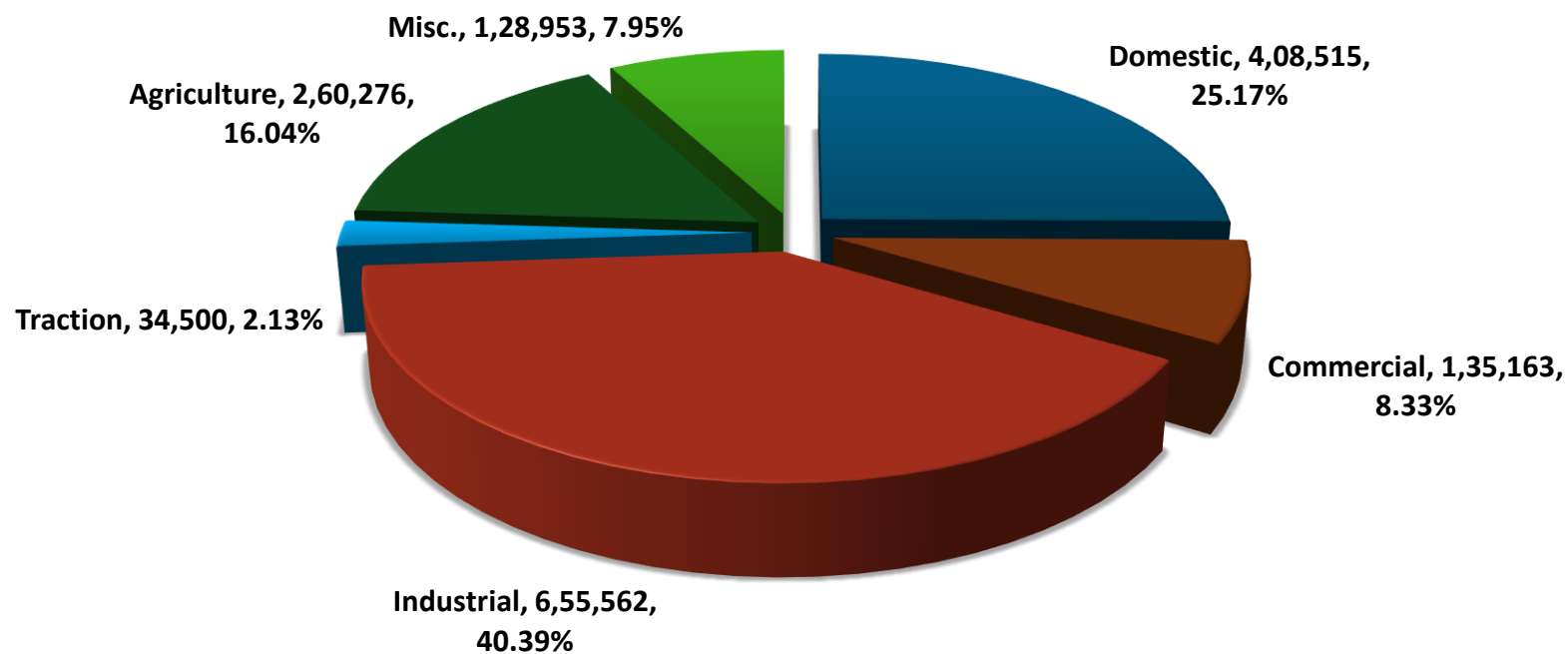


Trend of Energy Consumption - Category-wise 2003-04 TO 2024-25

Chart : 25A



**Electricity Consumption Sector-wise
(Utilities & Non- Utilities)
2024-25**



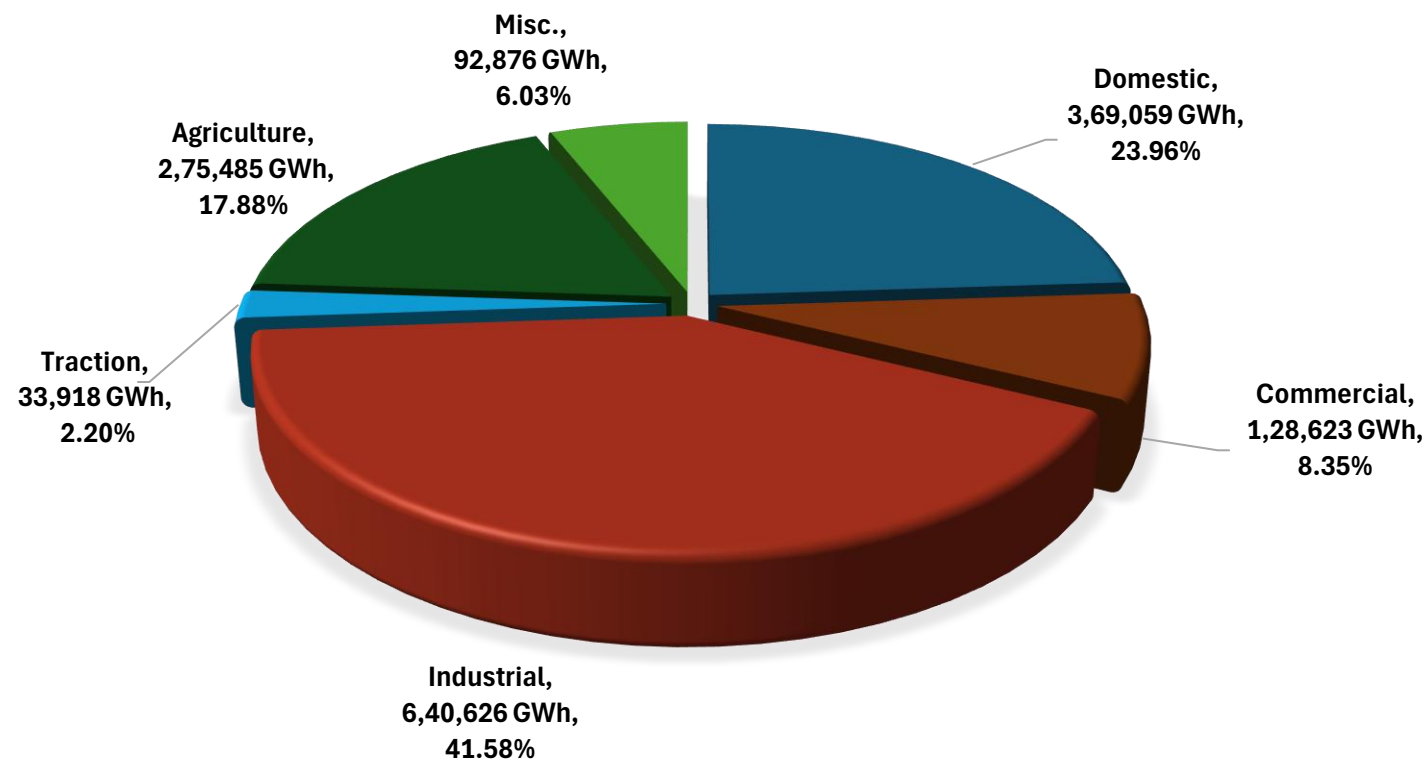
TOTAL = 1622969 GWh

@ Provisional

■ Domestic
 ■ Commercial
 ■ Industrial
 ■ Traction
 ■ Agriculture
 ■ Misc.

Chart : 26A

**Electricity Consumption Sector-wise
(Utilities & Non-Utilities)
2023-24**

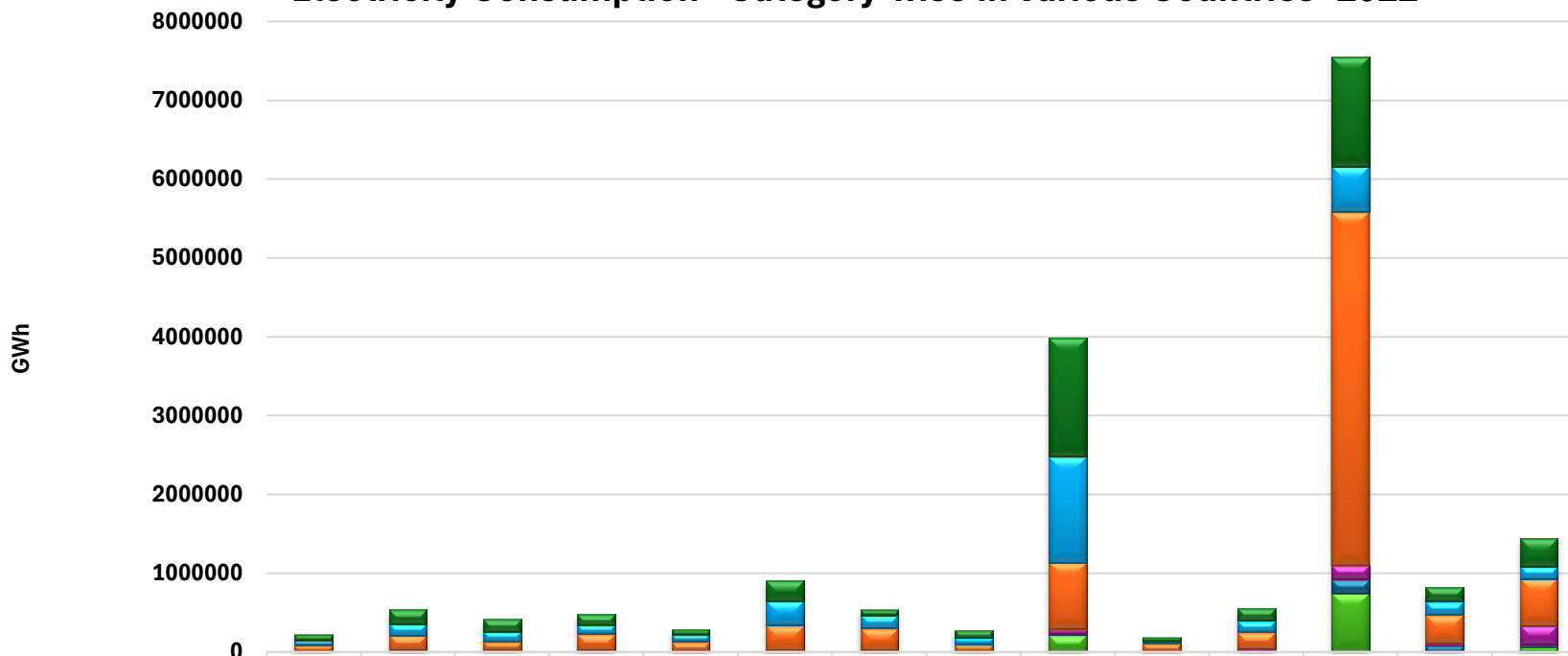


TOTAL 1540587 GWh

■ Domestic ■ Commercial ■ Industrial ■ Traction ■ Agriculture ■ Misc.

Chart : 27

Electricity Consumption - Category-wise in Various Countries -2022

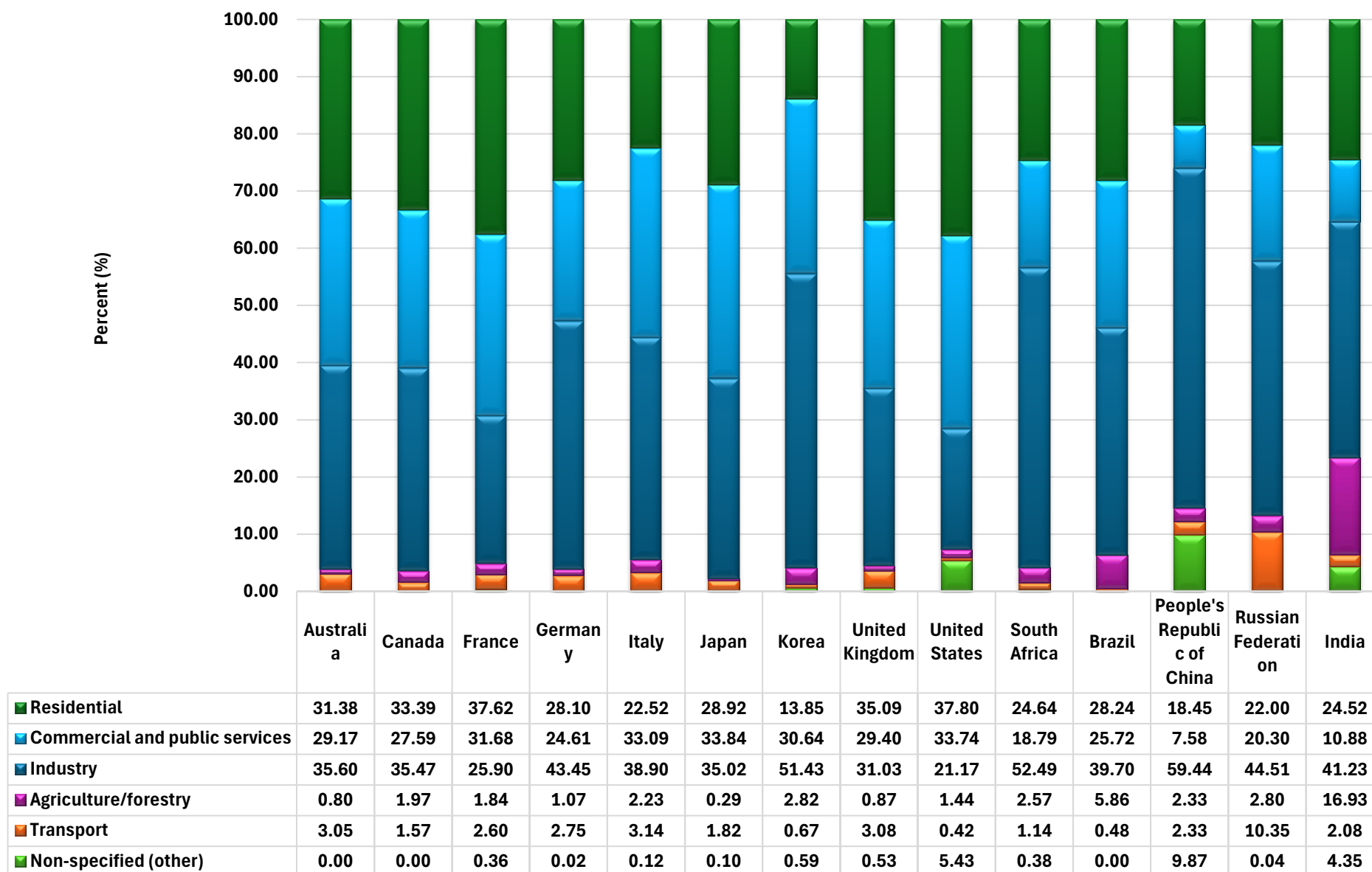


	Australia	Canada	France	Germany	Italy	Japan	Korea	United Kingdom	United States	South Africa	Brazil	People's Republic of China	Russian Federation	India
Residential	69291	179161	156370	134267	64640	262210	75013	96247	1508585	45886	155599	1393586	181000	353156
Commercial and public services	64411	148032	131705	117561	94967	306879	165987	80628	1346520	34989	141676	572373	167031	156714
Industry	78630	190276	107647	207567	111638	317568	278607	85103	844692	97769	218743	4488746	366270	593895
Agriculture/forestry	1764	10591	7646	5130	6387	2608	15280	2377	57321	4785	32288	175672	23000	243852
Transport	6745	8449	10823	13147	9009	16530	3655	8439	16739	2123	2636	175934	85159	30028
Non-specified (other)	0	0	1509	73	347	929	3175	1466	216687	705	0	745589	359	62666

Source: International Energy Agency (IEA) except India

Chart : 28

Electricity Consumption (Category-wise % Share) in Various Countries -2022



Source: - International Energy Agency (IEA) except India

Table 5
Per Capita Electricity Consumption and T&D Losses
of various Countries in 2021 & 2022

Sl. No.	Per Capita Consumption (kWh)			Sl. No.	T & D Losses (%)		
	Name of the Country	2021	2022		Name of the Country	2021	2022
1	Canada	15,004	14,906	1	Korea	3.62	3.70
2	USA	12,546	12,986	2	Japan	5.26	5.10
3	Australia	9,843	9,925	3	Germany	5.37	5.51
4	Japan	7,971	7,811	4	Italy	6.51	6.67
5	France	6,948	6,660	5	Australia	5.82	5.94
6	Germany	6,583	6,270	6	South Africa	12.83	12.82
7	Korea	11,732	11,716	7	France	9.03	8.49
8	UK	4,567	4,314	8	China	4.30	4.06
9	Russia	7,252	7,219	9	USA	6.36	5.12
10	Italy	5,290	5,216	10	Canada	6.09	5.94
11	South Africa	3,643	3,546	11	UK	8.86	9.36
12	Brazil	2,663	2,723	12	Russia	12.11	11.97
13	China	5,851	6,113	13	Brazil	20.35	18.88
14	India*	1,255	1,331	14	India	19.27	17.68
15	World	3,366	3,247	15	World	7.50	7.45

Note :-

Basic data obtained from IEA Website (except) India.

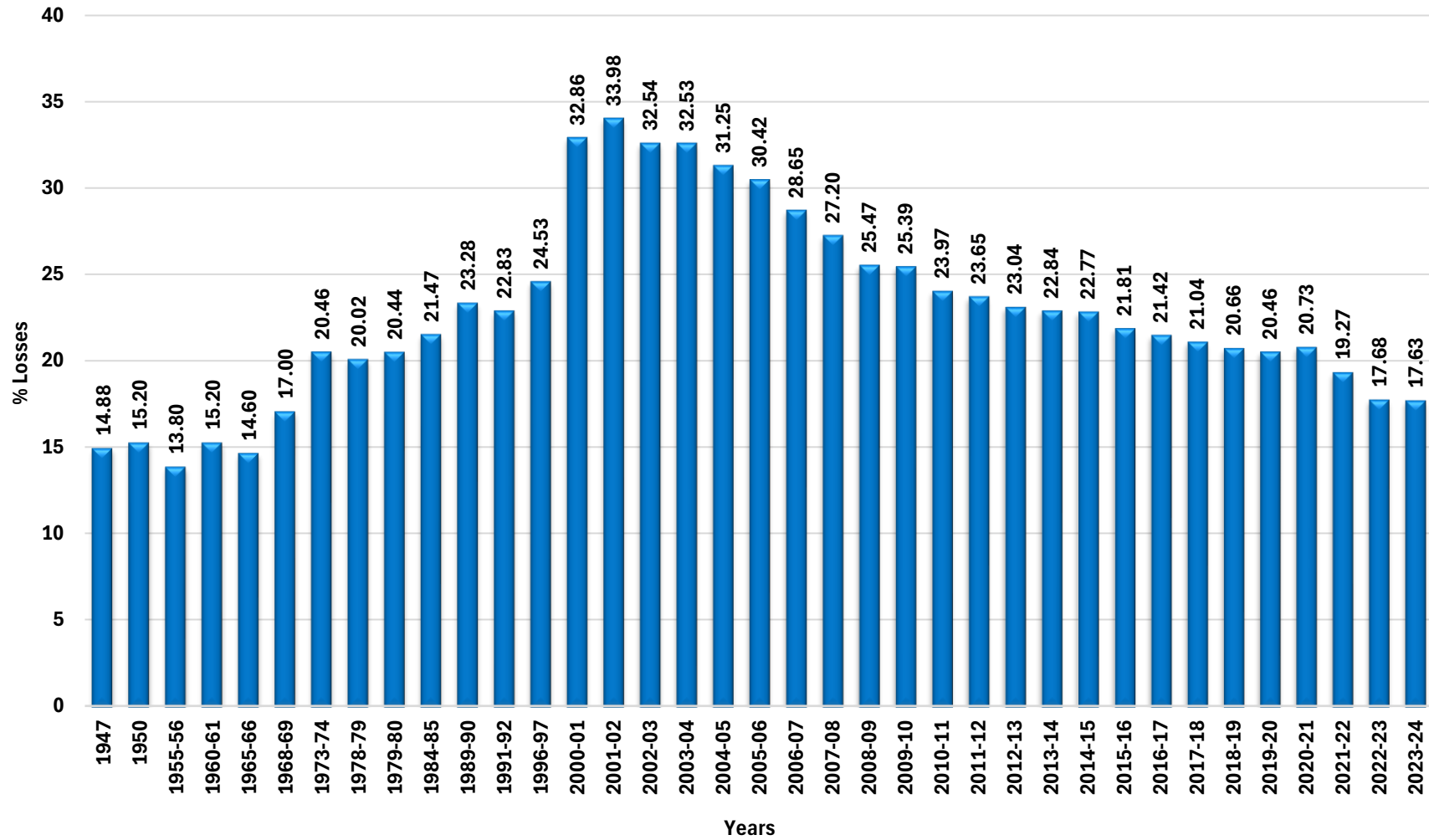
* Per Capita Consumption= (Gross Electrical Energy Availability/Midyear Population).

Per Capita Consumption of India during the year 2022-23 & 2023-24 is 1331 & 1400 respectively.

T&D Losses of India during the year 2023-24 is 17.63%

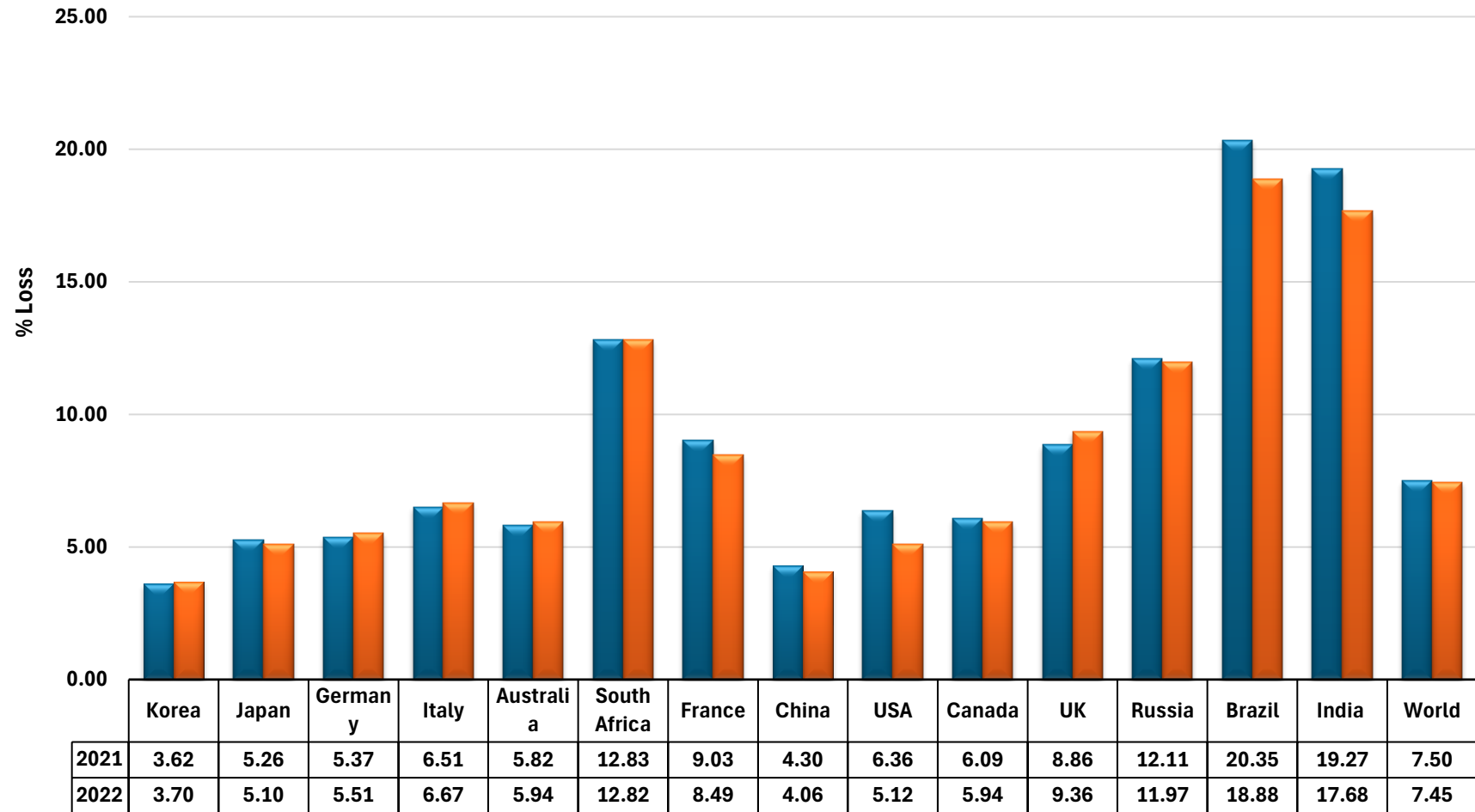
All India Transmission and Distribution Losses (%)

Chart : 29



**T & D Losses (%) of Various Countries
(2021 & 2022)**

Chart : 30



Source: - International Energy Agency (IEA) (except India)

■ 2021 ■ 2022

Table 6
Installed Generation Capacity of Captive Power Plants in Industries having
Demand of 0.5 MW & Above - Mode-wise

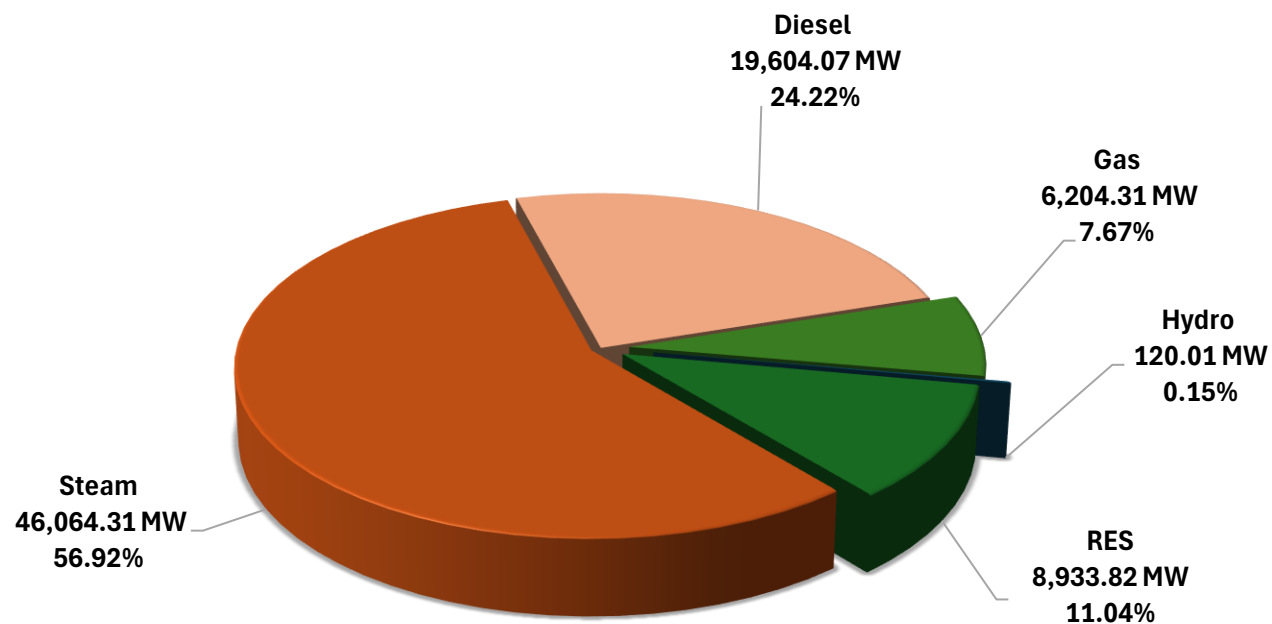
(MW)

Sl. No.	As on	Steam	Diesel	Gas	Hydro	RES	Railways	Total
1	2	3	4	5	6	7	8	9
								(3+4+5+6+7+8)
1	31.12.1947	**	**	**	**	**	0.00	410.00
2	31.12.1950	**	**	**	**	**	0.00	587.85
3	31.03.1956 (End of the 1st Plan)	**	**	**	**	**	0.00	759.65
4	31.03.1961 (End of the 2nd Plan)	**	**	**	**	**	0.00	1,001.37
5	31.03.1966(End of the 3rd Plan)	**	**	**	**	**	0.00	1,082.36
6	31.03.1969(End of the 3 Annual Plans)	**	**	**	**	**	0.00	1,277.47
7	31.03.1974(End of the 4th Plan)	**	**	**	**	**	0.00	1,732.70
8	31.03.1979(End of the 5th Plan)	1,949.23	559.17	44.27	2.61	0.00	62.31	2,617.59
9	31.03.1980(End of the 2 Annual Plans)	2,021.61	720.58	54.27	2.61	0.00	60.44	2,859.51
10	31.03.1985 (End of the 6th Plan)	2,803.18	2,077.06	155.31	2.91	0.00	81.80	5,120.26
11	31.03.1990 (End of the 7th Plan)	4,822.85	2,754.48	425.51	3.60	0.00	109.29	8,115.73
12	31.03.1992 (End of the 2 Annual Plans)	5,377.10	3,291.06	495.72	4.30	0.00	133.23	9,301.41
13	31.03.1997(End of the 8th Plan)	6,151.69	4,529.91	1,166.38	3.50	64.83	162.62	12,078.93
14	31.03.2002(End of the 9th Plan)	8,354.03	6,521.95	2,125.91	51.10	92.21	0.00	17,145.20
15	31.03.2007 (End of 10th Plan)	11,397.52	7,723.35	2,976.00	60.50	177.67	0.00	22,335.04
16	31.03.2012(End of 11th Plan)	22,615.39	9,955.23	5,884.95	47.59	872.21	0.00	39,375.37
17	31.03.2017 (End of 12th Plan)	30,571.95	13,349.69	6,109.21	65.29	1,433.06	0.00	51,529.19
18	31.03.2018	32,854.41	13,144.60	7,156.37	50.96	1,726.17	0.00	54,932.51
19	31.03.2019	47,679.04	15,570.97	8,787.13	103.39	3,066.59	0.00	75,207.13
20	31.03.2020	51,542.77	12,774.66	7,315.87	130.77	4,475.29	0.00	76,239.36
21	31.03.2021	47,759.52	17,562.86	7,361.11	130.70	5,693.78	0.00	78,507.97
22	31.03.2022	45,302.73	18,648.78	5,685.25	134.60	6,961.08	0.00	76,732.43
22	31.03.2023	46,782.17	18,077.84	6,360.26	132.26	7,047.29	0.00	78,399.82
23	31.03.2024	46,064.31	19,604.07	6,204.31	120.01	8,933.36	0.00	80,926.07
24	31.03.2025 @	46,500.00	19,700.00	6,250.00	120.01	9,130.00	0.00	81,700.01

** Breakup Not Available

@ Estimated

Installed Generation Capacity of Captive Power Plants in Industries
Demand of 0.5 MW & Above - Mode-wise
(As on 31.03.2025)



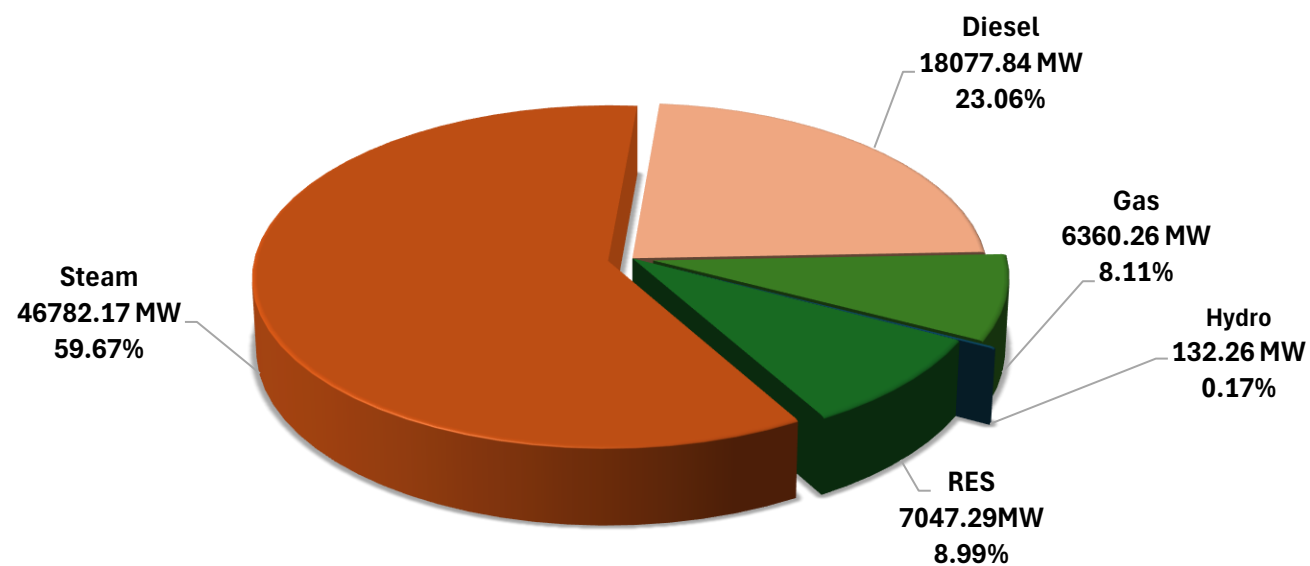
@-Estimated

TOTAL = 81700.01 MW

Steam Diesel Gas Hydro RES

Chart : 31A

**Installed Generation Capacity of Captive Power Plants in Industries
Demand of 0.5MW & Above - Mode-wise
(As on 31.03.2024)**

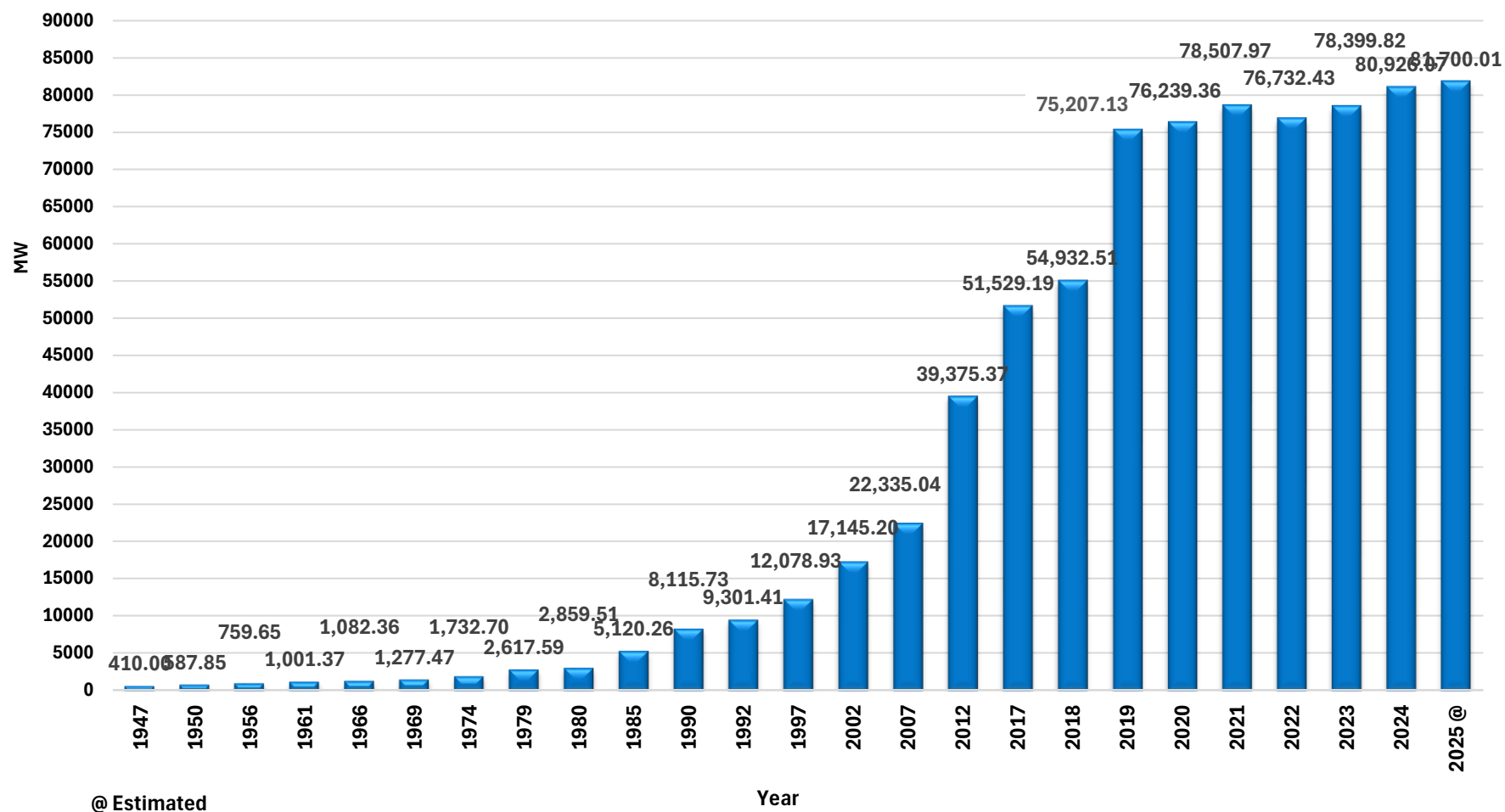


TOTAL = 80,926.07 MW

Steam Diesel Gas Hydro RES

Installed Generating Capacity of Captive Power Plants Industries Demand of 0.5MW & Above

Chart : 32



Note: 1. Figures for the years 1947 & 1950 are as on 31st December.
2. Figures for the year 1955-56 & onward are as on 31st March.

Table 7
Energy Generated by Captive Power Plants Industries having Demand of 0.5 MW & Above
Mode-wise

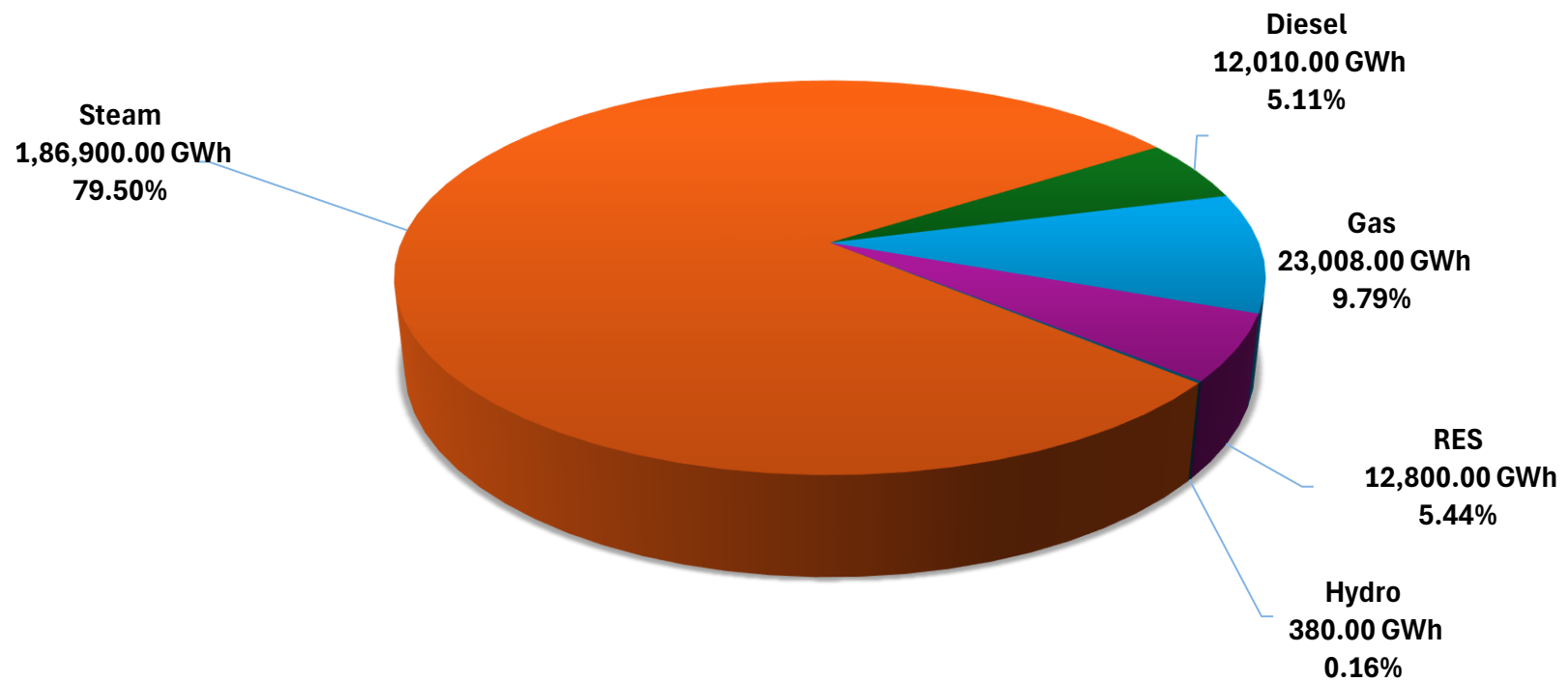
								(GWh)
Sl. No.	During financial year ending with	Hydro	Steam	Diesel	Gas	RES	Railways	Total
1	2	3	4	5	6	7	8	9 (3+4+5+6+7+8)
1	1947	**	**	**	**	**	**	856.91
2	1950	**	**	**	**	**	0.00	1,467.81
3	1955-56 (End of the 1st Plan)	**	**	**	**	**	0.00	2,184.82
4	1960-61 (End of the 2nd Plan)	**	**	**	**	**	0.00	3,186.10
5	1965-66(End of the 3rd Plan)	**	**	**	**	**	0.00	3,733.19
6	1968-69(End of the 3 Annual Plans)	**	**	**	**	**	0.00	4,136.06
7	1973-74(End of the 4th Plan)	**	**	**	**	**	0.00	6,067.47
8	1978-79(End of the 5th Plan)	13.22	7,211.30	299.13	48.97	-	34.47	7,607.09
9	1979-80(End of the 2 Annual Plans)	15.06	7,225.60	825.71	90.67	-	35.67	8,192.71
10	1984-85 (End of the 6th Plan)	18.29	9,966.55	2,001.18	317.03	-	42.80	12,345.85
11	1989-90 (End of the 7th Plan)	15.83	19,110.46	2,757.27	1,313.29	-	29.07	23,225.92
12	1991-92(End of the 2 Annual Plans)	17.57	23,409.35	3,247.81	1,905.24	-	22.52	28,602.49
13	1996-97(End of the 8th Plan)	28.84	29,127.94	6,573.38	5,039.22	45.63	25.07	40,840.08
14	2001-02(End of the 9th Plan)	116.59	41,853.85	10,719.66	8,835.19	155.53	-	61,680.82
15	2006-07(End of 10th Plan)	217.81	56,184.04	9,974.85	15,207.17	215.88	-	81,799.75
16	2011-12 (End of 11th Plan)	131.28	1,04,862.65	6,244.32	21,971.57	1,178.17	-	1,34,387.99
17	2016-17 (End of 12th Plan)	143.64	1,37,588.31	9,181.74	22,855.31	2,277.02	-	1,72,046.03
18	2017-18	112.48	1,43,867.68	8,106.89	25,362.17	2,328.17	-	1,79,777.39
19	2018-19	270.04	1,84,250.36	5,334.32	19,545.09	3,673.92	-	2,13,073.73
20	2019-20	348.04	2,05,545.78	1,919.32	25,443.15	6,310.47	-	2,39,566.76
21	2020-21	339.12	1,93,142.60	2,504.19	21,683.55	7,157.94	-	2,24,827.40
22	2021-22	356.66	1,79,235.45	2,104.51	20,801.17	6,812.98	-	2,09,310.76
23	2022-23	290.70	1,79,831.20	2,035.36	21,086.63	8,687.92	-	2,11,931.81
24	2023-24	273.57	1,83,093.29	10,157.69	21,677.33	8,700.81	-	2,23,902.69
25	2024-25 @	380.00	1,86,900.00	12,010.00	23,008.00	12,800.00	-	2,35,098.00

** Breakup Not Available

@ Estimated

Electrical Energy Generated by Captive Power Plants Industries
Demand of 0.5 MW & above - Mode-wise
2024-25

Chart : 33

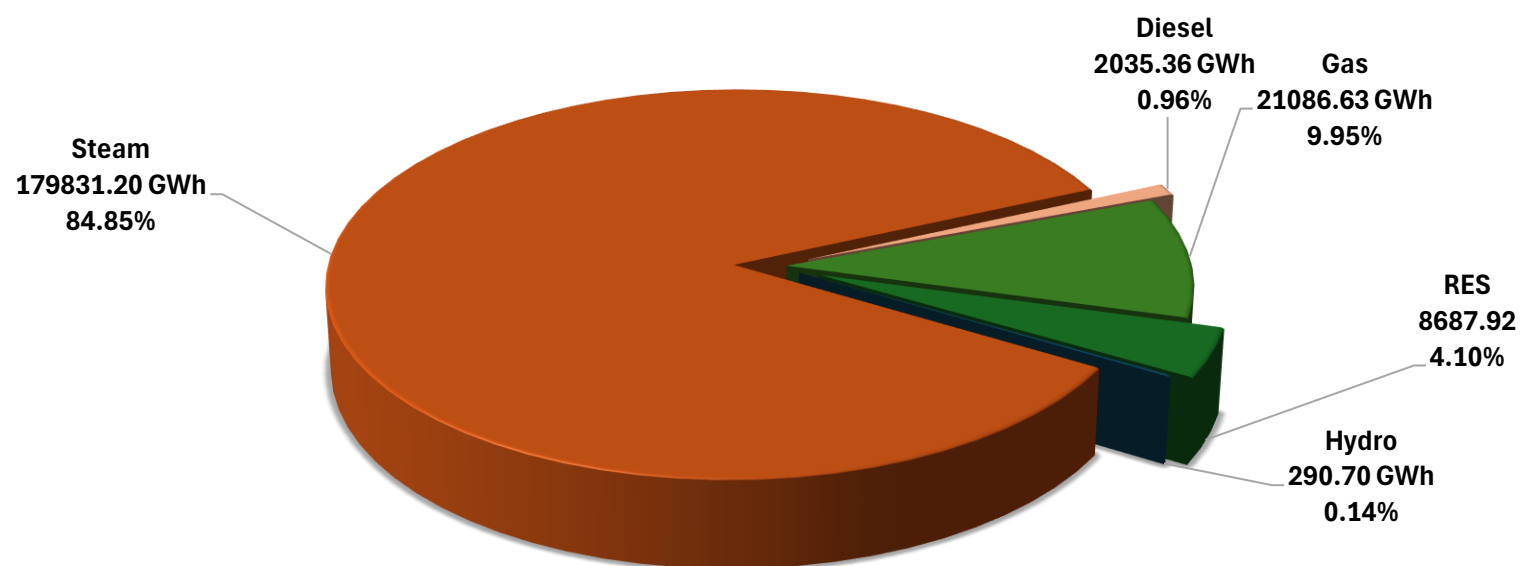


@ Estimated

Total =2,35,098 GWh

■ Hydro ■ Steam ■ Diesel ■ Gas ■ RES

Electrical Energy Generated by Captive Power Plants Industries Demand of 0.5 MW & above - Mode- wise 2023-24



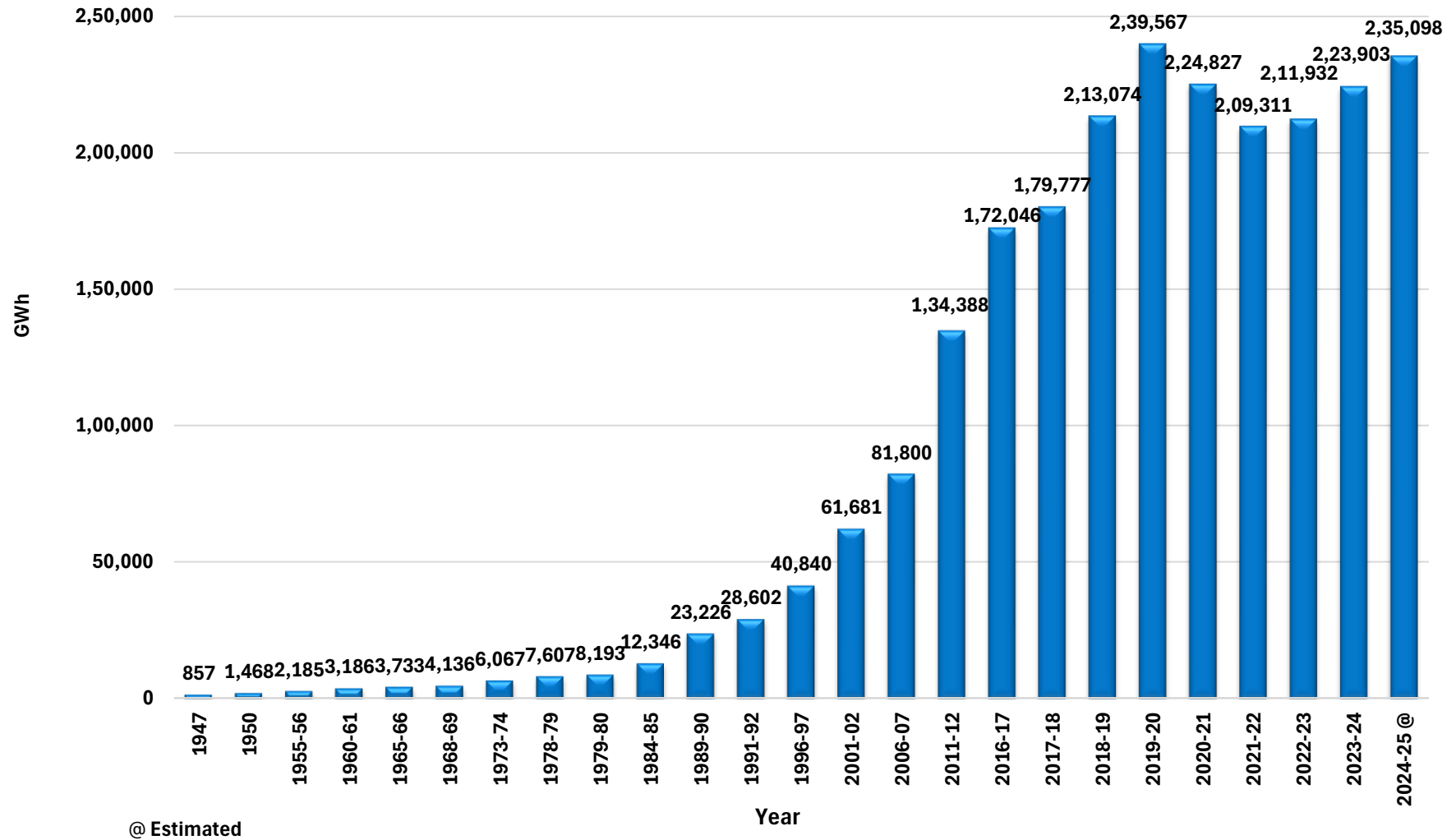
Total = 211931.81 GWh

Hydro Steam Diesel Gas RES

Energy Generated by Captive Power Plants INdustries

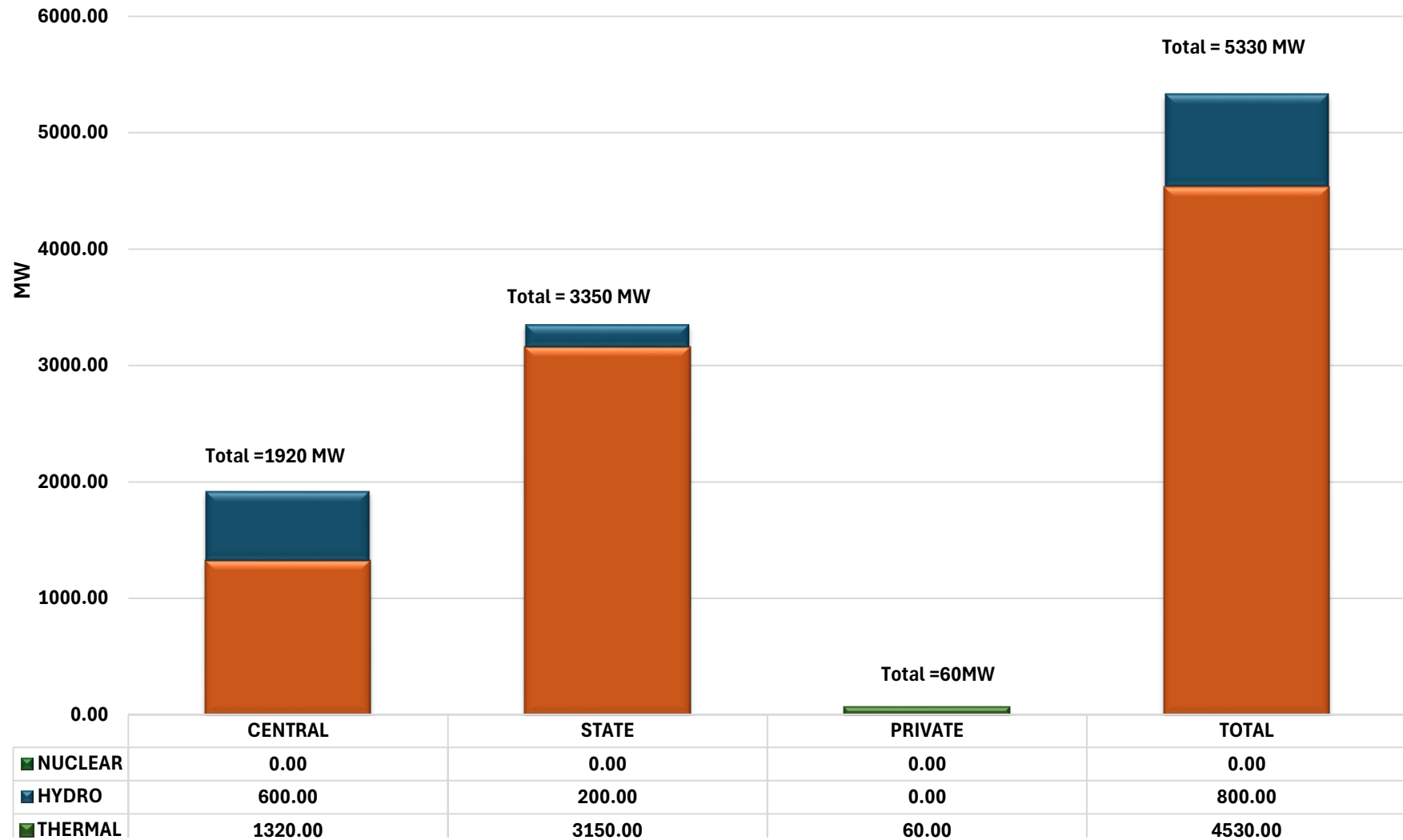
Demand of 0.5 MW & above

Chart : 34



Capacity Addition Programme 2024-25

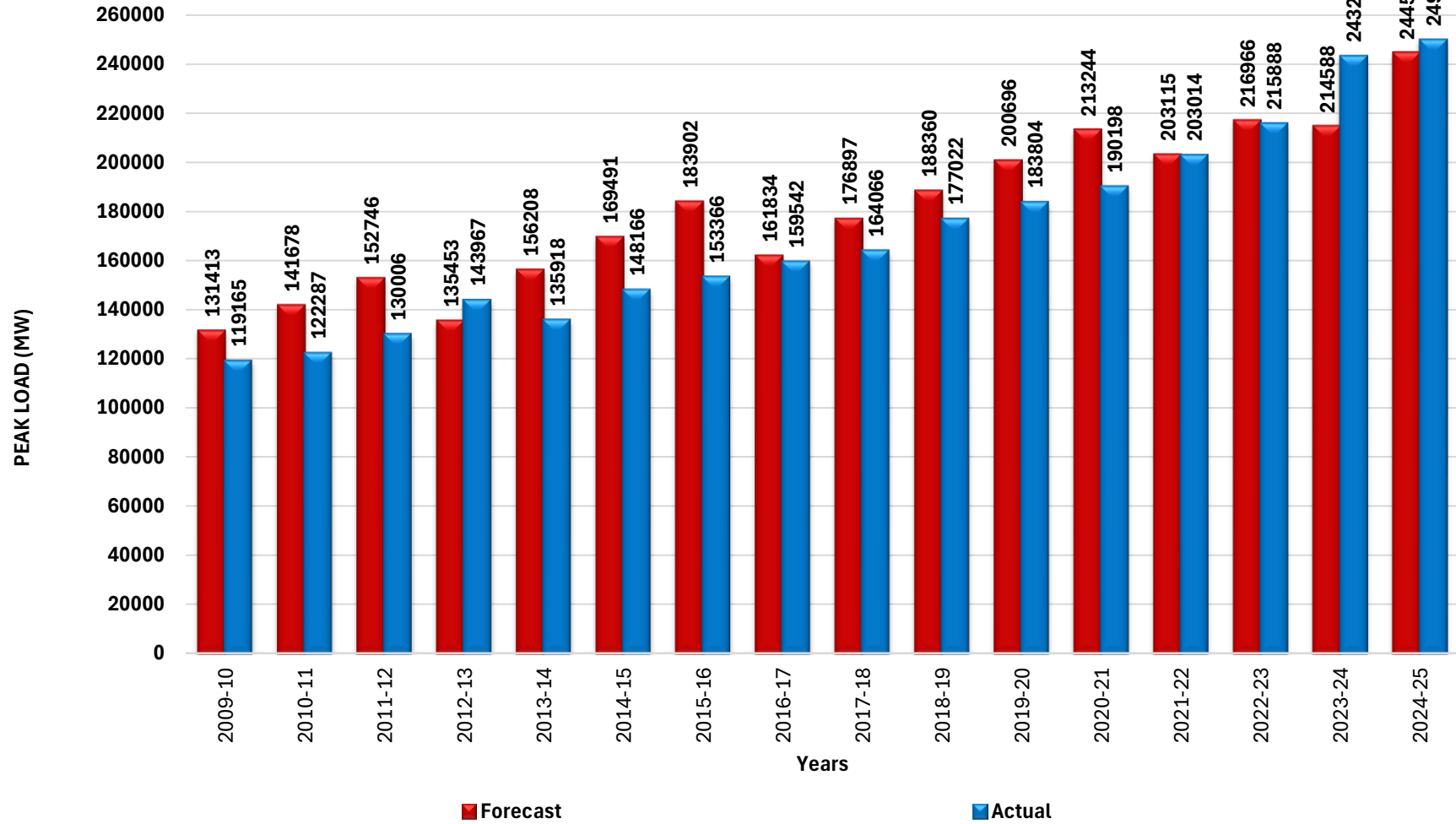
Chart : 35



Forecast of All India Peak Electricity Demand (Utilities)

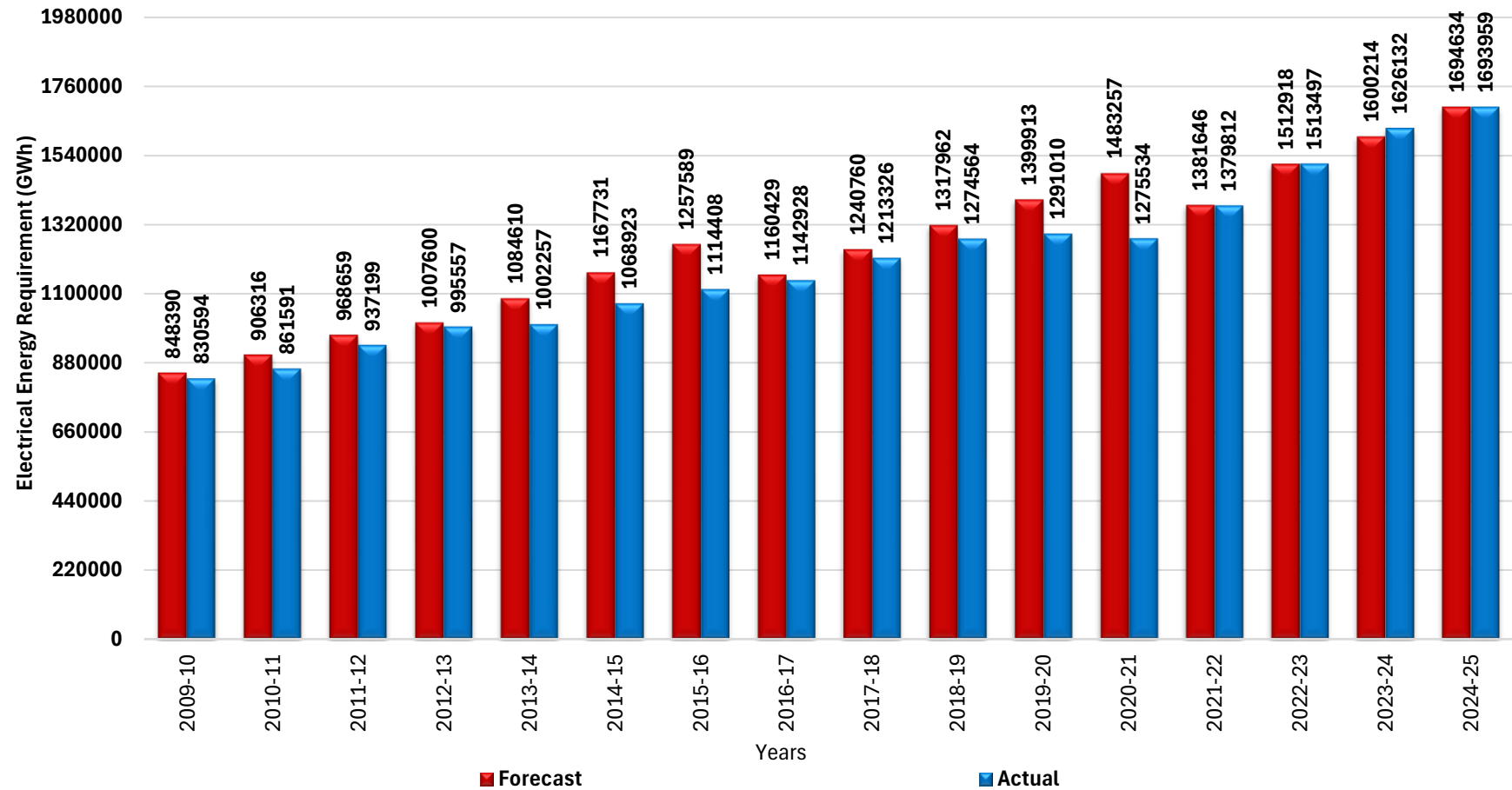
Chart : 36

As Per 20th Electric Power Survey in MW



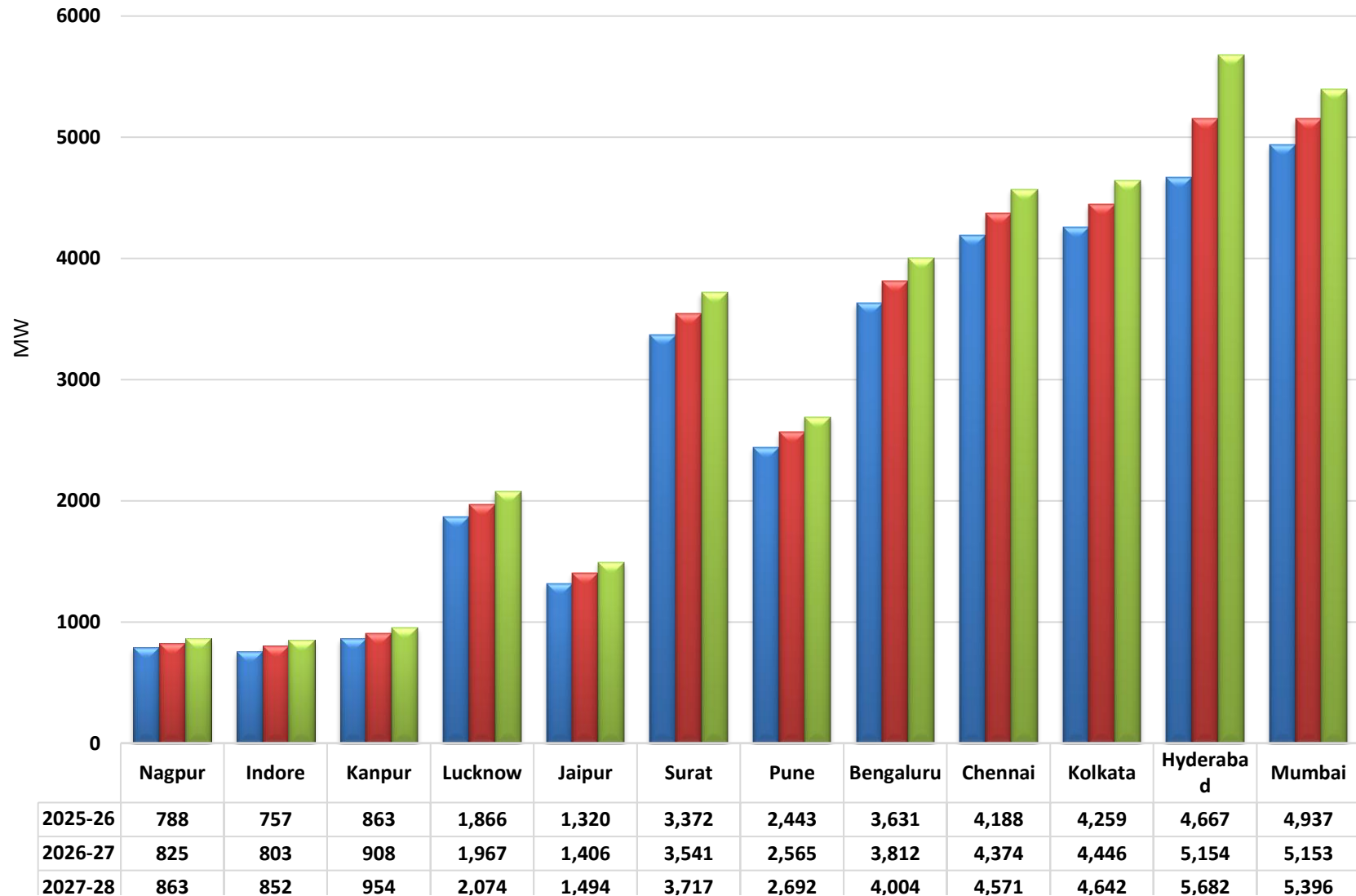
**Forecast of All India Energy Requirement
(Utilities)
As Per 20th Electric Power Survey
in GWh**

Chart : 37



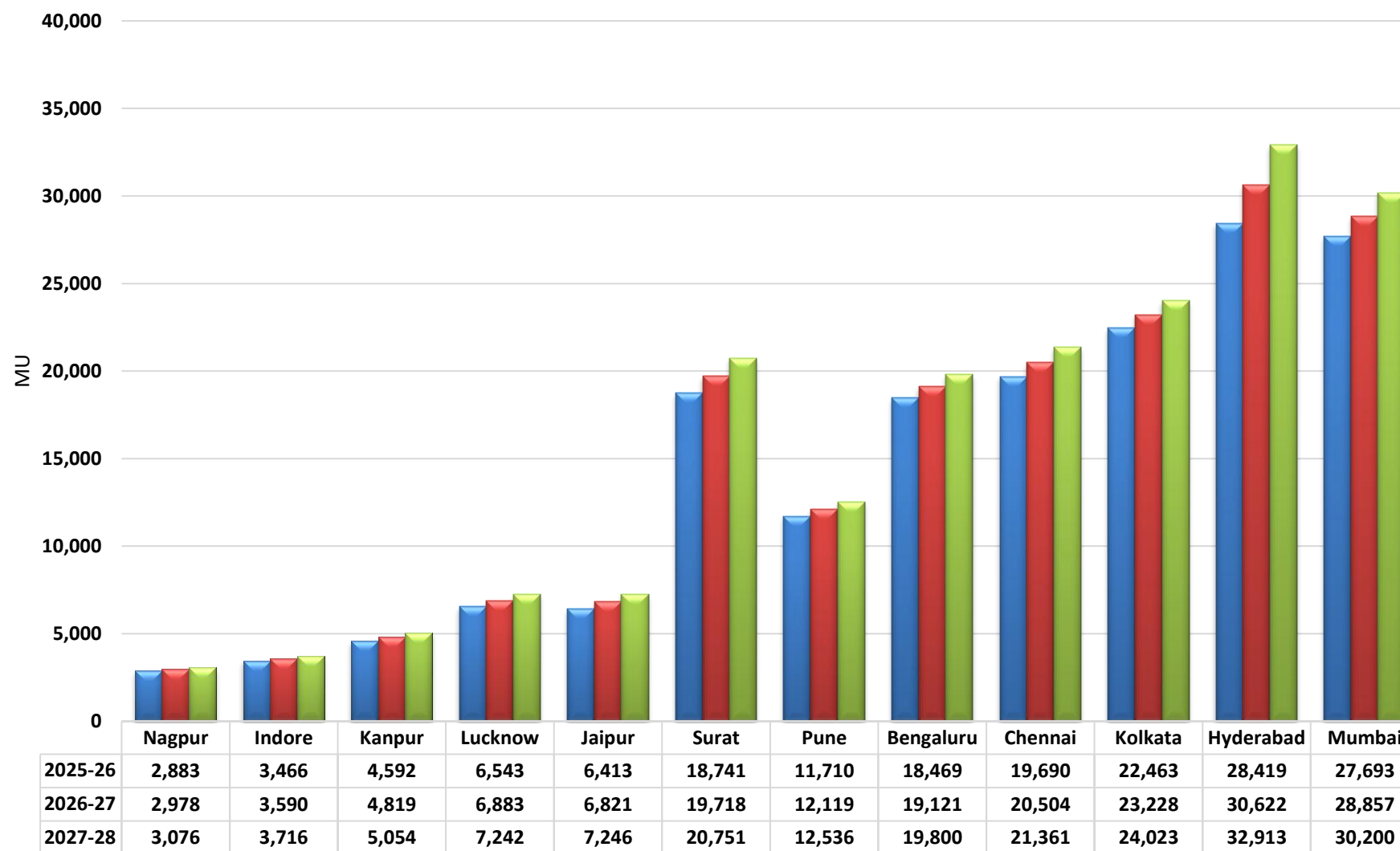
20th Electric Power Survey of India (Volume - III)
Annual Electric Peak Load (in MW) of Mega Cities at Power Station Bus Bar
For the year 2025-26, 2026-27 & 2027-28

Chart :37A



20th Electric Power Survey of India (Volume - III)
Electrical Energy Requirement (MU) - Mega Cities
(2025-26, 2026-27 & 2027-28)

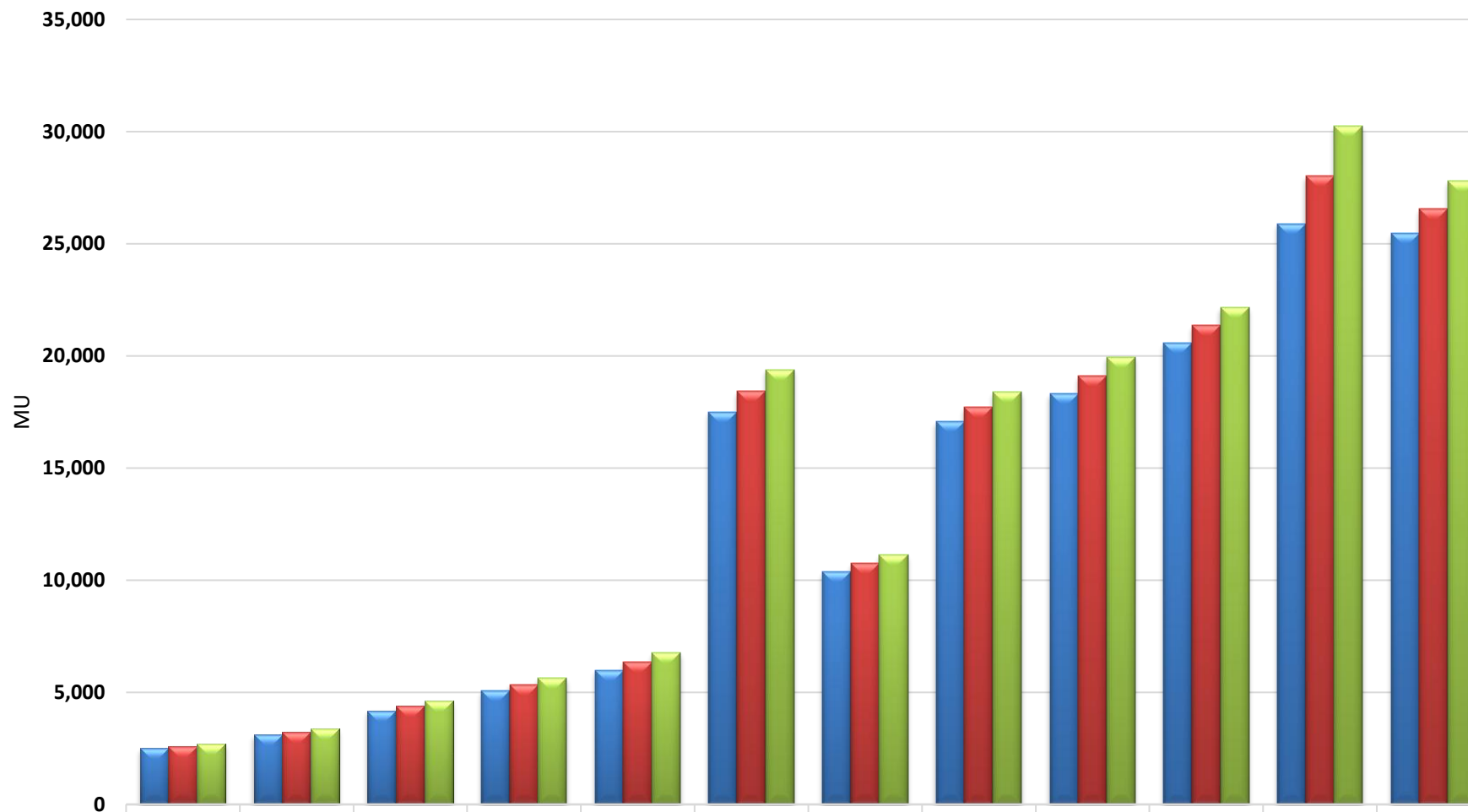
Chart : 37B



Axis Title

20th Electric Power Survey of India (Volume - III)
Electrical Energy Consumption (MU) Mega Cities
(2025-26, 2026-27 & 2027-28)

Chart : 37C



	Nagpur	Indore	Kanpur	Lucknow	Jaipur	Surat	Pune	Bengaluru	Chennai	Kolkata	Hyderabad	Mumbai
2025-26	2,506	3,119	4,171	5,068	5,980	17,506	10,365	17,069	18,305	20,588	25,896	25460
2026-27	2,597	3,242	4,387	5,345	6,367	18,418	10,744	17,715	19,102	21,356	28,013	26,550
2027-28	2,691	3,367	4,611	5,638	6,771	19,382	11,131	18,390	19,944	22,153	30,227	27,803

Table No.8
All India Power Supply Position Energy-wise & Peak-wise (Utilities)
1985-86 to 2024-25

Year	Energy				Peak Demand			
	Requirement	Availability	Deficit	Deficit	Demand	Availability	Deficit	Deficit
	(GWh)	(GWh)	(GWh)	(%)	(MW)	(MW)	(MW)	(%)
1985-86	1,70,746	1,57,262	13,484	7.90	28,090	24,215	3,875	13.79
1986-87	1,92,356	1,74,276	18,080	9.40	30,850	26,924	3,926	12.73
1987-88	2,10,993	1,87,976	23,017	10.91	31,990	28,242	3,748	11.72
1988-89	2,23,194	2,05,909	17,285	7.74	36,245	31,713	4,532	12.50
1989-90	2,47,762	2,28,151	19,611	7.92	40,385	33,658	6,727	16.66
1990-91	2,67,632	2,46,560	21,072	7.87	44,005	37,171	6,834	15.53
1991-92	2,88,974	2,66,432	22,542	7.80	48,055	39,027	9,028	18.79
1992-93	3,05,266	2,79,824	25,442	8.33	52,805	41,984	10,821	20.49
1993-94	3,23,252	2,99,494	23,758	7.35	54,875	44,830	10,045	18.31
1994-95	3,52,260	3,27,281	24,979	7.09	57,530	48,066	9,464	16.45
1995-96	3,89,721	3,54,045	35,676	9.15	60,981	49,836	11,145	18.28
1996-97	4,13,490	3,65,900	47,590	11.51	63,853	52,376	11,477	17.97
1997-98	4,24,505	3,90,330	34,175	8.05	65,435	58,042	7,393	11.30
1998-99	4,46,584	4,20,235	26,349	5.90	67,905	58,445	9,460	13.93
1999-00	4,80,430	4,50,594	29,836	6.21	72,669	63,691	8,978	12.35
2000-01	5,07,216	4,67,409	39,807	7.85	74,872	65,628	9,244	12.35
2001-02	5,22,537	4,83,350	39,187	7.50	78,441	69,189	9,252	11.79
2002-03	5,45,674	4,97,589	48,085	8.81	81,492	71,547	9,945	12.20
2003-04	5,59,264	5,19,398	39,866	7.13	84,574	75,066	9,508	11.24

Table No.8 (Contd.)
All India Power Supply Position Energy-wise & Peak-wise (Utilities)
1985-86 to 2024-25

Year	Energy				Peak Demand			
	Requirement	Availability	Deficit	Deficit	Demand	Availability	Deficit	Deficit
	(GWh)	(GWh)	(GWh)	(%)	(MW)	(MW)	(MW)	(%)
2004-05	5,91,373	5,48,115	43,258	7.31	87,906	77,652	10,254	11.66
2005-06	6,31,757	5,78,819	52,938	8.38	93,255	81,792	11,463	12.29
2006-07	6,90,587	6,24,495	66,092	9.57	1,00,715	86,818	13,897	13.80
2007-08	7,39,343	6,66,007	73,336	9.92	1,08,866	90,793	18,073	16.60
2008-09	7,77,039	6,91,038	86,001	11.07	1,09,809	96,785	13,024	11.86
2009-10	8,30,594	7,46,644	83,950	10.11	1,19,166	1,04,009	15,157	12.72
2010-11	8,61,591	7,88,355	73,236	8.50	1,22,287	1,10,256	12,031	9.84
2011-12	9,37,199	8,57,886	79,313	8.46	1,30,006	1,16,191	13,815	10.63
2012-13	9,95,557	9,08,652	86,905	8.73	1,35,453	1,23,294	12,159	8.98
2013-14	10,02,257	9,59,829	42,428	4.23	1,35,918	1,29,815	6,103	4.49
2014-15	10,68,923	10,30,785	38,138	3.57	1,48,166	1,41,160	7,006	4.73
2015-16	11,14,408	10,90,850	23,558	2.11	1,53,366	1,48,463	4,903	3.20
2016-17	11,42,928	11,35,332	7,596	0.66	1,59,542	1,56,934	2,608	1.63
2017-18	12,13,326	12,04,697	8,629	0.71	1,64,066	1,60,752	3,314	2.02
2018-19	12,74,595	12,67,526	7,070	0.55	1,77,022	1,75,528	1,494	0.84
2019-20	12,91,010	12,84,444	6,566	0.51	1,83,804	1,82,533	1,271	0.69
2020-21	12,75,534	12,70,663	4,871	0.38	1,90,198	1,89,395	802	0.42
2021-22	13,79,812	13,74,024	5,787	0.42	2,03,014	2,00,539	2,475	1.22
2022-23	15,13,497	15,05,914	7,583	0.50	2,15,888	2,07,231	8,657	4.01
2023-24	16,26,132	16,22,020	4,112	0.30	2,43,271	2,39,931	3,340	1.40
2024-25	16,93,969	16,92,369	1,590	0.10	2,49,856	2,49,854	2	0.00

**All India Peak and Energy Deficit
(Utilities)
1985-86 to 2024-25**

Chart : 38

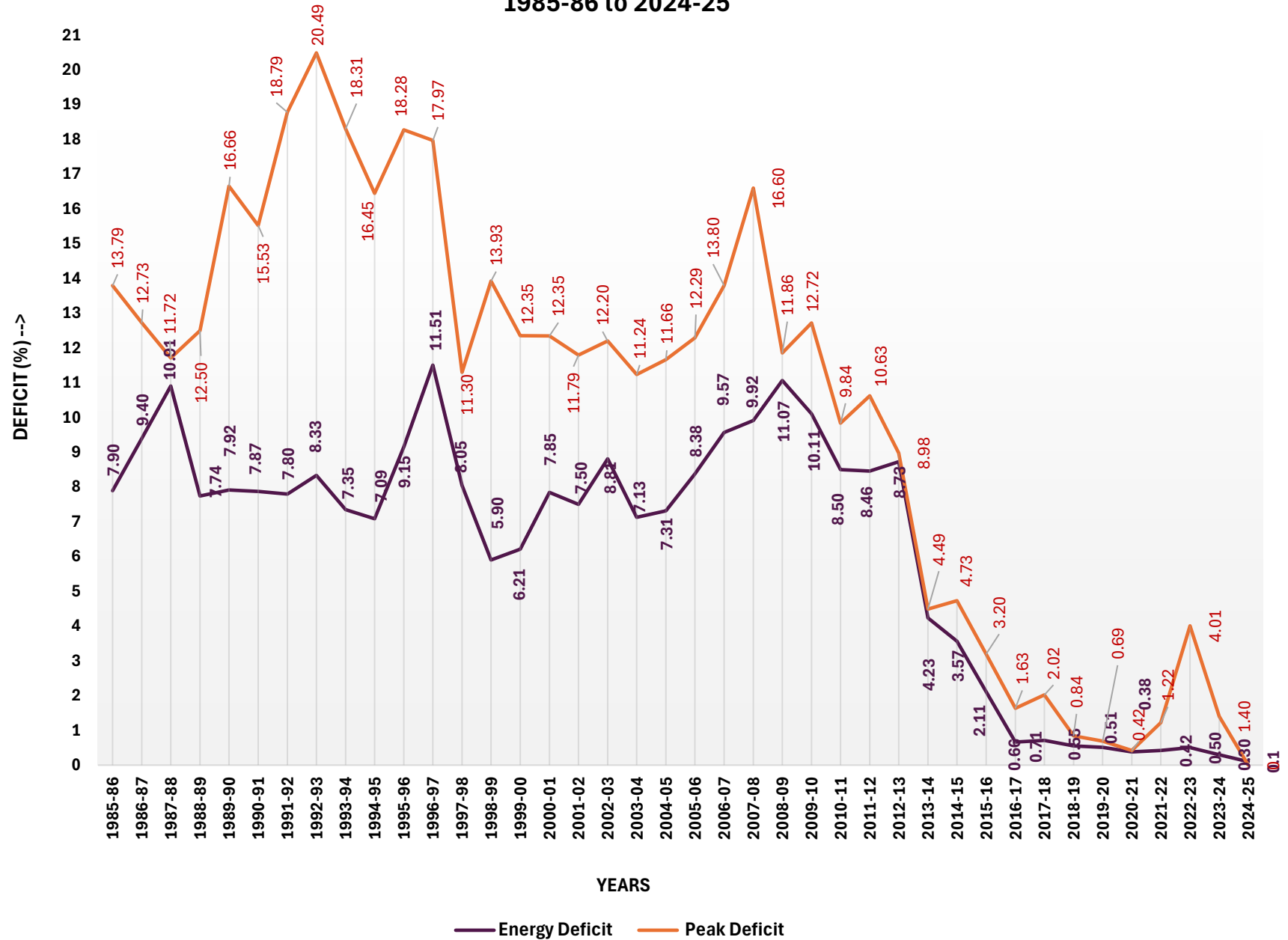


Table 8A
Electrical Energy Requirement (MU) & Peak Electricity Demand (MW) - Region-Wise
projections for the Year 2025-26 and 2026-27 As Per 20th EPS (Volume-I)

Region/Year	Electrical Energy Requirement (MU)		Peak Electricity Demand (MW)	
	2025-26	2026-27	2025-26	2026-27
Northern	5,59,218	5,92,312	92,476	97898
Western	5,60,416	5,96,793	83,986	89457
Southern	4,34,511	4,60,853	75,861	80864
Eastern	2,18,961	2,32,971	34,857	37265
North East	23,521	24,904	4,556	4855
All India Total	17,96,627	19,07,835	2,60,118	2,77,201

NOTE : EPS - Electric Power Survey

Table 8B
Forecast of Electrical Energy Consumption - Category-Wise
(2025-26 & 2026-27)
As Per 20th EPS (Volume-I)

Category	Year	Year
	2025-26	2026-27
Domestic	4,59,565	4,89,009
Commercial	1,28,855	1,36,081
Irrigation	3,04,367	3,20,374
Industries	4,32,956	4,71,281
Others	1,83,414	1,93,309
Total	15,09,157	16,10,053

Table 8C
Forecast of Electrical Energy Requirement (in MU) - NCR
(2025-26 2026-27)
As Per 20th EPS (Volume-II)

		GWh
NCR Sub Region	2025-26	2026-27
Delhi	40,771	42,566
Haryana	51,605	55,666
Rajasthan	14,757	16,245
Uttar Pradesh	39,922	43,113
NCR - Total	1,47,559	1,58,169

Chart : 39

Forecast of Electrical Energy Requirement (in MU) of NCR (2025-26 & 2026-27)

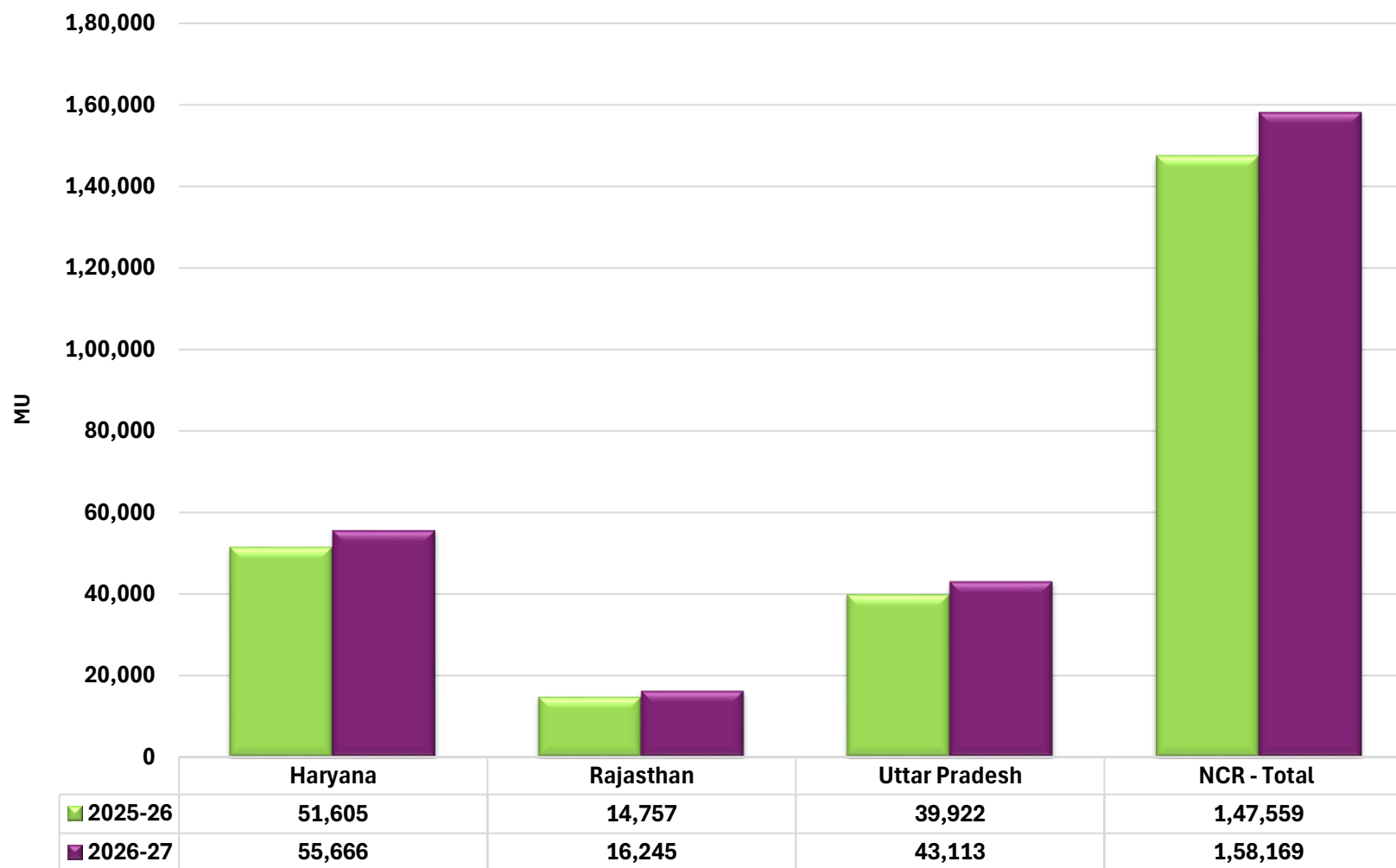
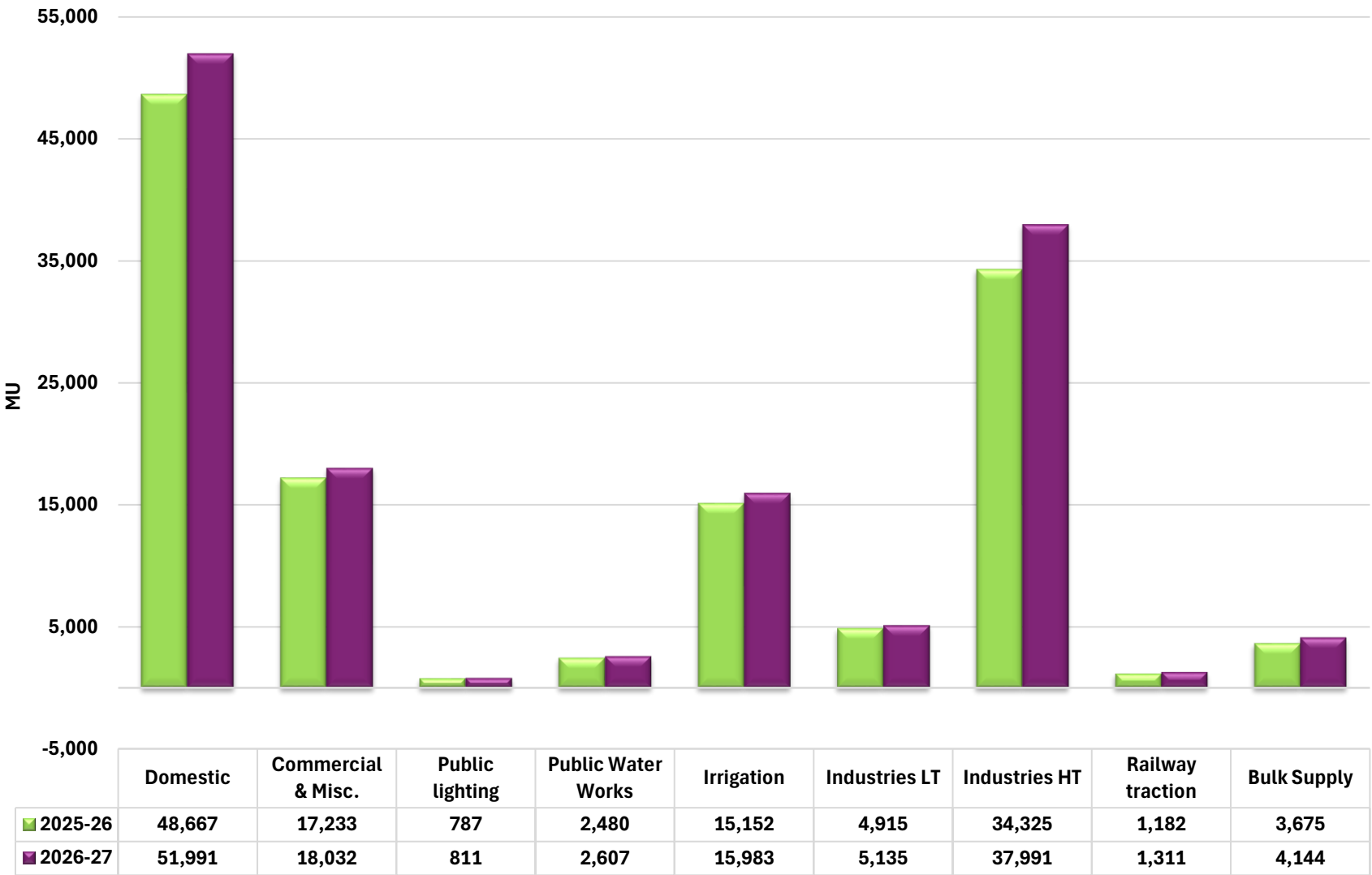


Table 8D
Forecast of Electrical Energy Consumption (in MU) of NCR
(2025-26 & 2026-27)
As Per 20th EPS (Volume-II)

GWh		
Consumption Category	2025-26	2026-27
Domestic	48,667	51,991
Commercial & Misc.	17,233	18,032
Public lighting	787	811
Public Water Works	2,480	2,607
Irrigation	15,152	15,983
Industries LT	4,915	5,135
Industries HT	34,325	37,991
Railway traction	1,182	1,311
Bulk Supply	3,675	4,144
Total (Energy Consumption)	1,28,416	1,38,005

Forecast of Electrical Energy Consumption (in MU) of NCR
(2025-26 & 2026-27)



MAP OF INDIA

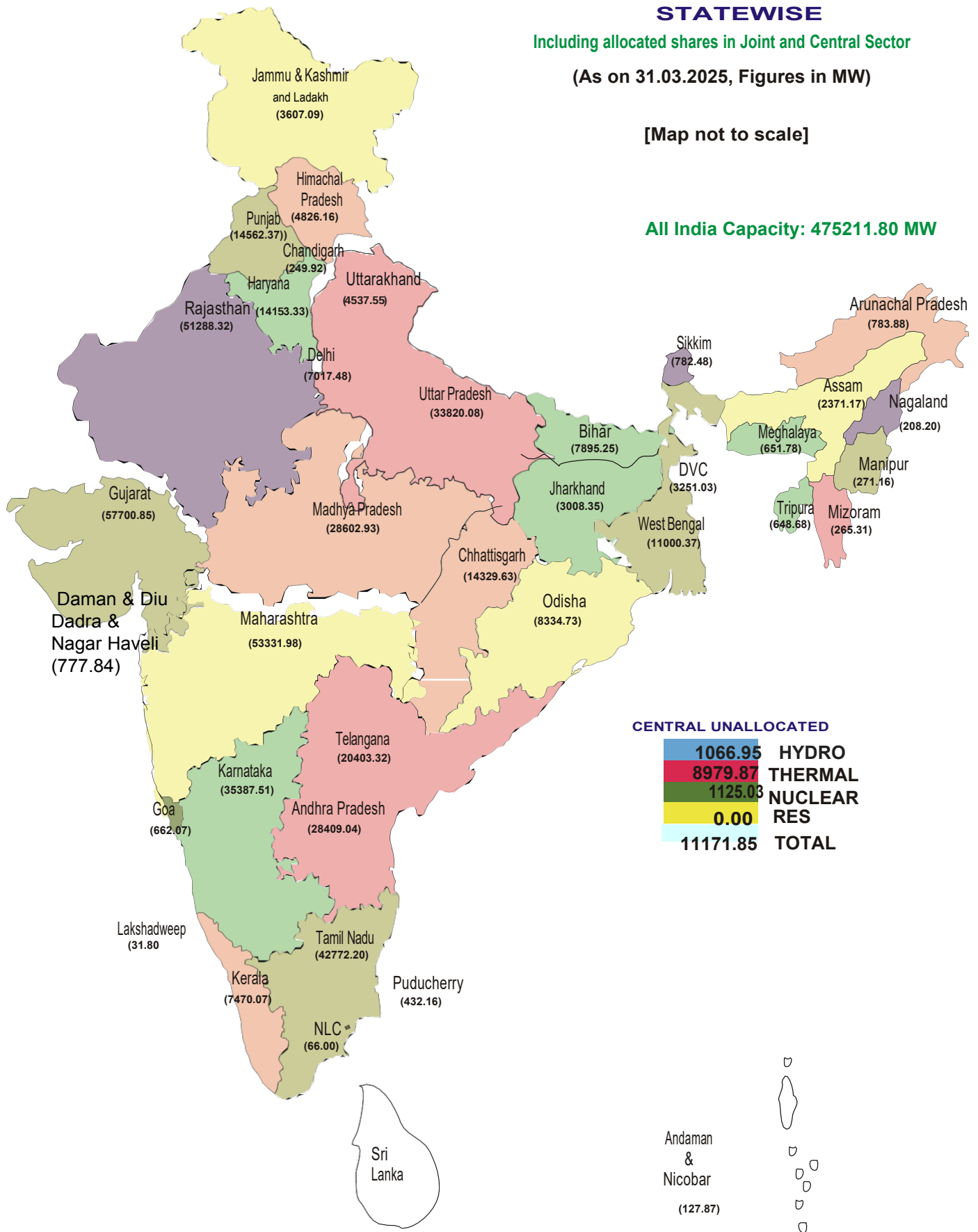
SHOWING INSTALLED GENERATING CAPACITY STATEWISE

Including allocated shares in Joint and Central Sector

(As on 31.03.2025, Figures in MW)

[Map not to scale]

All India Capacity: 475211.80 MW



MAP OF INDIA

SHOWING

INSTALLED GENERATING CAPACITY STATEWISE/MODEWISE

Including allocated shares in Joint and Central Sector
(As on 31.03.2025, Figures in MW)

[Map not as per scale]

All India Capacity: 475211.80 MW

